REVERSIBLE RULE 110 HAS NO HIDDEN SYMMETRIES: SIX FAMILIES OF SECOND-ORDER CELLULAR AUTOMATA SETTLED BY EDGE MOVES

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ABSTRACT. For an elementary rule f, the second-order automaton $F_f(x,y)_i = (y_i,\, x_i \oplus f(y_{i-1}, y_i, y_{i+1}))$ on $(\{0,1\}^2)^{\mathbb{Z}}$ is reversible, and the automorphism group of its spacetime subshift is the centralizer of F_f among the automorphisms of the full 4-shift. We prove that this group is $\langle \sigma, F_f \rangle \cong \mathbb{Z}^2$, generated by the shift and F_f, for the fourteen rules of the classes of 110, 14, 46 and 79 under reflection and complementation, and that it is $\langle \sigma, F_f \rangle \times \langle \kappa \rangle \cong \mathbb{Z}^2 \times \mathbb{Z}/2$, with κ the cellwise complement, for the four rules of the classes of 24 and 189; none of these rules is permutive in any variable. An automorphism is normalised by its offsets on kinks, and a block map commuting with F_f that fixes the kinks is shown to be the identity. The proof works at the stuck right edge of configurations with a constant right tail: kinks glued to such a configuration act as moves of its edge cell that commute with every such block map, and a controllability theorem for these moves, proved by induction, reads off the block map one column at a time. For the classes of 24 and 189 a factor map onto rule 46 carries the kinks and the edge moves to those of 46, and rule 189, which shares the fourth power of its automaton with rule 66, is handled by block maps commuting with a power of F. All proofs are by hand; no computer is used in any proof.

1. INTRODUCTION

Let $\mathcal{A}$ be a finite alphabet and σ the left shift on $\mathcal{A}^{\mathbb{Z}}$. By the Curtis–Hedlund–Lyndon theorem [16] the endomorphisms of the full shift are the block maps, and its automorphisms are the invertible block maps [21, 4]. The automorphism group $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ is very large: it contains every finite group and the free group on two generators, it is residually finite [2], it embeds in the automorphism group of every nontrivial mixing shift of finite type [19], and its center is generated by the shift [26]; see also [8, 15]. Centralizers of single block maps were studied by Coven, Hedlund and Rhodes [6]. A reversible cellular automaton $F \in \mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ determines a two-dimensional subshift, its *spacetime* $X_F \subset \mathcal{A}^{\mathbb{Z}^2}$, whose rows are the configurations $F^n(c)$, $n \in \mathbb{Z}$; the entropy geometry of such spacetime configurations was studied by Milnor [23], and the spacetime of a general endomorphism was studied by Cyr, Franks and Kra [7], and the joint action of a block map and the shift by Nasu [25] and, through its expansive directions, by Boyle and Lind [3]. The automorphism group of X_F is the centralizer of F in $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ (Lemma 2.2), and it always contains $\langle \sigma, F \rangle$. We ask whether it contains anything else, for specific nonlinear reversible automata.

The automata we consider are the second-order rules of Fredkin [22, 30, 32]; for reversible automata in general see [29, 18, 24]. As a family, the second-order elementary rules were introduced and studied systematically by Takesue [27, 28]. For an elementary rule $f\colon \{0,1\}^3 \to \{0,1\}$, the second-order automaton F_f acts on $(\{0,1\}^2)^{\mathbb{Z}}$ by the recursion $y_{t+1} = y_{t-1} \oplus f(y_t)$ (equation (1)); it is reversible for every f, and, following Takesue [27], one writes fR for it. The elementary rule 110 is computationally universal [5]; whether the second-order automaton F_{110} is universal is, as

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far as we know, open, and nothing below depends on it. Reversible second-order automata with complex behaviour, such as 37R, are discussed in [32, pp. 437–457].

Theorem 1.1. (a) *Let f be one of the rules*

$$62,\ 110,\ 118,\ 124 \qquad 14,\ 42,\ 84,\ 112 \qquad 46,\ 116 \qquad 79,\ 93,\ 186,\ 242.$$

Then $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle$, *and* $(a, b) \mapsto \sigma^a F_f^b$ *is an isomorphism* $\mathbb{Z}^2 \to \mathrm{Aut}(X_{F_f})$.

(b) *Let f be one of the rules*

$$24,\ 66 \qquad 189,\ 231.$$

Then the cellwise complement $\kappa(x, y) = (1-x, 1-y)$ *commutes with* σ *and* F_f, $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \times \langle \kappa \rangle$, *and* $(a, b, j) \mapsto \sigma^a F_f^b \kappa^j$ *is an isomorphism* $\mathbb{Z}^2 \times \mathbb{Z}/2 \to \mathrm{Aut}(X_{F_f})$.

The groups of rules in (a) are the classes of 110, 14, 46 and 79 under reflection and complementation, and those in (b) are the classes of 24 and 189; these operations conjugate the automorphism groups to each other (Section 2.3). The theorem combines Theorems 4.14, 5.10, 5.15, 6.18 and 7.22. None of the eighteen rules is permutive in any variable, so a local difference does not spread at a fixed speed.

Context. Known techniques for determining automorphism groups of $\mathbb{Z}^2$-subshifts do not apply here: the complexity arguments [9, 14] are for one-dimensional subshifts whose word complexity grows at most linearly or is not superlinear, while every configuration in $\mathcal{A}^{\mathbb{Z}}$ is a row of X_F; algebraic rigidity, which gives $\mathbb{Z}^2$ for Ledrappier's shift [20] ([31], [1, Theorem 7]), needs a group shift; and Hochman's theorem [17] needs positive $\mathbb{Z}^2$-entropy, while X_F has entropy zero. For context only, and not used below: the author announces in [10] the analogous result for the nonlinear rules permutive in an outer variable, such as rule 30, by a method that follows the front at which a local difference spreads at exact speed 1, and in [11] that for rule 54, the other elementary rule for which Wolfram reports signs of class 4 behaviour [32, pp. 693–697], the second-order automaton has words fixed in every context, which embed every finite group in $\mathrm{Aut}(X_{F_{54}})$. A SAT enumeration announced in [12] finds, for rule 110 and radius at most 4, no block map commuting with F_{110} other than the maps $\sigma^a F_{110}^b$ and the two constant maps onto $(0, 0)^{\mathbb{Z}}$ and $(1, 1)^{\mathbb{Z}}$; Section 8 states what follows rigorously for endomorphisms.

The method. The proof has two parts.

Normalisation. For $f = 110$ the fixed points of σF other than the two constants are the kinks $\eta_p = (0, 0)^{(-\infty,p)}(1, 0)_p(1, 1)^{(p,\infty)}$, and those of $\sigma^{-1}F$ are their images $\rho\eta_p$ under the time reversal $\rho(x, y) = (y, x)$ (Lemma 3.1). Every automorphism g shifts the two families of kinks by integers $e^+(g)$ and $e^-(g)$ (Proposition 3.2); a parity argument on the four constant configurations shows that $e^+ \equiv e^-$ (mod 2), so g is brought to offsets $(0, 0)$ by an element of $\langle \sigma, F \rangle$ (Proposition 3.4). It remains to prove *Lemma K* (Section 4): a block map h that commutes with F and fixes η_0 and $\rho\eta_0$ is the identity.

The stuck edge and edge moves. Consider a configuration c that vanishes to the right of a last nonzero cell r. Since $f(a, 0, 0) = 0$, the edge r never moves, and the edge column evolves autonomously. Gluing a kink $\rho\eta_N$ far to the right of c and following the glued configuration in time, the kink bounces off the edge and leaves behind a kink η moving to the right whose position records the value of the edge column at one time (Lemma 4.1). Since h fixes the kinks, it preserves that position; this shows that h does not change the edge column (Proposition 4.2). The same gluing, done at an arbitrary time and followed by passage to the configuration left behind, gives a *probe*; it changes c only in its edge cell at one time, by an explicit rule depending on the two cells at the edge, and it commutes with h (Lemma 4.4). So h commutes with a family of *edge moves*, operators defined by F alone. If h is known to act as the identity on the last k columns of every such configuration, a composite of moves that sets two cells of row 0 just inside

those columns to 1 makes the column next to them readable through the relation $f(a,1,1)=\neg a$ (Lemma 4.6), and induction on k gives Lemma K.

Controllability. What is needed is therefore that the moves at the times $|m|\le k-1$ can reach such a configuration from every starting point. The controllability theorem (Theorem 4.10) describes the orbits of the moves exactly: they reach every configuration that agrees with c outside the last k cells and has a nonzero edge cell, with one exception, a frozen tip next to an exterior cell $(0,0)$. The proof is by induction on k, starting at $k=1$: the moves at the times $|m|\le k-1$ contain the moves of level $k-1$ applied to c, which change the cells behind the tip, and the same moves conjugated by $F^{\pm1}$, which change the tip by an explicit formula (Lemma 4.9); four elementary steps, each a one-line computation with the values of f, connect all the allowed tips.

For the class of 14 the same steps apply to the rule 42, with different moves and a frozen exterior cell $(1,1)$ (Theorem 5.4); the automorphisms of $X_{F_{14}}$ are normalised by offsets on three families of fixed configurations, related by a collision of kinks (Proposition 5.9), and transported to 42 by reflection and complementation. For 46 only the moves $01\leftrightarrow10$ remain, and the orbits are cut out exactly (Theorem 5.13) by a conserved bit on each pair of adjacent cells, which is the local conservation law of 46R found by Takesue [28, (4.4)–(4.6)]. For the class of 79 the background to the right of the edge is not a constant but a vacuum of period 2 in space and 6 in time, the kinks move at speed 1/3, and the moves switch the phase of the vacuum tail (Section 7). For the classes of 24 and 189 the complement κ commutes with F_{66} and F_{189}; the block map $\delta(c)_i=c_i\oplus c_{i+1}$ satisfies $\delta F_{66}=F_{46}\delta=\delta F_{189}$, maps the kinks of 66 to those of 46 and the edge moves of 66, which complement the constant tail of a configuration, to the moves of 46; so the controllability theorem for 46 applies without change. Since $F_{189}^4=F_{66}^4$, while $\sigma^{\pm1}F_{189}$ have no fixed points, Lemma K is proved for block maps commuting with a power of F_{66} (Section 6).

Proofs by hand. All proofs are by hand, for every radius; no computer is used in any proof. For the class of 79 the induction of the controllability theorem starts at level 4, from a description of the orbits of the moves at level 3 (Proposition 7.19), which is derived from a table of the moves at the times $|m|\le3$. Computer checks of statements proved in the paper are specified in Appendix A; code and output accompany the paper.

Organisation. Section 2 fixes notation, proves that $\mathrm{Aut}(X_F)$ is the centralizer of F, and recalls the diagram of a configuration and the conjugacies between rules. Section 3 defines the offsets on kinks and reduces the main theorem to the triviality of zero-offset maps, for a family of rules containing 110, 42 and 46. Section 4 proves Lemma K for 110 and the theorem for its class. Section 5 treats the classes of 14 and 46, Section 6 the classes of 24 and 189, by a factor map onto 46, and Section 7 the class of 79. Section 8 discusses the scope of the method, endomorphisms, reversing symmetries and open problems. Appendix A specifies computer checks that confirm statements proved in the paper; none of them is used in a proof.

2. Preliminaries

2.1. **Second-order automata and their spacetime.** Throughout, $\mathcal{A}=\{0,1\}^2$, and a cell $(x,y)\in\mathcal{A}$ is also written as the integer $2x+y\in\{0,1,2,3\}$, or as the word xy; thus $3=(1,1)$ and $0=(0,0)$. For a symbol s, a cell or a bit, and an interval $I\subseteq\mathbb{Z}$, s^I denotes the sequence indexed by I with every entry equal to s, and s_p the symbol s at the site p; juxtaposition concatenates such pieces. For a cell s, $s^{\mathbb{Z}}$ is the constant configuration with value s. For example, $(0,0)^{(-\infty,p)}(1,0)_p(1,1)^{(p,\infty)}$ is the configuration equal to $(0,0)$ left of p, to $(1,0)$ at p and to $(1,1)$ right of p. The left shift is $\sigma(c)_i=c_{i+1}$. A *block map of radius* R is a map $g\colon\mathcal{A}^{\mathbb{Z}}\to\mathcal{A}^{\mathbb{Z}}$ of the form $g(c)_i=\gamma(c_{i-R},\dots,c_{i+R})$ for a local rule $\gamma\colon\mathcal{A}^{2R+1}\to\mathcal{A}$. A block map of radius R is also a block map of every larger radius. By the Curtis–Hedlund–Lyndon theorem [16], $\mathrm{End}(\mathcal{A}^{\mathbb{Z}})$ is the

monoid of block maps and $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ the group of invertible ones; every automorphism has finite radius.

Let $f\colon \{0,1\}^3 \to \{0,1\}$ be an elementary rule, identified with its Wolfram number, the integer $\sum f(a,b,c)\,2^{4a+2b+c}$ over $(a,b,c) \in \{0,1\}^3$. Define F_f on $\mathcal{A}^{\mathbb{Z}}$ by

$$F_f(c)_i = \big(y_i,\ x_i \oplus f(y_{i-1}, y_i, y_{i+1})\big), \qquad c_i = (x_i, y_i). \tag{1}$$

Reading y as the current row and x as the previous one, this is the recursion $y_{t+1} = y_{t-1} \oplus f(y_t)$. It is reversible, with $F_f^{-1}(c)_i = \big(y_i \oplus f(x_{i-1}, x_i, x_{i+1}),\ x_i\big)$. It has radius 1, and $\sigma^a F_f^b$ has radius at most $|a| + |b|$. We write the values of f as the list $f(0,0,0), f(0,0,1), \ldots, f(1,1,1)$.

Definition 2.1. The *spacetime subshift* of a reversible cellular automaton F on $\mathcal{A}^{\mathbb{Z}}$ is

$$X_F = \big\{(F^n(c))_{n\in\mathbb{Z}} :\ c \in \mathcal{A}^{\mathbb{Z}}\big\} \subset \mathcal{A}^{\mathbb{Z}^2},$$

with the $\mathbb{Z}^2$-action generated by the horizontal shift σ and the vertical shift τ, $(\tau z)_n = z_{n+1}$. An *automorphism* of X_F is a homeomorphism of X_F commuting with both.

Lemma 2.2. *For reversible F the map $c \mapsto (F^n(c))_{n\in\mathbb{Z}}$ is a homeomorphism $\mathcal{A}^{\mathbb{Z}} \to X_F$ conjugating (σ, F) to (σ, τ), and it induces an isomorphism*

$$\mathrm{Aut}(X_F) \ \cong\ \{g \in \mathrm{Aut}(\mathcal{A}^{\mathbb{Z}}) :\ gF = Fg\}.$$

Under this isomorphism $\langle \sigma, \tau\rangle$ corresponds to $\langle \sigma, F\rangle$.

Proof. The map is continuous, injective (the 0-th row recovers c), and surjective onto X_F by definition, hence a homeomorphism of compact spaces; it carries σ to σ and F to τ since $F^n(F(c)) = F^{n+1}(c)$. A homeomorphism of $\mathcal{A}^{\mathbb{Z}}$ commuting with σ is a block map by Curtis–Hedlund–Lyndon, and commuting with τ means commuting with F. □

We identify $\mathrm{Aut}(X_F)$ with the group of invertible block maps commuting with F. Block maps commuting with F that are not invertible also occur, for instance the constant maps onto fixed points of F; Lemma K of Section 4 applies to them as well.

2.2. **Diagrams and time reversal.** Fix f and $F = F_f$. For $c \in \mathcal{A}^{\mathbb{Z}}$ and $t \in \mathbb{Z}$ let $y^t \in \{0,1\}^{\mathbb{Z}}$ be the y-row of $F^t(c)$. By (1) the x-row of $F(d)$ is the y-row of d, so the x-row of $F^t(c)$ is y^{t-1}, and in particular $c_i = (y_i^{-1}, y_i^0)$. The *diagram* of c is $Y_c\colon \mathbb{Z}^2 \to \{0,1\}$, $Y_c(t,i) = y_i^t$; so

$$(F^t c)_i = \big(Y_c(t-1,i),\ Y_c(t,i)\big),$$

and we call the configuration $F^t c$ the *state at time t*; its two rows are the rows $t-1$ and t of Y_c. By (1), every diagram satisfies the *spacetime relation*

$$Y(t+1,i) \oplus Y(t-1,i) = f\big(Y(t,i-1), Y(t,i), Y(t,i+1)\big) \qquad \text{for all } (t,i) \in \mathbb{Z}^2. \tag{2}$$

Each instance determines either time neighbour $Y(t \pm 1, i)$ from the other four values, so any two consecutive rows determine all rows, $c \mapsto Y_c$ is a bijection from $\mathcal{A}^{\mathbb{Z}}$ onto the set of solutions of (2), and two diagrams that agree on two consecutive rows are equal. Translations act by $Y_{\sigma^j F^k c}(t,i) = Y_c(t+k,\ i+j)$. If g is a block map commuting with F, then $F^t g(c) = g(F^t c)$, so rows $t-1$ and t of Y_{gc} are the two rows of $g(F^t c)$.

Time reversal. Let $\rho\colon \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ exchange the two bits of every cell, $\rho(c)_i = (y_i, x_i)$. That this exchange reverses time is noted in [27, p. 2500]; we record the computation. From (1), $F(\rho c)_i = \big(x_i,\ y_i \oplus f(x_{i-1}, x_i, x_{i+1})\big)$, so $\rho F \rho(c)_i = \big(y_i \oplus f(x_{i-1}, x_i, x_{i+1}),\ x_i\big) = F^{-1}(c)_i$. Thus $\rho F \rho = F^{-1}$ for every f, and $\rho\sigma = \sigma\rho$. Consequently $\rho g \rho$ is a block map commuting with F whenever g is, and $g \mapsto \rho g \rho$ is an automorphism of $\mathrm{Aut}(X_F)$. Since $F^t \rho = \rho F^{-t}$ and the y-row of ρd is the x-row of d,

$$Y_{\rho c}(t,i) = Y_c(-1-t,\ i). \tag{3}$$

2.3. **Conjugacies between rules.** The cellwise complement $\kappa(x,y)=(1-x,1-y)$ conjugates F_f to $F_{\kappa f}$ with $\kappa f(a,b,c)=f(1-a,1-b,1-c)$, as (1) shows, and it commutes with σ. Unlike the complementation in Wolfram's equivalence of elementary rules, which takes f to $1-f(1-a,1-b,1-c)$, the map $f\mapsto\kappa f$ does not complement the output, so the classes below are not Wolfram's classes [28, p. 377 and footnote 3]; for instance, Wolfram's class of 110 is $\{110,124,137,193\}$. Under κ and the reflection the 256 rules form the 88 classes of Takesue [27], [28, Table I]. The reflection $\mathrm{rev}(c)_i=c_{-i}$ conjugates F_f to F_{f^r} with $f^r(a,b,c)=f(c,b,a)$: from (1), $\mathrm{rev}\,F_f\,\mathrm{rev}(c)_i=F_f(\mathrm{rev}\,c)_{-i}=(y_i,\,x_i\oplus f(y_{i+1},y_i,y_{i-1}))$; and $\mathrm{rev}\,\sigma\,\mathrm{rev}=\sigma^{-1}$. Both conjugacies preserve the radius of block maps. Hence, for $\theta\in\{\kappa,\mathrm{rev}\}$ and f' the conjugate rule, $g\mapsto\theta g\theta^{-1}$ maps $\mathrm{Aut}(X_{F_f})$ isomorphically onto $\mathrm{Aut}(X_{F_{f'}})$, and it maps $\langle\sigma,F_f\rangle$ onto $\langle\sigma,F_{f'}\rangle$. So the statement $\mathrm{Aut}(X_{F_f})=\langle\sigma,F_f\rangle\cong\mathbb{Z}^2$ depends only on the class of f under these two operations. Moreover κ commutes with rev, so $g\mapsto\theta g\theta^{-1}$ maps κ to itself. If $\kappa f=f$, then κ commutes with σ and F_f, and the same holds for every rule f' of the class of f, since $\kappa f^r=(\kappa f)^r$; so the statement $\mathrm{Aut}(X_{F_f})=\langle\sigma,F_f\rangle\times\langle\kappa\rangle\cong\mathbb{Z}^2\times\mathbb{Z}/2$ also depends only on the class of f. The classes of Theorem 1.1 are $\{62,110,118,124\}$, $\{14,42,84,112\}$, $\{46,116\}$ and $\{79,93,186,242\}$ in (a), and $\{24,66\}$ and $\{189,231\}$ in (b).

2.4. **Fixed points of $\sigma^{\pm1}F$.**

Lemma 2.3. *For an elementary rule f let $\Phi_f\subseteq\{0,1\}^{\mathbb{Z}}$ be the set of sequences φ with*

$$\varphi(k-1)\oplus\varphi(k+1)=f\big(\varphi(k-1),\varphi(k),\varphi(k+1)\big)\qquad\text{for all } k\in\mathbb{Z}.$$

The maps $\varphi\mapsto c$, $c_i=(\varphi(i+1),\varphi(i))$, and $\varphi\mapsto d$, $d_i=(\varphi(i-1),\varphi(i))$, are bijections from Φ_f onto the fixed points of σF_f and of $\sigma^{-1}F_f$ respectively.

Proof. If $c_i=(\varphi(i+1),\varphi(i))$, then by (1), $F_f(c)_i=\big(\varphi(i),\,\varphi(i+1)\oplus f(\varphi(i-1),\varphi(i),\varphi(i+1))\big)$, which is $(\varphi(i),\varphi(i-1))=c_{i-1}$ exactly when the displayed equation holds at $k=i$. So $F_fc=\sigma^{-1}c$, that is $\sigma F_fc=c$. Conversely, if $\sigma F_fc=c$, then $\varphi(i)=y(c_i)$ satisfies $x(c_i)=y(F_f^{-1}c)_i=y(\sigma c)_i=\varphi(i+1)$, and the equation holds by the same computation. The case of $\sigma^{-1}F_f$ is identical with $F_fd=\sigma d$. □

We call a triple $(a_1,a_2,a_3)\in\{0,1\}^3$ *allowed* for f if $a_1\oplus a_3=f(a_1,a_2,a_3)$; so Φ_f is the set of sequences all of whose triples of consecutive values are allowed.

3. Kinks and offsets

In this section f is an elementary rule with

$$f(0,0,0)=f(1,0,0)=f(1,1,1)=0,\qquad f(0,0,1)=f(0,1,1)=f(1,0,1)=1,\tag{4}$$

and $F=F_f$. The rules 110 (values $0,1,1,1,0,1,1,0$), 42 (values $0,1,0,1,0,1,0,0$) and 46 (values $0,1,1,1,0,1,0,0$) satisfy (4). By (4) the triples 000, 001, 011 and 111 are allowed, and 100 and 101 are not, so no allowed triple begins with 10.

Lemma 3.1 (kinks)**.** *For $p\in\mathbb{Z}$ let η_p be the configuration with $(\eta_p)_i=(0,0)$ for $i<p$, $(\eta_p)_p=(1,0)$ and $(\eta_p)_i=(1,1)$ for $i>p$. The fixed points of σF are $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ and the η_p, and $F\eta_p=\eta_{p+1}$. The fixed points of $\sigma^{-1}F$ are $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ and the $\rho\eta_p$, and $F\rho\eta_p=\rho\eta_{p-1}$.*

Proof. By Lemma 2.3 it suffices to determine Φ_f. No allowed triple begins with 10, so a sequence in Φ_f never contains 10 and is nondecreasing: it is $0^{\mathbb{Z}}$, $1^{\mathbb{Z}}$, or $\varphi(k)=[k>p]$ for some p, and all of these are in Φ_f, since they use only the triples 000, 001, 011 and 111. Under $c_i=(\varphi(i+1),\varphi(i))$ these give $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ and η_p, and $F\eta_p=\sigma^{-1}\eta_p=\eta_{p+1}$. For $\sigma^{-1}F$, $d_i=(\varphi(i-1),\varphi(i))$ gives $(0,0)$ for $i\le p$, $(0,1)$ at $p+1$ and $(1,1)$ beyond, which is $\rho\eta_{p+1}$; and $F\rho\eta_p=\sigma\rho\eta_p=\rho\eta_{p-1}$. □

Here and below, $[S] \in \{0,1\}$ is 1 if the statement S holds and 0 otherwise.

Proposition 3.2 (offsets)**.** *Every $g \in \mathrm{Aut}(X_F)$ fixes $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, and there are integers $e^+(g)$, $e^-(g)$ with*

$$g\,\eta_p = \eta_{p+e^+(g)}, \qquad g\,\rho\eta_p = \rho\eta_{p+e^-(g)} \qquad (p \in \mathbb{Z}).$$

The maps $e^{\pm}\colon \mathrm{Aut}(X_F) \to \mathbb{Z}$ are homomorphisms, $(e^+, e^-)(\sigma^a F^b) = (b-a, -a-b)$, and $|e^{\pm}(g)| \le R$ if g has radius R. Consequently $(a,b) \mapsto \sigma^a F^b$ is an isomorphism $\mathbb{Z}^2 \to \langle \sigma, F\rangle$; in particular F has infinite order.

Proof. Since $f(0,0,0) = f(1,1,1) = 0$, F maps a constant configuration $(x,y)^{\mathbb{Z}}$ to $(y,x)^{\mathbb{Z}}$, so the constants fixed by both σ and F are $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, and g permutes these two. It also permutes the fixed points of σF, hence, by Lemma 3.1, the set $\{\eta_p\}$. If g exchanged the two constants, then $g\eta_0$, which agrees with $g((0,0)^{\mathbb{Z}})$ far to the left because g is a block map, would be left-asymptotic to $(1,1)^{\mathbb{Z}}$; but every η_q is left-asymptotic to $(0,0)^{\mathbb{Z}}$. So g fixes both constants, $g\eta_0 = \eta_e$ for some $e = e^+(g)$, and $g\eta_p = g\sigma^{-p}\eta_0 = \sigma^{-p}\eta_e = \eta_{p+e}$. For g of radius R, $(g\eta_0)_i = (0,0)$ for $i < -R$ and $(1,1)$ for $i > R$, so $|e| \le R$. Clearly e^+ is a homomorphism. The map $\rho g \rho$ is again an invertible block map commuting with σ, and with F because $\rho F \rho = F^{-1}$; put $e^-(g) = e^+(\rho g \rho)$, so that $g\rho\eta_p = \rho(\rho g\rho)\eta_p = \rho\eta_{p+e^-(g)}$. The values on $\sigma^a F^b$ follow from $\sigma\eta_p = \eta_{p-1}$, $F\eta_p = \eta_{p+1}$, $\sigma\rho\eta_p = \rho\eta_{p-1}$ and $F\rho\eta_p = \rho\eta_{p-1}$. Finally, $\sigma^a F^b = \mathrm{id}$ gives $b - a = -a - b = 0$. □

The image of $\langle \sigma, F\rangle$ under (e^+, e^-) is $\{(m,n) : m \equiv n \pmod 2\}$.

Definition 3.3. For $R' \ge 0$, let $(\mathrm{Z}_{R'})$ be the statement: *every $h \in \mathrm{Aut}(X_F)$ of radius at most R' with $e^+(h) = e^-(h) = 0$ is the identity.*

Proposition 3.4. *Let $R \ge 0$. If (Z_{4R+4}) holds, then every $g \in \mathrm{Aut}(X_F)$ of radius at most R lies in $\langle \sigma, F\rangle$. If only (Z_{2R}) holds, the same conclusion holds for those g with $e^+(g) \equiv e^-(g) \pmod 2$. If $(\mathrm{Z}_{R'})$ holds for every R', then $\mathrm{Aut}(X_F) = \langle \sigma, F\rangle \cong \mathbb{Z}^2$.*

Proof. Let g have radius R, so $|e^{\pm}(g)| \le R$ by Proposition 3.2. If $e^+(g) + e^-(g)$ is even, put $a = -\frac{1}{2}(e^+ + e^-)$ and $b = \frac{1}{2}(e^+ - e^-)$; then $|a| + |b| = \max(|a+b|, |a-b|) = \max(|e^-|, |e^+|) \le R$, and $h = F^{-b}\sigma^{-a}g$ has offsets $(0,0)$ and radius at most $2R$, so $h = \mathrm{id}$ by (Z_{2R}), and $g = \sigma^a F^b$.

Suppose $e^+ + e^-$ is odd. As $F((x,y)^{\mathbb{Z}}) = (y,x)^{\mathbb{Z}}$, F permutes the four constant configurations as the transposition exchanging $(0,1)^{\mathbb{Z}}$ and $(1,0)^{\mathbb{Z}}$. Every automorphism g' commutes with σ, so it permutes the four fixed points of σ, which are the constant configurations, and the induced permutation commutes with that of F. The centralizer in $\mathrm{Sym}(4)$ of a transposition $(\alpha\,\beta)$ is $\langle(\alpha\,\beta)\rangle \times \langle(\alpha'\,\beta')\rangle$, where $\{\alpha', \beta'\}$ is the complement of $\{\alpha, \beta\}$; every element of this group has order at most 2, so g'^2 fixes every constant configuration. Put $a = -\frac{1}{2}(e^+ + e^- + 1)$ and $b = \frac{1}{2}(e^+ - e^- - 1)$, which are integers with $|a| + |b| = \max(|e^- + 1|, |e^+|) \le R + 1$ and $(e^+, e^-)(\sigma^a F^b) = (e^+(g), e^-(g) + 1)$. So $g_1 = F^{-b}\sigma^{-a}g$ has offsets $(0,-1)$ and radius at most $2R+1$, and $g_1^2(\sigma F)^{-1}$ has offsets $(0,0)$, since $(e^+, e^-)(\sigma F) = (0,-2)$, and radius at most $2(2R+1) + 2 = 4R + 4$. By (Z_{4R+4}), $g_1^2 = \sigma F$. But g_1^2 fixes every constant configuration, while σF exchanges $(0,1)^{\mathbb{Z}}$ and $(1,0)^{\mathbb{Z}}$. So the odd case does not occur. The last statement follows, since every automorphism has finite radius, and $\langle \sigma, F\rangle \cong \mathbb{Z}^2$ by Proposition 3.2. □

4. Rule 110R: Lemma K

In this section $f = 110$ and $F = F_{110}$. The rule is $f(a,b,c) = (b \oplus c) \vee (b \wedge \neg a)$, with values $0,1,1,1,0,1,1,0$ on $000, \ldots, 111$; it satisfies (4). It is permutive in no variable, and in the spacetime relation $Y(t+1,i) \oplus Y(t-1,i) = f(Y(t,i-1), Y(t,i), Y(t,i+1))$ only the two time neighbours enter permutively in every context: f is permutive in a exactly when $(b,c) = (1,1)$,

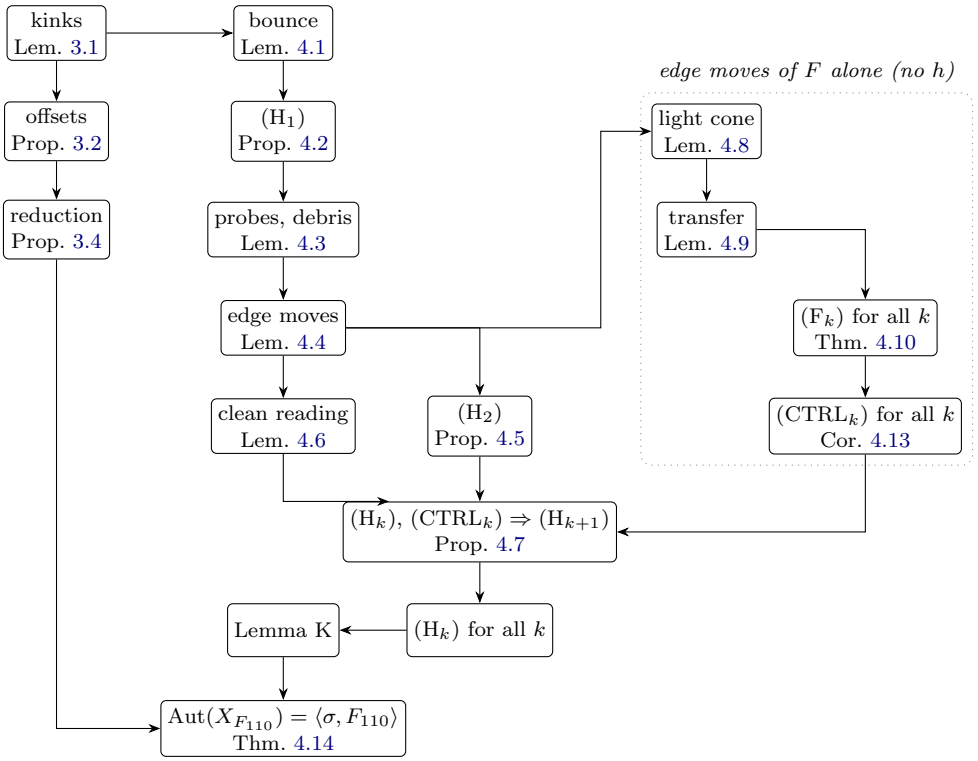


FIGURE 1. Dependencies in the proof for the class of 110. Every step is proved by hand, for every k and every radius. The boxes in the dotted frame concern the edge moves of F alone and do not involve the block map h. For 42 the same diagram applies with Theorem 5.4; for 46 the controllability theorem is Theorem 5.13; for 66 and 189 the steps are those of Section 6, with the controllability theorem for 46 transported by the factor map δ (Lemma 6.14); for 93 the probes, moves and (H_2), (H_3) are replaced as described in Section 7, and the induction starts from Proposition 7.19, which describes the orbits of the moves at level 3.

in c exactly when $(a, b) \neq (0, 1)$, and in b exactly when $(a, c) \neq (0, 1)$. So the front of a difference does not move at a fixed speed. We use the following values of f, read off its table:

$$f(a,0,0) = 0, \quad f(a,0,1) = 1, \quad f(a,1,0) = 1, \quad f(a,1,1) = \neg a, \quad f(1,b,c) = b \oplus c.$$

In particular $f(a, b, 0) = b$ and $f(0, b, c) = b \vee c$. By Propositions 3.2 and 3.4, it remains to prove $(\mathrm{Z}_{R'})$ for every R'. A zero-offset automorphism fixes η_0 and $\rho\eta_0$, so $(\mathrm{Z}_{R'})$ follows from the next statement.

Lemma K. *Let h be a block map of radius R that commutes with F_{110} and fixes η_0 and $\rho\eta_0$. Then $h = \mathrm{id}$.*

Lemma K holds for every R (Theorem 4.14); the rest of this section proves it. It does not assume that h is invertible. The structure of the proof is shown in Figure 1. If h commutes with F and c is a configuration, put $G = Y_{hc} \oplus Y_c$. Both diagrams satisfy the spacetime relation, so

$$(5)\ G(t+1,i) \oplus G(t-1,i) = f\big(Y'(t,i-1), Y'(t,i), Y'(t,i+1)\big) \oplus f\big(Y(t,i-1), Y(t,i), Y(t,i+1)\big),$$

where $Y = Y_c$ and $Y' = Y_{hc} = Y \oplus G$.

4.1. **Configurations that vanish to the right.** Call $c \in \mathcal{A}^{\mathbb{Z}}$ 0-*tailed* if $c \neq (0,0)^{\mathbb{Z}}$ and $c_i = (0,0)$ for all large i, and let $e(c)$ be the largest i with $c_i \neq (0,0)$, the *edge* of c. Recall that $(F^t c)_i = (Y_c(t-1,i), Y_c(t,i))$. Since $f(a,0,0) = 0$ and $f(a,b,0) = b$, the columns of Y_c to the right of $e = e(c)$ stay 0 and column e obeys $Y_c(t+1,e) = Y_c(t-1,e) \oplus Y_c(t,e)$; as this recurrence is reversible, $e(F^t c) = e(c)$ for all $t \in \mathbb{Z}$, and column e runs through the cycle $\ldots 011011 \ldots$, never 0 in two consecutive rows. Let $\mathcal{H}$ be the set of block maps h, of any radius and not necessarily invertible, that commute with F and fix η_0 and $\rho\eta_0$. Such an h commutes with σ, so it fixes every η_p and $\rho\eta_p$, and it fixes $(0,0)^{\mathbb{Z}}$, which η_0 contains far to the left; and $\rho h \rho \in \mathcal{H}$, because $\rho F \rho = F^{-1}$. For $k \geq 1$ consider

(H_k) *for every* $h \in \mathcal{H}$ *and every* 0*-tailed* c, $Y_{hc}(t,i) = Y_c(t,i)$ *for all* $t \in \mathbb{Z}$ *and all* $i \geq e(c) - k + 1$.

So (H_k) says that $hc = c$ on the columns $i \geq e(c) - k + 1$ at all times; in particular hc is 0-tailed with $e(hc) = e(c)$. (H_{R+1}) implies Lemma K for radius R: if $h \in \mathcal{H}$ has radius R and $w \in \mathcal{A}^{2R+1}$ is not $(0,0)^{2R+1}$, let c be w on $[-R,R]$ and $(0,0)$ elsewhere; then $e(c) \leq R$, so $h(w) = (hc)_0 = c_0 = w_0$; and $h((0,0)^{2R+1}) = (0,0)$ because h fixes $(0,0)^{\mathbb{Z}}$.

For d with $d_i = (0,0)$ for $i > r$ and a configuration Λ with $\Lambda_i = (0,0)$ for $i \leq r$, write $d \sqcup \Lambda$ for the configuration equal to d on $(-\infty, r]$ and to Λ on (r, ∞).

4.2. **Fronts and the edge column.** Since $f(1,1,1) = 0$, F and F^{-1} map a configuration that is $(1,1)$ far to the right to another such configuration, so every row of its diagram is 1 far to the right. We say that such a c has a *front* at (t_0, q) if $Y_c(t_0, q) = 0$, $Y_c(t_0, i) = 1$ for $i > q$, and $Y_c(t_0 - 1, i) = 1$ for $i \geq q$. The kink η_p has $Y_{\eta_p}(t,i) = [i > p + t]$, so it has a front at $(t, p+t)$ for every t; the kink $\rho\eta_N$ has $Y_{\rho\eta_N}(t,i) = [i \geq N - t]$, so the rightmost 0 of its row t is at $N - 1 - t$.

Lemma 4.1 (zero wake and bounce)**.** (a) *If c has a front at (t_0, q), then for every $t \geq t_0$ row t of Y_c is 0 on $[q, q + t - t_0]$ and 1 on $(q + t - t_0, \infty)$, and $F^t c$ agrees with η_{q+t-t_0} on $[q, \infty)$. So c has a front at $(t, q + t - t_0)$ for every $t \geq t_0$, and the integer $U(c) = q - t_0$ does not depend on the front chosen: it is the eventual value of $\mu_t - t$, where μ_t is the largest i with $Y_c(t,i) = 0$. The kink $\rho\eta_N$ has no front.*

(b) *If c has a front and $h \in \mathcal{H}$, then hc has a front and $U(hc) = U(c)$.*

(c) (Bounce.) *Let d be 0-tailed with $e(d) = e$, let $N \geq e + 1$, and let $c = d \sqcup \rho\eta_N$, glued at $r = e$. Then c has a front, and $U(c) = 2e - N + \varphi$, where $\varphi = Y_d(N - e - 1, e)$.*

Proof. (a) Write $Y = Y_c$ and $s = q + t - t_0$, and argue by induction on $t \geq t_0$; at $t = t_0$ the statement about row t is the hypothesis, and row $t_0 - 1$ is 1 on $[q, \infty)$. Assume the statement for rows $t - 1$ and t, and compute row $t + 1$ from $Y(t+1,i) = Y(t-1,i) \oplus f(Y(t,i-1), Y(t,i), Y(t,i+1))$. For $i > s + 1$ it is $1 \oplus f(1,1,1) = 1$; for $i = s + 1$ it is $1 \oplus f(0,1,1) = 0$; for $i = s$ it is $1 \oplus f(Y(t,s-1), 0, 1) = 0$, since $f(a,0,1) = 1$; and for $q \leq i < s$ it is $0 \oplus f(Y(t,i-1), 0, 0) = 0$, since $f(a,0,0) = 0$. As $Y_{\eta_p}(-1,i) = [i \geq p]$ and $Y_{\eta_p}(0,i) = [i > p]$, rows $t - 1$ and t are those of η_s on $[q, \infty)$. So $\mu_t = t + q - t_0$ for $t \geq t_0$, which gives the independence of U. For $\rho\eta_N$, $\mu_t = N - 1 - t$ is decreasing, so it is not of the form $t + U$ for all large t.

(b) Let R be the radius of h. Since h fixes η_0, which is $(1,1)$ far to the right, it maps the window $(1,1)^{2R+1}$ to $(1,1)$, so hc is $(1,1)$ far to the right. Let c have a front at (t_0, q), let $t \geq t_0 + R$ and $p = q + t - t_0 \geq q + R$. By (a), $F^t c$ agrees with η_p on $[q, \infty)$, so $F^t hc = hF^t c$ agrees with $h\eta_p = \eta_p$ on $[q + R, \infty)$, which contains p. Hence hc has a front at (t,p), and $U(hc) = p - t = U(c)$. No inverse of h is used.

(c) Put $Y = Y_c$ and $t_0 = N - e - 1 \geq 0$. The configurations c and d agree on $(-\infty, N-1]$, and $F^{\pm 1}$ have radius 1, so the states of c and d at time $t \geq 0$ agree on $(-\infty, N - 1 - t]$; at $t = t_0$ this is $(-\infty, e]$. On (e, ∞), row t of Y is $[i \geq N - t]$, as for $\rho\eta_N$, for $-1 \leq t \leq t_0$: this holds for $t = -1, 0$, and for $1 \leq t + 1 \leq t_0$ the only entry of row $t + 1$ on (e, ∞) that reads a column $\leq e$ is the one at $e + 1$, which is $Y(t-1, e+1) \oplus f(Y(t,e), Y(t,e+1), Y(t,e+2))$ with

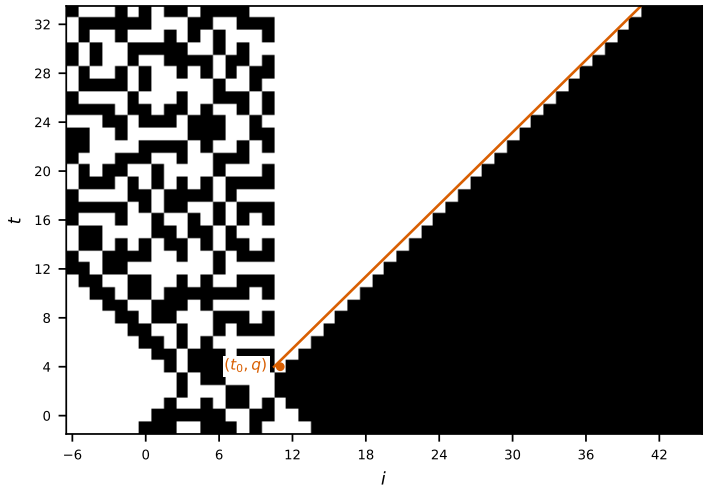


FIGURE 2. A front for rule 110 (Lemma 4.1(a)): the diagram of a configuration that is $(1,1)$ to the right of a random block. From the front at (t_0, q) on, row t is 0 on $[q, q+t-t_0]$ and 1 beyond, and the front moves to the right at speed 1. Time increases upwards; black cells are 1.

$Y(t, e+1) = 0$, as $e+1 < N-t$; and $f(a,0,c) = c$ does not read a. So row t_0 is 1 on (e,∞), row t_0-1 is 0 at $e+1$ and 1 on $[e+2,\infty)$, $Y(t_0,e) = \varphi$ and $Y(t_0-1,e) = Y_d(t_0-1,e)$. Row t_0+1 is $1 \oplus f(1,1,1) = 1$ on $[e+2,\infty)$ and $0 \oplus f(\varphi,1,1) = \neg\varphi$ at $e+1$.

If $\varphi = 1$, then c has a front at $(t_0+1, e+1)$, and $U(c) = e+1-(t_0+1) = 2e-N+1$. If $\varphi = 0$, then $Y(t_0-1,e) = 1$, because the edge column of d is never 0 in two consecutive rows, so $Y(t_0+1,e) = 1 \oplus f(Y(t_0,e-1),0,1) = 0$, and row t_0+1 is 1 on $[e+1,\infty)$. Then $Y(t_0+2,e+1) = 1 \oplus f(0,1,1) = 0$ and $Y(t_0+2,i) = 1 \oplus f(1,1,1) = 1$ for $i \geq e+2$, so c has a front at $(t_0+2, e+1)$, and $U(c) = e+1-(t_0+2) = 2e-N$. □

Proposition 4.2 ((H_1))**.** *For every $h \in \mathcal{H}$ and every 0-tailed d, the configuration hd is 0-tailed with $e(hd) = e(d)$, and $Y_{hd}(t,i) = Y_d(t,i)$ for all $t \in \mathbb{Z}$ and all $i \geq e(d)$. That is,* (H_1) *holds.*

Proof. Let R be the radius of h, $e = e(d)$, $N \geq e+2R+1$, and $c = d \sqcup \rho\eta_N$, glued at e. Since h fixes $(0,0)^{\mathbb{Z}}$, $(hd)_i = (0,0)$ for $i > e+R$. For $i \leq N-R-1$ the window of i lies in $(-\infty, N-1]$, where $c = d$, so $(hc)_i = (hd)_i$; for $i \geq e+R+1$ it lies in (e,∞), where $c = \rho\eta_N$, so $(hc)_i = (h\rho\eta_N)_i = (\rho\eta_N)_i$. These ranges cover $\mathbb{Z}$, and on their overlap both hd and $\rho\eta_N$ are $(0,0)$. So $hc = \rho\eta_N$ if $hd = (0,0)^{\mathbb{Z}}$; otherwise hd is 0-tailed with $e' = e(hd) \leq e+R < N$, and $hc = hd \sqcup \rho\eta_N$, glued at e'. By Lemma 4.1(c), c has a front, so by part (b) of that lemma hc has a front and $U(hc) = U(c)$; by part (a), $\rho\eta_N$ has none, so $hd \neq (0,0)^{\mathbb{Z}}$. Part (c), for c and for hc, now gives

$$2e' - N + Y_{hd}(N-e'-1, e') = 2e - N + Y_d(N-e-1, e).$$

So $2(e'-e) \in \{-1,0,1\}$, hence $e' = e$, and then $Y_{hd}(s,e) = Y_d(s,e)$ for every $s = N-e-1 \geq 2R$. Both edge columns obey the reversible recurrence $Y(t+1,e) = Y(t-1,e) \oplus Y(t,e)$, so they agree at all times. To the right of e both diagrams vanish, since $e(F^t hd) = e(F^t d) = e$ for all t. □

4.3. **Probes and edge moves.**

Lemma 4.3 (probes and debris)**.** *Let $h \in \mathcal{H}$ have radius R, let c be 0-tailed with $e(c) = r$, put $Y = Y_c$, and for $n \in \mathbb{Z}$ let $P_n(c) = F^{-n}(F^n c \sqcup \rho\eta_{r+2})$, the* probe *of c at time n.*

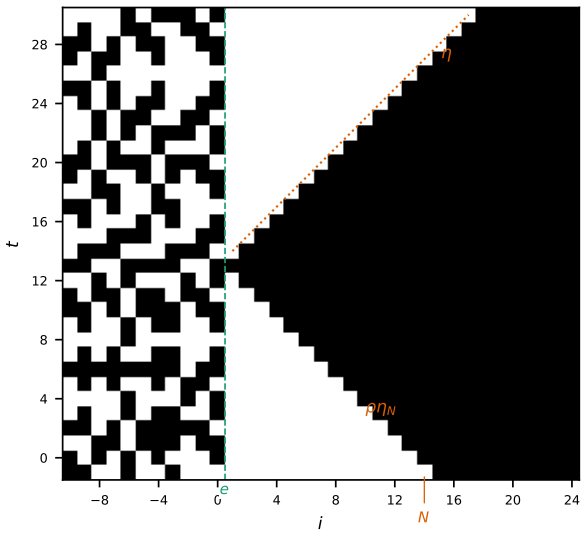


FIGURE 3. The bounce of Lemma 4.1(c) for rule 110: the diagram of $d \sqcup \rho\eta_N$, with d a random 0-tailed configuration with edge e (marked). The kink $\rho\eta_N$ enters from the right, meets the edge column at time $N - e - 1$ and leaves as a front of speed 1, whose position $U = 2e - N + \varphi$ records one value φ of the edge column. Time increases upwards; black cells are 1.

(a) $F^n P_n(c) = F^j(F^{n-j}c \sqcup \rho\eta_{r+2+j})$ *for every* $j \geq 0$, *and* $hP_n(c) = P_n(hc)$.
(b) *Let* $\varepsilon = 1$ *if* $Y(n+1, r) = 0$ *and* $\varepsilon = 0$ *otherwise. There is a unique* 0*-tailed configuration* $D = D_n(c)$, *the* debris, *with* $e(D) = r$ *such that*

$$F^s P_n(c) = F^s D \sqcup \eta_{r+s-n-1-\varepsilon} \qquad \text{for all } s \geq n + 2 + \varepsilon.$$

The diagrams of D *and of* $P = P_n(c)$ *agree on the columns* $\leq r$ *at all rows* $\geq n + 1 + \varepsilon$. *On the columns* $\leq r$, *the diagram of* P *equals* Y *at the rows* $n - 1$, n, $n + 1$; *at row* $n + 2$ *it equals* Y *on the columns* $\leq r - 1$. *If* $\varepsilon = 1$, *then* $Y_P(n + 2, r) = 0$ *and* $Y_P(n + 3, r) = 1$. *Finally* $hD_n(c) = D_n(hc)$.
(c) ρ *maps* 0*-tailed configurations to* 0*-tailed configurations with the same edge,* $Y_{\rho c}(t, i) = Y_c(-1 - t, i)$, *and* $\rho h \rho \in \mathcal{H}$.

The configuration $F^n c \sqcup \rho\eta_{r+2}$ is the state at time n of c with a kink glued next to its edge; by (b), after time $n + 2 + \varepsilon$ the glued configuration splits into the debris and an outgoing kink η, and the debris differs from c only near the edge (Figure 4). The statements about h use (H_1), which holds by Proposition 4.2.

Proof. Two gluing facts. (i) If $d_i = (0, 0)$ for $i > r$, then $F(d \sqcup \eta_N) = Fd \sqcup \eta_{N+1}$ for $N \geq r + 1$, and $F(d \sqcup \rho\eta_N) = Fd \sqcup \rho\eta_{N-1}$ for $N \geq r + 3$. Indeed the y-component at $r + 1$ is 0 in both glued configurations, so the cells $i \leq r$ evolve as in Fd; the cells $i \geq r + 2$ read only cells $\geq r + 1$, which are those of the kink; and the new cell $r + 1$ is $(0, 0)$, because $f(a, 0, 0) = 0$ and $f(a, 0, 1) = 1$ do not depend on a (for η_N the cell $r+1$ is $(1, 0)$ if $N = r+1$ and $(0, 0)$ otherwise; for $\rho\eta_N$, $N \geq r+3$, the cells $r + 1$, $r + 2$ are $(0, 0)$). Then use $F\eta_N = \eta_{N+1}$ and $F\rho\eta_N = \rho\eta_{N-1}$ (Lemma 3.1). (ii) If moreover $(hd)_i = (0, 0)$ for $i > r$ and $\Lambda \in \{\eta_N, \rho\eta_N\}$ with $N \geq r+2R+1$, then $h(d \sqcup \Lambda) = hd \sqcup \Lambda$: for $i \leq N - R - 1$ the window of i lies where $d \sqcup \Lambda = d$, and for $i \geq r + R + 1$ it lies where $d \sqcup \Lambda = \Lambda$, and $h\Lambda = \Lambda$; these ranges cover $\mathbb{Z}$, and on (r, N) both hd and Λ are $(0, 0)$.

(a) The identity is (i) applied j times. Take $j \geq 2R-1$. By Proposition 4.2, $hF^{n-j}c = F^{n-j}hc$ vanishes to the right of r, so by (ii), $h(F^{n-j}c \sqcup \rho\eta_{r+2+j}) = F^{n-j}hc \sqcup \rho\eta_{r+2+j}$; apply F^{j-n}, which commutes with h.

(b) Write $P = P_n(c)$. Rows $n-1$ and n of Y_P are the two rows of $F^nP = F^nc \sqcup \rho\eta_{r+2}$: on the columns $\leq r$ they are $Y(n-1,\cdot)$ and $Y(n,\cdot)$; row $n-1$ is 0 at $r+1, r+2$ and 1 beyond, and row n is 0 at $r+1$ and 1 beyond. By the relation, row $n+1$ of Y_P is $Y(n+1,\cdot)$ on the columns $\leq r$, because $Y_P(n, r+1) = 0 = Y(n, r+1)$, and it is 1 on the columns $> r$ (at $r+1$ it is $0 \oplus f(Y(n,r),0,1) = 1$, at $r+2$ it is $0 \oplus f(0,1,1) = 1$, and beyond $1 \oplus f(1,1,1) = 1$). Row $n+2$ is $1 \oplus f(1,1,1) = 1$ on the columns $\geq r+2$ and $0 \oplus f(Y(n+1,r),1,1) = \neg Y(n+1,r) = \varepsilon$ at $r+1$; on the columns $\leq r-1$ it reads only columns $\leq r$ of rows n, $n+1$, so it equals $Y(n+2,\cdot)$ there.

If $\varepsilon = 0$, the state $F^{n+2}P$ has the cell $(1,0)$ at $r+1$ and $(1,1)$ beyond, so $F^{n+2}P = d \sqcup \eta_{r+1}$, where d agrees with $F^{n+2}P$ on $(-\infty, r]$ and vanishes beyond; its cell r is $(Y(n+1,r),\cdot) = (1,\cdot) \neq (0,0)$. If $\varepsilon = 1$, then $Y(n+1,r) = 0$, so $Y(n,r) = 1$ because the edge column of c is never 0 twice in a row, and $Y_P(n+2,r) = Y(n,r) \oplus f(Y(n+1,r-1),0,1) = 0$. Then $Y_P(n+3,r+1) = Y_P(n+1,r+1) \oplus f(0,1,1) = 0$, $Y_P(n+3,i) = 1$ for $i \geq r+2$, and $Y_P(n+3,r) = Y_P(n+1,r) \oplus f(Y_P(n+2,r-1),0,1) = 1$. So $F^{n+3}P = d \sqcup \eta_{r+1}$ with cell r of d equal to $(0,1)$. In both cases put $D = F^{-(n+2+\varepsilon)}d$; it is 0-tailed with $e(D) = r$, and (i) gives $F^sP = F^sD \sqcup \eta_{r+s-n-1-\varepsilon}$ for $s \geq n+2+\varepsilon$. Rows $s-1$ and s of a diagram are the two rows of the state at time s, so the diagrams of D and P agree on the columns $\leq r$ at all rows $\geq n+1+\varepsilon$. If D' is 0-tailed with $e(D') = r$ and $F^sP = F^sD' \sqcup \eta_{N'}$ for some s and some $N' \geq r+1$, then by (i), applied to F^sD', which vanishes to the right of r, $F^{s'}P = F^{s'}D' \sqcup \eta_{N'+s'-s}$ for every $s' \geq s$; for $s' \geq \max(s, n+2+\varepsilon)$, $F^{s'}D'$ and $F^{s'}D$ both agree with $F^{s'}P$ on $(-\infty, r]$ and vanish beyond, so $F^{s'}D' = F^{s'}D$ and $D' = D$; this is the uniqueness.

By Proposition 4.2, $c' = hc$ is 0-tailed with $e(c') = r$ and $Y_{c'}(n+1,r) = Y(n+1,r)$, so c' has the same ε. For s so large that $N = r+s-n-1-\varepsilon \geq r+2R+1$, (a), (ii) and Proposition 4.2 for F^sD give $F^sD_n(c') \sqcup \eta_N = F^sP_n(c') = hF^sP = hF^sD \sqcup \eta_N$, hence $F^sD_n(c') = F^shD$ and $hD = D_n(c')$.

(c) ρ exchanges the two rows of every cell, which with $F^t\rho = \rho F^{-t}$ gives the formula for $Y_{\rho c}$; and $\rho h\rho$ commutes with F since $\rho F\rho = F^{-1}$, and fixes $\rho\eta_0$ and η_0. □

Edge moves. A probe followed by passage to its debris changes a configuration only at its edge cell, at a single time; so does its conjugate by ρ; and h commutes with both. For a 0-tailed b with $e(b) = r$ and $m \in \mathbb{Z}$ write the cells of the state F^mb as $(F^mb)_i = (x_i, y_i)$, so that $x_i = Y_b(m-1,i)$, $y_i = Y_b(m,i)$, and the edge cell is $(x_r, y_r) \neq (0,0)$.

Lemma 4.4 (edge moves)**.** *Let b be 0-tailed with $e(b) = r$, let $m \in \mathbb{Z}$, and let (x_i, y_i) be the cells of F^mb. Put $M_m(b) = D_{m-2}(b)$ and $\bar{M}_m(b) = \rho D_{-m-2}(\rho b)$.*

- (a) *$M_m(b)$ is 0-tailed with edge r, and $F^mM_m(b)$ agrees with F^mb at every cell except the edge cell, where (x_r, y_r) becomes $(1,0)$ if $(x_r, y_r) = (0,1)$, and becomes $(1, y_r \oplus x_{r-1})$ if $x_r = 1$.*
- (b) *$\bar{M}_m(b)$ is 0-tailed with edge r, and $F^m\bar{M}_m(b)$ agrees with F^mb at every cell except the edge cell, where (x_r, y_r) becomes $(0,1)$ if $(x_r, y_r) = (1,0)$, and becomes $(x_r \oplus y_{r-1}, 1)$ if $y_r = 1$.*
- (c) *$hM_m = M_mh$ and $h\bar{M}_m = \bar{M}_mh$ on 0-tailed configurations with edge r, for every $h \in \mathcal{H}$ and every $m \in \mathbb{Z}$.*

So, at any time m and in the notation x_ry_r for the edge cell, the *moves* are $01 \to 10$ and $10 \to 01$, always; $1y \to 1\bar{y}$ if $x_{r-1} = 1$; and $x1 \to \bar{x}1$ if $y_{r-1} = 1$. In the remaining cases (a) and (b) leave b unchanged. By (a) and (b), M_m and $\bar{M}_m$ are described by their effect on the state

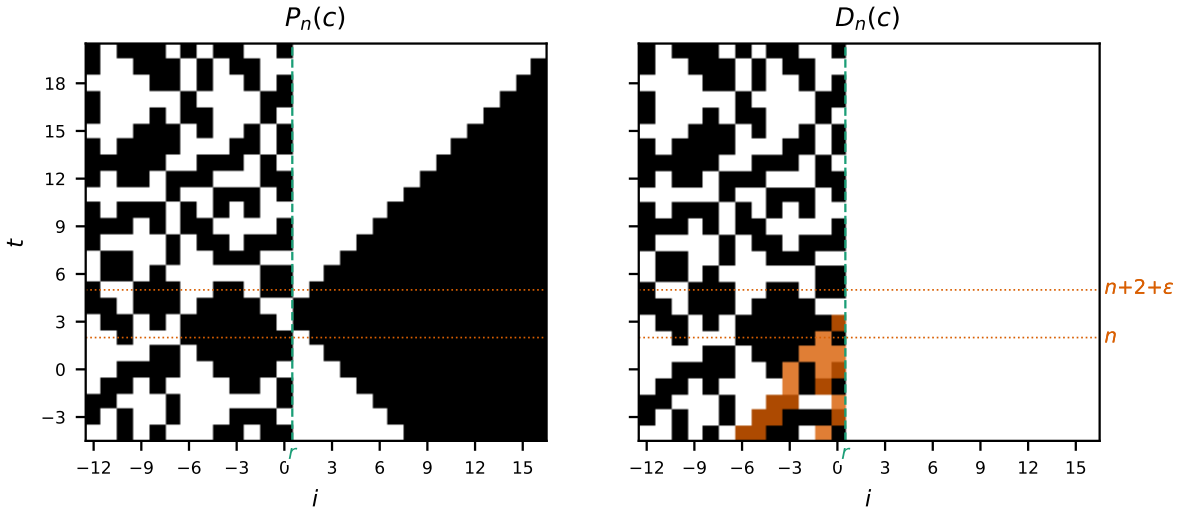


FIGURE 4. A probe and its debris for rule 110 (Lemma 4.3). Left: the diagram of the probe $P_n(c)$; the kink glued at time n next to the edge column r leaves as an outgoing kink η. Right: the debris $D_n(c)$. Here $n = 2$ and $\varepsilon = 1$; for $s \geq n+2+\varepsilon$, $F^s P_n(c) = F^s D_n(c) \sqcup \eta_{r+s-n-1-\varepsilon}$. On the columns $\leq r$ the two diagrams differ only in the shaded cells, all at rows $\leq n + \varepsilon$.

at time m alone, without reference to any h. Part (c) is an identity of operators: a composite O of moves is a fixed map, and $hO = Oh$, although the effect of each move depends on the configuration to which it is applied.

Proof. (a) Put $n = m - 2$, $Y = Y_b$, $P = P_n(b)$ and $D = D_n(b)$, and let ε be as in Lemma 4.3(b), so $\varepsilon = 1$ exactly when $x_r = Y(m-1, r) = 0$. By that lemma D is 0-tailed with $e(D) = r$, and the diagram of P equals Y on the columns $\leq r$ at the rows $m-3, m-2, m-1$, and at row m on the columns $\leq r-1$; at (m, r) it is $Y(m-2, r) \oplus f(x_{r-1}, x_r, 1)$, while $y_r = Y(m-2, r) \oplus f(x_{r-1}, x_r, 0) = Y(m-2, r) \oplus x_r$.

If $x_r = 1$, then $\varepsilon = 0$, so the diagrams of D and P agree on the columns $\leq r$ at the rows $\geq m-1$. Hence rows $m-1$, m of D are x and y on the columns $\leq r$, except that $Y_D(m, r) = Y(m-2, r) \oplus f(x_{r-1}, 1, 1) = Y(m-2, r) \oplus \neg x_{r-1} = y_r \oplus x_{r-1}$. If $x_r = 0$, then $y_r = 1$, $\varepsilon = 1$, and the diagrams of D and P agree on the columns $\leq r$ at the rows $\geq m$. Row m of D is that of P: it is y_i for $i \leq r-1$ and $Y_P(m, r) = 0$ at r. Row $m-1$ of D follows from rows m and $m+1$ by the relation solved for the earlier row: for $i \leq r-1$ the relation reads only columns $\leq r$ of these rows, so $Y_D(m-1, i) = Y_P(m-1, i) = x_i$; and, as $Y_D(m, r+1) = 0$, $Y_D(m-1, r) = Y_D(m+1, r) \oplus f(y_{r-1}, 0, 0) = Y_P(m+1, r) = 1$, the value at $(n+3, r)$ in Lemma 4.3(b). So the edge cell becomes $(1, 0)$ and nothing else changes. In both cases the cells $i > r$ are $(0, 0)$ since $e(D) = r$.

(b) By Lemma 4.3(c), ρb is 0-tailed with edge r and $Y_{\rho b}(t, i) = Y_b(-1-t, i)$, so the state of ρb at time $-m$ has the rows $Y_{\rho b}(-m-1, \cdot) = y$ and $Y_{\rho b}(-m, \cdot) = x$. Apply (a) to ρb at time $-m$, that is $M_{-m}(\rho b) = D_{-m-2}(\rho b)$, with the roles of x and y exchanged, and then ρ, which turns the state at time $-m$ back into the state at time m with its two rows exchanged.

(c) By Proposition 4.2, hb is again 0-tailed with edge r. Lemma 4.3(b) gives $hD_n(b) = D_n(hb)$ for every n, which is $hM_m = M_m h$. Since $\rho h \rho \in \mathcal{H}$, the same lemma gives $(\rho h \rho) D_n = D_n(\rho h \rho)$, and therefore $h\bar{M}_m(b) = \rho(\rho h \rho) D_{-m-2}(\rho b) = \rho D_{-m-2}(\rho h b) = \bar{M}_m(hb)$. □

4.4. **The second column and the inductive step.**

Proposition 4.5 ((H_2))**.** (H_2) *holds.*

Proof. Let $h \in \mathcal{H}$, let c be 0-tailed with $e(c) = r$, and put $Y = Y_c$ and $G = Y_{hc} \oplus Y$. By Proposition 4.2, G vanishes on the columns $\geq r$, and it remains to show that $G(t, r-1) = 0$ for every t.

(a) Suppose $Y(t, r) = 1$, and put $m = t + 1$, so that the state at time m has the rows t and $t + 1$ and its edge cell has $x_r = Y(t, r) = 1$. By Lemma 4.4(a), $b = M_m c$ has $Y_b(t + 1, r) = Y(t + 1, r) \oplus Y(t, r - 1)$. By Proposition 4.2, $Y_{hc}(t, r) = 1$ and $Y_{hc}(t + 1, r) = Y(t + 1, r)$, so the same lemma gives $Y_{M_m hc}(t + 1, r) = Y(t + 1, r) \oplus Y_{hc}(t, r - 1)$. By Lemma 4.4(c), $M_m hc = hb$, and by Proposition 4.2 for b, which is 0-tailed with edge r, $Y_{hb}(t+1, r) = Y_b(t+1, r)$. Comparing, $Y_{hc}(t, r - 1) = Y(t, r - 1)$, that is, $G(t, r - 1) = 0$.

(b) Suppose $Y(t, r) = 0$. Then $Y(t \pm 1, r) = 1$, since the edge column runs through the cycle $\ldots 011011 \ldots$, so $G(t \pm 1, r - 1) = 0$ by (a). Since $f(a, b, 0) = b$ and $Y_{hc}(t, r) = Y(t, r) = 0$, the relation at $(t, r - 1)$ reads $Y(t + 1, r - 1) \oplus Y(t - 1, r - 1) = Y(t, r - 1)$ in both diagrams, so $G(t, r - 1) = G(t + 1, r - 1) \oplus G(t - 1, r - 1) = 0$. □

Lemma 4.6 (clean reading)**.** *Let $k \geq 2$ and assume* (H_k)*. Let $h \in \mathcal{H}$, let b be 0-tailed with $e(b) = r$, and let O be a finite composite of moves M_m, $\bar{M}_m$ with $|m| \leq k - 1$. Then $Y_{Ob}(0, i) = Y_b(0, i)$ and $Y_{Ohb}(0, i) = Y_{hb}(0, i)$ for $i \leq r - k$. If moreover $Y_{Ob}(0, r - k + 1) = Y_{Ob}(0, r - k + 2) = 1$, then $Y_{hb}(0, r - k) = Y_b(0, r - k)$.*

Proof. A move at time m changes the state at time m only at column r. Since F and F^{-1} have radius 1, the states at time 0 before and after the move differ only on columns $\geq r - |m| \geq r - k + 1$, so row 0 is unchanged on columns $\leq r - k$. Applying this to each factor of O in turn, starting from b or from hb, which is 0-tailed with edge r by Proposition 4.2, gives the first statement. Put $Y = Y_{Ob}$ and $G' = Y_{hOb} \oplus Y$. By Lemma 4.4(c), $hOb = Ohb$, so $G'(0, r - k) = Y_{Ohb}(0, r - k) \oplus Y_{Ob}(0, r - k) = Y_{hb}(0, r - k) \oplus Y_b(0, r - k)$. Since Ob is 0-tailed with edge r, (H_k) gives $G' = 0$ on columns $\geq r - k + 1$ at all times. The relation (5) for G' at $(0, r - k + 1)$ then reads

$$0 = f\big(Y(0, r - k) \oplus G'(0, r - k), 1, 1\big) \oplus f\big(Y(0, r - k), 1, 1\big),$$

where we used $Y(0, r - k + 1) = Y(0, r - k + 2) = 1$ and $r - k + 2 \leq r$. As $f(a, 1, 1) = \neg a$, $G'(0, r - k) = 0$. □

For $k \geq 1$ consider the following statement about F_{110} alone; by the remark after Lemma 4.4 it involves no h.

(CTRL_k) *for every 0-tailed c, with $r = e(c)$, there is a finite composite O of moves M_m, $\bar{M}_m$ with $|m| \leq k - 1$ such that $Y_{Oc}(0, r - k + 1) = Y_{Oc}(0, r - k + 2) = 1$.*

Proposition 4.7. *For $k \geq 2$,* (CTRL_k) *and* (H_k) *imply* (H_{k+1})*.*

Proof. Let $h \in \mathcal{H}$, let c be 0-tailed with $e(c) = r$, and let $t \in \mathbb{Z}$. Then $F^t c$ is 0-tailed with edge r, and (CTRL_k) applied to $F^t c$ and Lemma 4.6 with $b = F^t c$ give $Y_{hF^t c}(0, r - k) = Y_{F^t c}(0, r - k)$. Since $hF^t = F^t h$ and $Y_{F^t c}(0, i) = Y_c(t, i)$, this is $Y_{hc}(t, r - k) = Y_c(t, r - k)$. Together with (H_k) this is (H_{k+1}). □

4.5. **The controllability theorem.** The orbits of the moves can be determined exactly. Recall that $c_i = (Y_c(-1, i), Y_c(0, i))$: the state at time 0 is c itself. For $k \geq 1$ and a 0-tailed c with $e(c) = r$, let $\mathcal{O}_k(c)$ be the set of configurations Oc, where O ranges over the finite composites of moves M_m, $\bar{M}_m$ with $|m| \leq k - 1$; we call c_{r-k+1} the *tip* and c_{r-k} the *exterior cell* of c at level k.

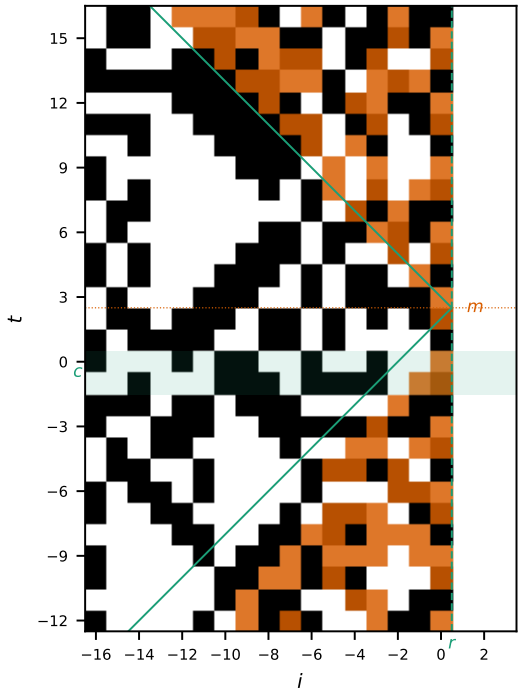


FIGURE 5. The light cone of a move (Lemma 4.8(a)) for rule 110: the diagrams of b and of $M_m b$ for $m = 3$ differ only at the highlighted cells, which lie in the cone $\{t - m \geq r - i\} \cup \{m - 1 - t \geq r - i\}$ below the edge column r; at time 0 the change is confined to $[r - |m|, r]$.

Lemma 4.8 (light cone)**.** *Let b be 0-tailed with $e(b) = r$, let $m \in \mathbb{Z}$, and let b' be the image of b under M_m or $\bar{M}_m$.*

(a) *b' is 0-tailed with $e(b') = r$, and for $i \leq r$, $Y_{b'}(t, i) \neq Y_b(t, i)$ only if $t - m \geq r - i$ or $m - 1 - t \geq r - i$. In particular b' agrees with b outside $[r - |m|, r]$.*

(b) *b is the image of b' under M_m or $\bar{M}_m$. If (x_r, y_r) is the edge cell of the state at time m and (x_{r-1}, y_{r-1}) the cell to its left, the moves at time m always connect $x_r y_r = 01$ with 10, and they connect 11 with 01 or 10 if and only if $(x_{r-1}, y_{r-1}) \neq (0, 0)$.*

Proof. (a) By Lemma 4.4, the move changes the state at time m only in its edge cell, which stays $\neq (0, 0)$, so b' is 0-tailed with edge r. Since $F^{\pm 1}$ have radius 1, the state at time $m + s$ changes only on the columns $\geq r - |s|$; row t is the second row of the state at time t and the first row of the state at time $t + 1$, which gives the cone, and at $t = -1, 0$ the columns $\geq r - |m|$ (Figure 5).

(b) This is the description of the moves after Lemma 4.4: $01 \to 10$ under M_m and $10 \to 01$ under $\bar{M}_m$ are inverse to each other, and $1y \to 1\bar{y}$ (when $x_{r-1} = 1$) and $x1 \to \bar{x}1$ (when $y_{r-1} = 1$) are involutions, since a move does not change the cell (x_{r-1}, y_{r-1}). □

By (b), $\mathcal{O}_k(c)$ is the class of c under the equivalence relation generated by single moves at the times $|m| \leq k - 1$; in particular $\mathcal{O}_k(b) = \mathcal{O}_k(c)$ for $b \in \mathcal{O}_k(c)$, and $\mathcal{O}_{k-1}(c) \subseteq \mathcal{O}_k(c)$. By (a), every element of $\mathcal{O}_k(c)$ agrees with c outside $[r - k + 1, r]$.

The moves at the times $|m| \leq k - 1$ contain three families of moves of level $k - 1$: those at the times $|m| \leq k - 2$, applied to b itself, which change only the cells $r - k + 2, \ldots, r$; and the same moves applied to Fb and to $F^{-1}b$, which can change the tip of b. The next lemma says how. It uses nothing about f beyond Lemma 4.4(a),(b).

Lemma 4.9 (transfer)**.** *Let $k \geq 2$, let b be 0-tailed with $e(b) = r$, write $b_{r-k+j} = (u_j, v_j)$ for $j = 0, 1, 2$, and put $v_3 = Y_b(0, r-k+3)$, so that $v_3 = 0$ if $k = 2$. By Section 4.1, Fb is 0-tailed with $e(Fb) = r$, so that $\mathcal{O}_{k-1}(Fb)$ is defined.*

(a) *$F^j M_m = M_{m-j} F^j$ and $F^j \bar{M}_m = \bar{M}_{m-j} F^j$ for all $j, m \in \mathbb{Z}$, and $\rho M_m \rho = \bar{M}_{-m}$. Consequently $F^{-1}\mathcal{O}_{k-1}(Fb) \subseteq \mathcal{O}_k(b)$ and $\rho\, \mathcal{O}_k(b) = \mathcal{O}_k(\rho b)$.*

(b) *The exterior cell of Fb and its tip at level $k-1$ are (v_1, α) and (v_2, β), where*

$$\alpha = u_1 \oplus f(v_0, v_1, v_2), \qquad \beta = u_2 \oplus f(v_1, v_2, v_3).$$

(c) *Let $\tilde{b}$ agree with Fb outside $[r-k+2, r]$, and let w be the x-bit of $\tilde{b}_{r-k+2}$. Then $b' = F^{-1}\tilde{b}$ agrees with b on $(-\infty, r-k]$, and*

$$b'_{r-k+1} = \big(\alpha \oplus f(v_0, v_1, w),\ v_1\big).$$

Proof. (a) By Lemma 4.4(a), $F^m M_m(b)$ is obtained from $F^m b$ by a change of the edge cell that is a function of the cells $r-1$ and r of $F^m b$ alone. Applied to $F^j b$ at the time $m-j$, the same description says that $F^{m-j} M_{m-j}(F^j b)$ is obtained in the same way from $F^{m-j} F^j b = F^m b$; so $F^{m-j} M_{m-j} F^j b = F^m M_m b$, that is, $M_{m-j} F^j = F^j M_m$. The same argument with Lemma 4.4(b) applies to $\bar{M}_m$. By definition, $\bar{M}_{-m}(b) = \rho D_{m-2}(\rho b) = \rho M_m(\rho b)$. If $\tilde{b} = O' F b$ for a composite O' of moves at the times $|m| \leq k-2$, then $F^{-1}\tilde{b} = F^{-1} O' F b$ is the image of b under the corresponding composite of moves at the times $m+1$, and $|m+1| \leq k-1$. Finally, conjugation by ρ maps the set of moves at the times $|m| \leq k-1$ onto itself.

(b) The cell i of Fb is $(Y_b(0,i), Y_b(1,i))$, and $Y_b(1,i) = Y_b(-1,i) \oplus f(Y_b(0,i-1), Y_b(0,i), Y_b(0,i+1))$; take $i = r-k+1$ and $i = r-k+2$.

(c) Since $\tilde{b} = Fb'$, the cell i of $\tilde{b}$ is $(Y_{b'}(0,i), Y_{b'}(1,i))$. For $i \leq r-k+1$ it equals the cell i of Fb, so $Y_{b'}(0,i) = Y_b(0,i)$ and $Y_{b'}(1,i) = Y_b(1,i)$; and $Y_{b'}(0, r-k+2) = w$. Hence $Y_{b'}(-1,i) = Y_{b'}(1,i) \oplus f\big(Y_{b'}(0,i-1), Y_{b'}(0,i), Y_{b'}(0,i+1)\big)$ equals $Y_b(-1,i)$ for $i \leq r-k$, and at $i = r-k+1$ it is $Y_b(1, r-k+1) \oplus f(v_0, v_1, w) = \alpha \oplus f(v_0, v_1, w)$, by (b). □

Theorem 4.10. *Let $k \geq 1$ and let c be 0-tailed with $e(c) = r$. Then $\mathcal{O}_k(c)$ is the set $\mathcal{S}_k(c)$ of configurations c' with*

(i) *$c'_i = c_i$ for every $i \notin [r-k+1, r]$, and $c'_r \neq (0,0)$; and*

(ii) *if the exterior cell c_{r-k} is $(0,0)$, then $c'_{r-k+1} = (1,1)$ if and only if $c_{r-k+1} = (1,1)$.*

In words: the moves at the times $|m| \leq k-1$ reach every configuration that agrees with c outside the k cells ending at the edge and whose edge cell is not $(0,0)$, except that a tip $(1,1)$, and the absence of one, cannot be changed when the exterior cell is $(0,0)$.

We call the statement of the theorem for a given k (F_k). The constraints (i) and (ii) are invariant (Lemma 4.11); (F_1) is immediate; and (F_{k-1}) implies (F_k) for every $k \geq 2$, by applying (F_{k-1}) to b, to Fb and to $F^{-1}b$, the last two through Lemma 4.9.

Lemma 4.11 (invariance)**.** *Every $c' \in \mathcal{O}_k(c)$ satisfies (i) and (ii).*

Proof. (i) is Lemma 4.8(a). For (ii), let $c_{r-k} = (0,0)$ and consider one move at a time $|m| \leq k-1$, applied to b with $b_{r-k} = (0,0)$; write $b_{r-k+1} = (u, v)$. If $k = 1$, the claim is Lemma 4.8(b) with $m = 0$, since then $(x_{r-1}, y_{r-1}) = b_{r-1} = (0,0)$. Let $k \geq 2$. By the cone of Lemma 4.8(a), the cell b_{r-k} never changes, $u = Y_b(-1, r-k+1)$ can change only if $m = k-1$, and $v = Y_b(0, r-k+1)$ only if $m = -(k-1)$. A move at time $m = k-1$ changes neither v nor $Y_b(1, r-k+1)$, and $Y_b(1, r-k+1) = u \oplus f(0, v, Y_b(0, r-k+2)) = u \oplus (v \vee Y_b(0, r-k+2))$, as $f(0, b, c) = b \vee c$. So if $v = 1$, then $u = 1 \oplus Y_b(1, r-k+1)$ does not change. Likewise a move at time $-(k-1)$ changes neither u nor $Y_b(-2, r-k+1) = v \oplus (u \vee Y_b(-1, r-k+2))$, so if $u = 1$, then v does not change. Hence $u \wedge v$ is invariant. □

Proof of Theorem 4.10. By Lemma 4.11, $\mathcal{O}_k(c) \subseteq \mathcal{S}_k(c)$. We prove the converse by induction on k. For $k = 1$, (i) allows only a change of the edge cell, and by Lemma 4.8(b) with $m = 0$, $(x_{r-1}, y_{r-1}) = c_{r-1}$, the three cells 01, 10, 11 form one class if $c_{r-1} \neq (0,0)$, and the classes $\{01, 10\}$ and $\{11\}$ if $c_{r-1} = (0,0)$: this is (F_1).

Let $k \geq 2$ and assume (F_{k-1}) for all 0-tailed configurations. If $b \in \mathcal{S}_k(c)$, then $\mathcal{S}_k(b) = \mathcal{S}_k(c)$, so $\mathcal{O}_k(b) \subseteq \mathcal{S}_k(c)$ by Lemma 4.11. For $b \in \mathcal{S}_k(c)$ write $b_{r-k+j} = (u_j, v_j)$ for $j = 0, 1, 2$, so that $(u_0, v_0) = c_{r-k}$ and (u_1, v_1) is the tip, and let α be as in Lemma 4.9(b). For $t \in \{0,1\}^2$ let A_t be the set of $b \in \mathcal{S}_k(c)$ with tip t. Each A_t allowed by (ii) is nonempty, because the cells $r-k+2, \ldots, r$ can be chosen freely subject to $b_r \neq (0,0)$. We use $f(a,0,c) = c$, $f(0,b,c) = b \vee c$ and $f(1,b,c) = b \oplus c$.

Step 1 (moves of level $k-1$ on b). Let $b, b' \in A_t$, and suppose that $t \neq (0,0)$ or $[b_{r-k+2} = (1,1)] = [b'_{r-k+2} = (1,1)]$. Then $b' \in \mathcal{O}_k(b)$. Indeed b' agrees with b outside $[r-k+2, r]$ and $b'_r \neq (0,0)$; the exterior cell of b at level $k-1$ is t; so $b' \in \mathcal{S}_{k-1}(b) = \mathcal{O}_{k-1}(b) \subseteq \mathcal{O}_k(b)$ by (F_{k-1}). In particular each of A_{01}, A_{10}, A_{11} lies in a single orbit.

Step 2 (tip $(u_1,0) \to (\neg u_1, 0)$). If $b \in A_{(u_1,0)}$, then $\mathcal{O}_k(b)$ meets $A_{(\neg u_1,0)}$. If $u_1 = 1$ and $b_{r-k+2} = (0,1)$, we first replace the cell b_{r-k+2} by $(1,0)$; by Step 1, with $t = (1,0)$, this stays in $\mathcal{O}_k(b)$ (for $k = 2$ it is the move $01 \to 10$ at time 0). So we may assume that $u_1 = 0$ or $b_{r-k+2} \neq (0,1)$. As $v_1 = 0$, Lemma 4.9(b) gives $\alpha = u_1 \oplus f(v_0, 0, v_2) = u_1 \oplus v_2$ and $\beta = u_2 \oplus (v_2 \vee v_3)$. The tip (v_2, β) of Fb is $(1,1)$ exactly when $(u_2, v_2) = (0,1)$, and then $\alpha = u_1 \oplus 1 = 1$ by our assumption. So if the exterior cell $(0, \alpha)$ of Fb at level $k-1$ is $(0,0)$, the tip of Fb is not $(1,1)$. Let $\tilde{b}$ be Fb with its cell $r-k+2$ replaced by $(\neg v_2, v_2)$, which is neither $(0,0)$ nor $(1,1)$. Then $\tilde{b} \in \mathcal{S}_{k-1}(Fb)$: it agrees with Fb outside $[r-k+2, r]$; $\tilde{b}_r \neq (0,0)$, since for $k = 2$ the cell r is the new cell and for $k \geq 3$ it is the edge cell of Fb; and if $(0,\alpha) = (0,0)$, neither Fb nor $\tilde{b}$ has tip $(1,1)$. By (F_{k-1}), $\tilde{b} \in \mathcal{O}_{k-1}(Fb)$, so $b' = F^{-1}\tilde{b} \in \mathcal{O}_k(b)$ by Lemma 4.9(a), and by Lemma 4.9(c) with $w = \neg v_2$,

$$b'_{r-k+1} = \big(\alpha \oplus f(v_0, 0, \neg v_2), 0\big) = (u_1 \oplus v_2 \oplus \neg v_2,\ 0) = (\neg u_1, 0).$$

Step 3 (tip $(0, v_1) \to (0, \neg v_1)$). If $b \in A_{(0,v_1)}$, then $\mathcal{O}_k(b)$ meets $A_{(0,\neg v_1)}$. Since ρ exchanges the two bits of every cell, it maps $\mathcal{S}_k(c)$ onto $\mathcal{S}_k(\rho c)$, and ρb has tip $(v_1, 0)$. Step 2 for ρb gives an element of $\mathcal{O}_k(\rho b)$ with tip $(\neg v_1, 0)$; its image under ρ lies in $\rho\,\mathcal{O}_k(\rho b) = \mathcal{O}_k(b)$ (Lemma 4.9(a)) and has tip $(0, \neg v_1)$.

Step 4 (tip $(u_1,1) \to (\neg u_1, 1)$ when $v_0 = 1$). Let $v_0 = 1$ and $b \in A_{(u_1,1)}$. Then $\alpha = u_1 \oplus f(1,1,v_2) = u_1 \oplus \neg v_2$, and the exterior cell $(1, \alpha)$ of Fb at level $k-1$ is not $(0,0)$. Let $\tilde{b}$ be Fb with its cell $r-k+2$ replaced by $(\neg v_2, 1)$. As in Step 2, $\tilde{b} \in \mathcal{S}_{k-1}(Fb)$, here because (ii) is void, so $b' = F^{-1}\tilde{b} \in \mathcal{O}_k(b)$, and $b'_{r-k+1} = (\alpha \oplus f(1,1,\neg v_2), 1) = (u_1 \oplus \neg v_2 \oplus v_2, 1) = (\neg u_1, 1)$. If $u_0 = 1$, the same argument applied to ρb, whose exterior cell is (v_0, u_0), shows as in Step 3 that every $b \in A_{(1,v_1)}$ has an element of $A_{(1,\neg v_1)}$ in its orbit.

Conclusion. If $c_{r-k} = (0,0)$ and $c_{r-k+1} = (1,1)$, then $\mathcal{S}_k(c) = A_{11}$, which lies in one orbit by Step 1. Otherwise $\mathcal{S}_k(c) \supseteq A_{00} \cup A_{01} \cup A_{10}$. By Step 2 every element of A_{00} is in the orbit of an element of A_{10}, and by Step 3 every element of A_{01} is in the orbit of an element of A_{00}; as A_{01} and A_{10} each lie in one orbit (Step 1), $A_{00} \cup A_{01} \cup A_{10}$ lies in one orbit. If $c_{r-k} = (0,0)$, this union is $\mathcal{S}_k(c)$. If $c_{r-k} \neq (0,0)$, then $v_0 = 1$ or $u_0 = 1$, $\mathcal{S}_k(c)$ contains also A_{11}, which lies in one orbit, and by Step 4 that orbit meets A_{01} (if $v_0 = 1$) or A_{10} (if $u_0 = 1$). In all cases $\mathcal{S}_k(c)$ lies in one orbit, which contains c; so $\mathcal{S}_k(c) \subseteq \mathcal{O}_k(c)$, which is ($\mathrm{F}_k$). □

Remark 4.12. The inductive step applies (F_{k-1}) only to b and to $F^{\pm 1}b$, and changes $F^{\pm 1}b$ only in the cells $r-k+2, \ldots, r$. It is therefore valid as soon as $r-k+2 \leq r$, that is for every $k \geq 2$, and the induction needs no base case other than (F_1).

Corollary 4.13. (CTRL_k) *holds for every $k \geq 2$, and* (H_k) *holds for every $k \geq 1$.*

Proof. Let c be 0-tailed with $e(c) = r$, and recall that $Y_{c'}(0, i)$ is the y-bit of c'_i. Let c' agree with c outside $[r-k+1, r]$, with $c'_{r-k+1} = (1,1)$ if $c_{r-k} = (0,0)$ and $c_{r-k+1} = (1,1)$, $c'_{r-k+1} = (0,1)$ otherwise, and $c'_i = (0,1)$ for $r-k+2 \le i \le r$. Then $c'_r \neq (0,0)$ (for $k = 2$ the cell $r-k+2$ is the edge cell), and c' satisfies (i) and (ii). By Theorem 4.10, $c' = Oc$ for a composite O of moves at times $|m| \le k-1$, and $Y_{Oc}(0, r-k+1) = Y_{Oc}(0, r-k+2) = 1$. This is ($\mathrm{CTRL}_k$), including the case in which the tip $(1,1)$ is frozen. (H_1) and (H_2) are Propositions 4.2 and 4.5, and Proposition 4.7 gives (H_{k+1}) from (H_k) for every $k \ge 2$. □

4.6. **Conclusion for the class of 110.**

Theorem 4.14. *Lemma K holds for every R, ($\mathrm{Z}_{R'}$) holds for every R', and*

$$\mathrm{Aut}(X_{F_{110}}) = \langle \sigma, F_{110} \rangle \cong \mathbb{Z}^2.$$

The same holds for every rule of the class of 110: $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2$ for $f \in \{62, 110, 118, 124\}$.

Proof. By Corollary 4.13, (H_{R+1}) holds for every R, and it implies Lemma K at radius R, as shown after the definition of (H_k). An automorphism of radius at most R' with $e^+ = e^- = 0$ fixes η_0 and $\rho\eta_0$ by Proposition 3.2, so it is the identity by Lemma K; this is ($\mathrm{Z}_{R'}$). By Proposition 3.4, $\mathrm{Aut}(X_{F_{110}}) = \langle \sigma, F_{110} \rangle \cong \mathbb{Z}^2$; every automorphism has finite radius by Curtis–Hedlund–Lyndon, and in the proof of that proposition the case of odd $e^+ + e^-$ is excluded. For the other members of the class, reflection and complementation conjugate F_{110} to F_f and map $\langle \sigma, F_{110} \rangle$ onto $\langle \sigma, F_f \rangle$, as recalled in Section 2.3. □

No computer is used in the proof for the class of 110.

5. Rules 14R and 46R

The method of Section 4 decides two further classes, $\{14, 42, 84, 112\}$ and $\{46, 116\}$. In the class of 14 we work with the rule 42, which is obtained from 14 by reflection followed by complementation. Except in Lemma 5.6 to Proposition 5.9, where $f = 14$, in this section $f \in \{42, 46\}$, $F = F_f$, and $\rho(x, y) = (y, x)$, so that $\rho F \rho = F^{-1}$. The values on $000, \dots, 111$ are

$$f_{42}(a,b,c) = c \wedge \neg(a \wedge b) : \ 0,1,0,1,0,1,0,0; \qquad f_{46} : \ 0,1,1,1,0,1,0,0,$$

so 42 differs from 110 on 010 and 110, and 46 differs from 110 on 110 only. All three rules satisfy

$$f(a,0,c) = c, \qquad f(a,1,1) = \neg a, \tag{6}$$

and so $f(a,0,0) = 0$, $f(a,0,1) = 1$, $f(0,1,1) = 1$ and $f(0,0,0) = f(1,1,1) = 0$. In addition $f_{42}(a,b,0) = 0$, while $f_{46}(a,1,c) = \neg a$ and $f_{46}(a,b,0) = b \wedge \neg a$.

5.1. **The shared steps.**

Lemma 5.1 (the shared steps)**.** *Let $f \in \{42, 46\}$.*

1. *The allowed triples of f are $000, 001, 010, 011, 111$ for 42 and $000, 001, 011, 111$ for 46. None begins with 10, so $\Phi_f = \Phi_{110}$. Moreover f satisfies (4), so Lemma 3.1 and Propositions 3.2 and 3.4 hold for F, with ($\mathrm{Z}_{R'}$) stated for F.*
2. *If c is 0-tailed with $e(c) = e$, then the columns of Y_c to the right of e vanish, column e is never 0 in two consecutive rows, and $e(F^t c) = e$ for all $t \in \mathbb{Z}$. Column e obeys $Y_c(t+1, e) = Y_c(t-1, e)$ for 42 and $Y_c(t+1, e) = Y_c(t-1, e) \oplus \big(Y_c(t, e) \wedge \neg Y_c(t, e-1)\big)$ for 46.*
3. *Define $\mathcal{H}$, (H_k), the probes P_n, the debris D_n, the moves M_m, $\bar{M}_m$, (CTRL_k) and $\mathcal{O}_k(c)$ as in Section 4, with F_f in place of F_{110}. Then Lemma 4.1, Proposition 4.2, Lemma 4.3, and the implication from (H_{R+1}) to Lemma K at radius R hold for F.*

Proof. (1) A triple (a_1, a_2, a_3) is allowed if $a_1 \oplus a_3 = f(a_1, a_2, a_3)$; read off the values. Condition (4) concerns the values on 000, 001, 011, 100, 101 and 111, on which f_{42} and f_{46} agree with f_{110}.

(2) If rows $t-1$ and t vanish on the columns $i > e$, then so does row $t+1$, as $Y(t+1, i) = Y(t-1, i) \oplus f(Y(t, i-1), 0, 0) = 0$, and likewise row $t-2$; this gives the first statement by induction in both time directions. If $Y(t-1, e) = Y(t, e) = 0$ for some t, the same computation at column e, whose right neighbour vanishes, gives $Y(t+1, e) = Y(t-2, e) = 0$, so column e vanishes and $c_e = (Y(-1, e), Y(0, e)) = (0, 0)$, a contradiction. So the edge cell $(Y(t-1, e), Y(t, e))$ of $F^t c$ is never $(0, 0)$, and $e(F^t c) = e$. The recurrences are the relation at (t, e) with $f_{42}(a, b, 0) = 0$ and $f_{46}(a, b, 0) = b \wedge \neg a$.

(3) The proofs of Section 4 use the following properties of f, all of which hold by (6) and (2). Lemma 4.1: $f(1, 1, 1) = 0$, $f(0, 1, 1) = 1$, $f(a, 0, 1) = 1$, $f(a, 0, 0) = 0$, $f(a, 0, c) = c$, $f(a, 1, 1) = \neg a$, and that the edge column is never 0 twice in a row. Lemma 4.3(a) and (c): the gluing facts (i) and (ii), which use $f(a, 0, 0) = 0$ and $f(a, 0, 1) = 1$. Lemma 4.3(b): in addition $f(a, 1, 1) = \neg a$, $f(0, 1, 1) = 1$, $f(1, 1, 1) = 0$, and that the edge column is never 0 twice in a row, which is (2). The implication from (H_{R+1}) to Lemma K uses nothing about f. The proof of Proposition 4.2 uses Lemma 4.1 and, in its last step, the recurrence of the edge column, which for 46 is not autonomous: it reads the cell to the left of the edge. That step is replaced as follows. The proof shows $e(hd) = e$ and $Y_{hd}(s, e) = Y_d(s, e)$ for every $s \geq 2R$, for every 0-tailed d with $e(d) = e$. For $T \in \mathbb{Z}$ the configuration $F^{-T} d$ is 0-tailed with edge e, and $hF^{-T} d = F^{-T} hd$; applied to it, the identity gives $Y_{hd}(s - T, e) = Y_d(s - T, e)$ for all $s \geq 2R$ and all T, that is, at all times. □

5.2. **Edge moves for 42 and 46.** The moves differ from those of 110.

Lemma 5.2 (edge moves for 42 and 46)**.** *Let b be 0-tailed with $e(b) = r$, let $m \in \mathbb{Z}$, and let (x_i, y_i) be the cells of $F^m b$, as in Lemma 4.4. Then $M_m(b)$ and $\bar{M}_m(b)$ are 0-tailed with edge r, and $F^m M_m(b)$ and $F^m \bar{M}_m(b)$ agree with $F^m b$ at every cell except the edge cell, where (x_r, y_r) changes as follows.*

(a) *For 42: under M_m, 01 becomes 10, and $(1, y_r)$ becomes $(1, y_r \oplus 1 \oplus x_{r-1})$; under $\bar{M}_m$, 10 becomes 01, and $(x_r, 1)$ becomes $(x_r \oplus 1 \oplus y_{r-1}, 1)$.*

(b) *For 46: under M_m, 01 becomes 10, and $M_m(b) = b$ if $x_r = 1$; under $\bar{M}_m$, 10 becomes 01, and $\bar{M}_m(b) = b$ if $y_r = 1$.*

(c) *$hM_m = M_m h$ and $h\bar{M}_m = \bar{M}_m h$ on 0-tailed configurations with edge r, for every $h \in \mathcal{H}$ and $m \in \mathbb{Z}$. Every move is undone by a move at the same time. The moves at time m connect $x_r y_r = 01$ with 10; for 42 they connect 11 with 01 or 10 if and only if $(x_{r-1}, y_{r-1}) \neq (1, 1)$, and for 46 they never change an edge cell 11.*

Proof. Follow the proof of Lemma 4.4(a), with $n = m - 2$, $P = P_n(b)$, $D = D_n(b)$ and ε as there. The case $x_r = 0$ of that proof uses only $f(a, 0, 1) = 1$, $f(a, 0, 0) = 0$, the values $Y_P(n+2, r) = 0$ and $Y_P(n+3, r) = 1$ of Lemma 4.3(b), and that the edge column is never 0 twice in a row, and it gives $01 \to 10$ for both rules. If $x_r = 1$, then $\varepsilon = 0$, and rows $m-1$, m of D are x and y on the columns $\leq r$, except that $Y_D(m, r) = Y(m-2, r) \oplus f(x_{r-1}, 1, 1) = Y(m-2, r) \oplus \neg x_{r-1}$, while $y_r = Y(m-2, r) \oplus f(x_{r-1}, 1, 0)$, which is $Y(m-2, r)$ for 42 and $Y(m-2, r) \oplus \neg x_{r-1}$ for 46. This is the statement for M_m; the statement for $\bar{M}_m$ follows by ρ, as in Lemma 4.4(b), and the commutation in (c) is proved as in Lemma 4.4(c), from Lemma 4.3(b) and (c). The moves $01 \to 10$ and $10 \to 01$ are inverse to each other, and for 42 the moves $1y \to 1\bar{y}$ (when $x_{r-1} = 0$) and $x1 \to \bar{x}1$ (when $y_{r-1} = 0$) are involutions, since a move does not change the cell (x_{r-1}, y_{r-1}). The last statement is read off from (a) and (b). □

So the moves of 42 are those of 110 with the conditions on the left neighbour complemented, $1y \to 1\bar{y}$ if $x_{r-1} = 0$ and $x1 \to \bar{x}1$ if $y_{r-1} = 0$, and for 46 only the moves $01 \leftrightarrow 10$ remain. With Lemma 5.2 in place of Lemma 4.4, and Lemma 5.1(2) in place of Section 4.1, the proofs

of Lemma 4.6, Proposition 4.7, Lemma 4.8(a) and Lemma 4.9 apply to F without change; Lemma 4.6 uses $f(a,1,1)=\neg a$, and the rest nothing about f. As there, a move at time m is determined by its effect on the state at time m, so $F^jM_m = M_{m-j}F^j$ and $F^j\bar{M}_m = \bar{M}_{m-j}F^j$ for $j\in\mathbb{Z}$, and $\rho M_{-m}\rho = \bar{M}_m$, so that $\rho\,\mathcal{O}_k(c) = \mathcal{O}_k(\rho c)$.

Proposition 5.3 ((H_2) for 42 and 46)**.** *For $f\in\{42,46\}$, (H_2) holds.*

Proof. Let $h\in\mathcal{H}$, let c be 0-tailed with $e(c)=r$, and put $Y=Y_c$ and $G=Y_{hc}\oplus Y$. By Proposition 4.2, G vanishes on the columns $\geq r$; we show $G(t,r-1)=0$.

(a) Suppose $Y(t,r)=1$. For 42, argue as in the proof of Proposition 4.5(a), with the new edge value $Y(t+1,r)\oplus 1\oplus Y(t,r-1)$ of Lemma 5.2(a) in place of $Y(t+1,r)\oplus Y(t,r-1)$. For 46, the relation at (t,r) reads $Y(t+1,r)\oplus Y(t-1,r) = f(Y(t,r-1),1,0) = \neg Y(t,r-1)$ in both diagrams, and its left side is the same in both, so $G(t,r-1)=0$.

(b) Suppose $Y(t,r)=0$. Then $Y(t+1,r)=1$ by Lemma 5.1(2), so the edge cell of the state at time $t+1$ is $(0,1)$, for c and, by Proposition 4.2, for hc. The move M_{t+1} replaces it by $(1,0)$ and changes nothing else at time $t+1$. So $b=M_{t+1}c$ has $Y_b(t,r)=1$ and $Y_b(t,r-1)=Y(t,r-1)$, and $M_{t+1}hc$, which is hb by Lemma 5.2(c), has $Y_{hb}(t,r-1)=Y_{hc}(t,r-1)$. By (a) for b, $Y_{hb}(t,r-1)=Y_b(t,r-1)$. Hence $G(t,r-1)=0$. □

5.3. **Controllability for 42.** For 42 the orbits of the moves are not those of Theorem 4.10; the frozen exterior cell is $(1,1)$ instead of $(0,0)$. We prove the part of the orbit structure that (CTRL_k) needs. As in Section 4, c_{r-k+1} is the tip and c_{r-k} the exterior cell of c at level k.

Theorem 5.4. *Let $f=42$, $k\geq 1$, let c be 0-tailed with $e(c)=r$, and let $\mathcal{C}$ be the set of configurations c' with $c'_i=c_i$ for $i\notin[r-k+1,r]$ and $c'_r\neq(0,0)$.*

(A) *If $c_{r-k}\neq(1,1)$, then $\mathcal{O}_k(c)=\mathcal{C}$.*

(B) *If $c_{r-k}=(1,1)$ and $c_{r-k+1}\neq(1,1)$, then $\mathcal{O}_k(c)$ contains every $c'\in\mathcal{C}$ with $c'_{r-k+1}\neq(1,1)$.*

Proof. By Lemma 4.8(a), $\mathcal{O}_k(c)\subseteq\mathcal{C}$, and by Lemma 5.2(c) the orbits are the classes of an equivalence relation. We prove (A) and (B) together by induction on k. For $k=1$, $\mathcal{C}$ allows only a change of the edge cell, and by Lemma 5.2(c) with $m=0$ and $(x_{r-1},y_{r-1})=c_{r-1}$, the three values 01, 10, 11 are connected if $c_{r-1}\neq(1,1)$, and 01, 10 are connected in any case.

Let $k\geq 2$ and assume (A) and (B) at level $k-1$ for all 0-tailed configurations. For $b\in\mathcal{C}$ write $b_{r-k+j}=(u_j,v_j)$ for $j=0,1,2$, so that $(u_0,v_0)=c_{r-k}$, and let α, β be as in Lemma 4.9(b). For $t\in\{0,1\}^2$ let C_t be the set of $b\in\mathcal{C}$ with tip t; it is nonempty, as the cells $r-k+2,\dots,r$ are free subject to $b_r\neq(0,0)$. We use $f(a,0,c)=c$, $f(0,1,c)=c$ and $f(1,1,c)=0$.

Step 1. For $t\neq(1,1)$, C_t lies in one orbit: two elements of C_t agree outside $[r-k+2,r]$, and the exterior cell of either at level $k-1$ is $t\neq(1,1)$, so (A) at level $k-1$ applies.

Step 2 (tip $(u_1,0)\to(\neg u_1,0)$). If $b\in C_{(u_1,0)}$, then $\mathcal{O}_k(b)$ meets $C_{(\neg u_1,0)}$. The exterior cell of Fb at level $k-1$ is $(0,\alpha)\neq(1,1)$, with $\alpha=u_1\oplus f(v_0,0,v_2)=u_1\oplus v_2$. Let $\tilde{b}$ be Fb with its cell $r-k+2$ replaced by $(\neg v_2,1)$. It agrees with Fb outside $[r-k+2,r]$, and $\tilde{b}_r\neq(0,0)$ (for $k=2$ the cell r is the new cell, for $k\geq 3$ it is the edge cell of Fb). By (A) at level $k-1$, $\tilde{b}\in\mathcal{O}_{k-1}(Fb)$, so $b'=F^{-1}\tilde{b}\in\mathcal{O}_k(b)$ by Lemma 4.9(a), and by Lemma 4.9(c) with $w=\neg v_2$, $b'_{r-k+1}=(\alpha\oplus f(v_0,0,\neg v_2),0)=(u_1\oplus v_2\oplus\neg v_2,0)=(\neg u_1,0)$.

Step 3 (tip $(0,v_1)\to(0,\neg v_1)$). If $b\in C_{(0,v_1)}$, then $\mathcal{O}_k(b)$ meets $C_{(0,\neg v_1)}$: apply Step 2 to ρb, whose tip is $(v_1,0)$, and then ρ, using $\rho\,\mathcal{O}_k(\rho b)=\mathcal{O}_k(b)$; ρ maps $\mathcal{C}$ onto the corresponding set for ρc and fixes the cells $(0,0)$ and $(1,1)$.

Step 4 (tip $(1,1)$). If $v_0=0$, then for every $b\in C_{11}$ the orbit $\mathcal{O}_k(b)$ meets C_{01}; if $u_0=0$, it meets C_{10}. Let $v_0=0$. Then $\alpha=1\oplus f(0,1,v_2)=\neg v_2$. Let $\tilde{b}$ be Fb with its cell $r-k+2$ replaced by $(\neg v_2,v_2)$, which is not $(0,0)$. If $v_2=1$, the exterior cell $(1,\alpha)=(1,0)$ of Fb at level $k-1$ is not $(1,1)$, and (A) at level $k-1$ gives $\tilde{b}\in\mathcal{O}_{k-1}(Fb)$. If $v_2=0$, the exterior cell is $(1,1)$, and the

tip $(0,\beta)$ of Fb and the tip $(1,0)$ of $\tilde{b}$ are not $(1,1)$, so (B) at level $k-1$ gives $\tilde{b} \in \mathcal{O}_{k-1}(Fb)$. In both cases $b' = F^{-1}\tilde{b} \in \mathcal{O}_k(b)$, and $b'_{r-k+1} = (\alpha \oplus f(0,1,\neg v_2),1) = (\neg v_2 \oplus \neg v_2, 1) = (0,1)$. If $u_0 = 0$, apply this to ρb, whose exterior cell (v_0,u_0) has y-bit 0, and then ρ.

Conclusion. By Step 1, each of C_{00}, C_{01}, C_{10} lies in one orbit; by Step 2 the orbit of C_{00} meets C_{10}, and by Step 3 the orbit of C_{01} meets C_{00}. So $\mathcal{C} \setminus C_{11}$ lies in one orbit, and if $c \notin C_{11}$ this orbit is $\mathcal{O}_k(c)$; this gives (B). If $c_{r-k} \neq (1,1)$, then $v_0 = 0$ or $u_0 = 0$, so by Step 4 every element of C_{11} is in the orbit of an element of $C_{01} \cup C_{10}$; hence $\mathcal{C}$ is one orbit, which is (A). □

With exterior cell $(1,1)$ the inclusion (B) is in fact an equality, the classes being the level sets of the length of the run of cells $(1,1)$ that starts at the tip (this has been checked for $k \leq 4$ and is not used). Only the inclusion is needed.

Corollary 5.5. *For $f = 42$, (CTRL$_k$) holds for every $k \geq 2$, and (H$_k$) holds for every $k \geq 1$.*

Proof. Let c be 0-tailed with $e(c) = r$. If $c_{r-k} \neq (1,1)$, or $c_{r-k} = (1,1)$ and $c_{r-k+1} \neq (1,1)$, Theorem 5.4 at level k gives $c' = Oc$, for a composite O of moves at the times $|m| \leq k-1$, with $c'_i = (0,1)$ for $r-k+1 \leq i \leq r$ and $c' = c$ elsewhere. If $c_{r-k} = c_{r-k+1} = (1,1)$, the tip is frozen; then either $c_{r-k+2} = (1,1)$, and $O = \mathrm{id}$ will do, or (B) at level $k-1$, where the exterior cell is $c_{r-k+1} = (1,1)$, gives $c' \in \mathcal{O}_{k-1}(c) \subseteq \mathcal{O}_k(c)$ with $c'_{r-k+1} = (1,1)$, $c'_i = (0,1)$ for $r-k+2 \leq i \leq r$, and $c' = c$ elsewhere. (For $k = 2$ this is the move $10 \to 01$ at time 0.) In each case $Y_{c'}(0, r-k+1) = Y_{c'}(0, r-k+2) = 1$, which is (CTRL$_k$). With Propositions 4.2 and 5.3, Proposition 4.7 gives (H$_k$) for every k. □

5.4. **The class of 14.** The rule 14 is $f(a,b,c) = \neg a \wedge (b \vee c)$, with values $0,1,1,1,0,0,0,0$ on $000,\ldots,111$. So $f(0,0,1) = f(0,1,c) = 1$, and $f(1,b,c) = f(0,0,0) = 0$. Its automorphisms are normalised by offsets on three families of fixed configurations, related by a collision of kinks. In Lemma 5.6 to Proposition 5.9, $f = 14$ and $F = F_{14}$. For $p \in \mathbb{Z}$ let η_p be as in Lemma 3.1, and put

$$\lambda_p = (1,1)^{(-\infty,p)}(0,0)^{[p,\infty)}, \qquad \zeta_p = (0,0)^{(-\infty,p]}(0,1)_{p+1}(1,1)^{[p+2,\infty)},$$

where the subscript $p+1$ marks the position of the cell $(0,1)$; so $\zeta_p = \rho\eta_{p+1}$, with $\rho(x,y) = (y,x)$ the time reversal, which satisfies $\rho F \rho = F^{-1}$ for every f.

Lemma 5.6 (fixed sets for 14)**.** *The fixed points of F are $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ and the λ_p. The fixed points of σF are $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ and the η_p, and $F\eta_p = \eta_{p+1}$. The fixed points of $\sigma^{-1}F$ are $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ and the ζ_p, and $F\zeta_p = \zeta_{p-1}$.*

Proof. By (1), $F(c) = c$ means that $x_i = y_i$ and $f(y_{i-1}, y_i, y_{i+1}) = 0$ for all i. Since $f(0,b,c) = 0$ only for $b = c = 0$, a bit $y_{i-1} = 0$ forces $y_i = y_{i+1} = 0$, so y is $0^{\mathbb{Z}}$, $1^{\mathbb{Z}}$ or $1^{(-\infty,p)}0^{[p,\infty)}$; conversely these satisfy the condition, since $f(1,b,c) = f(0,0,0) = 0$. With $x = y$ they give the two constants and the λ_p. For $\sigma^{\pm 1}F$ we use Lemma 2.3. The allowed triples (a_1,a_2,a_3), those with $a_1 \oplus a_3 = f(a_1,a_2,a_3)$, are 000, 001, 011, 101 and 111. No allowed triple has the form $a10$, so in a sequence of Φ_{14} the word 10 has no predecessor and never occurs: the sequence is nondecreasing, that is $0^{\mathbb{Z}}$, $1^{\mathbb{Z}}$ or $\varphi(k) = [k > p]$, and these use only the triples 000, 001, 011 and 111. So $\Phi_{14} = \Phi_{110}$ as a set of sequences, although the allowed triples differ (14 allows 101 where 110 allows 110, and the word 10 has no allowed predecessor for 14, but no allowed continuation for 110). Since the bijections of Lemma 2.3 depend only on Φ_f, the computation in the proof of Lemma 3.1 applies verbatim: the fixed points of σF are the constants and the η_p, with $F\eta_p = \sigma^{-1}\eta_p = \eta_{p+1}$, and those of $\sigma^{-1}F$ are the constants and the configurations $d_i = (\varphi(i-1), \varphi(i))$ for $\varphi(k) = [k > p]$, which is ζ_p, with $F\zeta_p = \sigma\zeta_p = \zeta_{p-1}$. □

Proposition 5.7 (offsets for 14)**.** *Every $g \in \mathrm{Aut}(X_F)$ fixes $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, and there are integers $o_\lambda(g)$, $o_\eta(g)$, $o_\zeta(g)$ with*

$$g\,\lambda_p = \lambda_{p+o_\lambda(g)}, \qquad g\,\eta_p = \eta_{p+o_\eta(g)}, \qquad g\,\zeta_p = \zeta_{p+o_\zeta(g)} \qquad (p \in \mathbb{Z}).$$

The maps $o_\lambda, o_\eta, o_\zeta \colon \mathrm{Aut}(X_F) \to \mathbb{Z}$ are homomorphisms, $|o_\lambda(g)|, |o_\eta(g)|, |o_\zeta(g)| \le R$ if g has radius R, and

$$\big(o_\lambda, o_\eta, o_\zeta\big)(\sigma^m F^n) = \big(-m,\ n-m,\ -m-n\big).$$

In particular $(m,n) \mapsto \sigma^m F^n$ is an isomorphism $\mathbb{Z}^2 \to \langle \sigma, F \rangle$.

Proof. Since g and g^{-1} commute with σ and F, g permutes the fixed points of F, of σF and of $\sigma^{-1}F$, and the two constants, which are the configurations fixed by both σ and F (F maps $(x,y)^{\mathbb{Z}}$ to $(y,x)^{\mathbb{Z}}$ because $f(0,0,0) = f(1,1,1) = 0$). If g exchanged the constants, then $g\lambda_0$, which agrees with $g((1,1)^{\mathbb{Z}})$ far to the left because g is a block map, would be left-asymptotic to $(0,0)^{\mathbb{Z}}$, while by Lemma 5.6 it is one of the λ_q, all left-asymptotic to $(1,1)^{\mathbb{Z}}$. So g fixes both constants and permutes each of the three families of Lemma 5.6. Put $g\lambda_0 = \lambda_u$, $g\eta_0 = \eta_v$ and $g\zeta_0 = \zeta_w$, and $(o_\lambda, o_\eta, o_\zeta)(g) = (u,v,w)$. Since $\sigma\lambda_p = \lambda_{p-1}$, $\sigma\eta_p = \eta_{p-1}$ and $\sigma\zeta_p = \zeta_{p-1}$, and g commutes with σ, $g\lambda_p = \lambda_{p+u}$, $g\eta_p = \eta_{p+v}$ and $g\zeta_p = \zeta_{p+w}$ for all p, and u, v, w are homomorphisms. If g has radius R, then $(g\lambda_0)_i$ is $(1,1)$ for $i < -R$ and $(0,0)$ for $i \ge R$, so $-R \le u \le R$; $(g\eta_0)_i$ is $(0,0)$ for $i < -R$ and $(1,1)$ for $i > R$, so $|v| \le R$; and $(g\zeta_0)_i$ is $(0,0)$ for $i \le -R$ and $(1,1)$ for $i \ge R+2$, so $-R \le w$ and $w+1 \le R+1$. The values on $\sigma^m F^n$ follow from $F\lambda_p = \lambda_p$, $F\eta_p = \eta_{p+1}$, $F\zeta_p = \zeta_{p-1}$ and the action of σ just recalled. The map $(m,n) \mapsto \sigma^m F^n$ is a homomorphism onto $\langle \sigma, F \rangle$, since σ and F commute; it is injective because $\sigma^m F^n = \mathrm{id}$ forces $o_\lambda(\sigma^m F^n) = -m = 0$ and then $o_\eta(\sigma^m F^n) = n = 0$. □

For $N \ge 1$ and $p < N$, and for $q \le N-2$, let

$$E(p,N) = (0,0)^{(-\infty,p)}(1,0)_p(1,1)^{(p,N)}(0,0)^{[N,\infty)},$$
$$Z(q,N) = (0,0)^{(-\infty,q]}(0,1)_{q+1}(1,1)^{[q+2,N)}(0,0)^{[N,\infty)}.$$

So $E(p,N)$ is a kink η_p followed by a static kink λ_N, and $Z(q,N)$ is a kink ζ_q followed by λ_N; $\sigma^{-k}E(p,N) = E(p+k,N+k)$ and $\sigma^{-k}Z(q,N) = Z(q+k,N+k)$.

Lemma 5.8 (collision)**.** *Let $N \ge 1$. The diagram of $E(0,N)$ is*

$$Y(t,i) = \big[\, N-1-|t-N+1| < i < N \,\big] \qquad (t,i \in \mathbb{Z}).$$

Consequently $F^t E(0,N) = E(t,N)$ for $t \le N-1$, and $F^t E(0,N) = Z(2N-2-t,N)$ for $t \ge N$: the moving kink η is reflected by the static kink λ_N as a kink ζ, and λ_N does not move.

Proof. Let $Q(t,i) = [t < i < N]$. We first show that Q satisfies the spacetime relation $Q(t+1,i) \oplus Q(t-1,i) = f(Q(t,i-1), Q(t,i), Q(t,i+1))$ at every (t,i) with $t \le N-2$. For $i < t$ and for $i > N$ all five values vanish, and $f(0,0,0) = 0$. For $i = t$ the left side is $0 \oplus 1$ and the triple is $(0,0,1)$, since $t+1 < N$. For $i = t+1$ the left side is $0 \oplus 1$ and the triple is $(0,1,\cdot)$. For $t+2 \le i \le N-1$ the left side is $1 \oplus 1$ and the triple begins with $Q(t,i-1) = 1$. For $i = N$ the left side is $0 \oplus 0$ and the triple is $(1,0,0)$. In each case the relation holds, by the values of f recalled above. The rows -1 and 0 of the diagram of $E(0,N)$, its x- and y-bits, are $[-1 < i < N]$ and $[0 < i < N]$, which are $Q(-1,\cdot)$ and $Q(0,\cdot)$. Each instance of the relation determines either time neighbour from the other four values, so, by induction forwards to $t = N-1$ and backwards to $t \to -\infty$, $Y = Q$ on all rows $t \le N-1$. In particular the row $N-1$ vanishes. Now $Y'(t,i) = Y(2N-2-t,i)$ also satisfies the relation, which is symmetric in $t \pm 1$; it agrees with Y on row $N-1$, and on row N because $Y(N,i) = Y(N-2,i) \oplus f(0,0,0) = Y(N-2,i)$. So $Y' = Y$, and $Y(t,i) = Q(2N-2-t,i)$ for $t \ge N-1$. Together these give the displayed formula. Reading off the cells $(Y(t-1,i), Y(t,i))$ of $F^t E(0,N)$ gives $E(t,N)$ for $t \le N-1$ and,

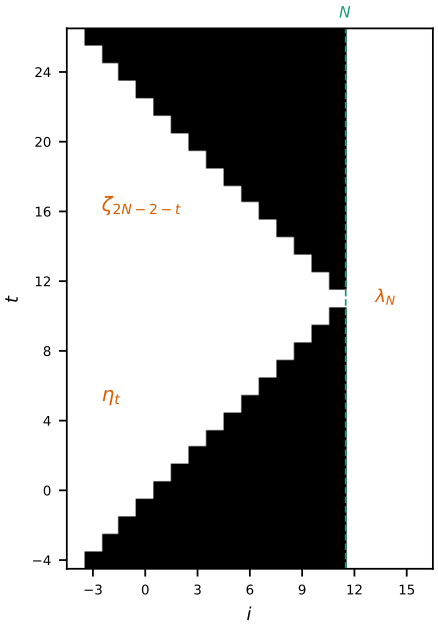


FIGURE 6. The collision of Lemma 5.8: the diagram of $E(0,N)$ for rule 14, $N=12$. The kink η moves to the right, is reflected by the static kink λ_N (dashed) at time $N-1$, and leaves as ζ_{2N-2-t}; λ_N does not move.

with $q = 2N-2-t \le N-2$, the x-bits $[q+1 < i < N]$ and y-bits $[q < i < N]$ of $Z(q,N)$ for $t \ge N$. □

Proposition 5.9 (relation of offsets)**.** *Every $g \in \mathrm{Aut}(X_F)$ satisfies $o_\zeta(g) = 2o_\lambda(g) - o_\eta(g)$.*

Proof. Let g have radius R, write $(u,v,w) = (o_\lambda, o_\eta, o_\zeta)(g)$, and let $N \ge 2R+1$. The configuration $E(0,N)$ agrees with η_0 on $(-\infty, N-1]$ and with λ_N on $[1,\infty)$. So $gE(0,N)$ agrees with $g\eta_0 = \eta_v$ on $(-\infty, N-1-R]$ and with $g\lambda_N = \lambda_{N+u}$ on $[R+1,\infty)$, and these half-lines cover $\mathbb{Z}$ because $N \ge 2R+1$. Since $|u|, |v| \le R$, $E(v, N+u)$ agrees with η_v on $(-\infty, N+u-1] \supseteq (-\infty, N-1-R]$ and with λ_{N+u} on $[v+1,\infty) \supseteq [R+1,\infty)$, and $N+u-v \ge N-2R \ge 1$. Hence $gE(0,N) = E(v,N+u) = \sigma^{-v}E(0,N')$ with $N' = N+u-v \ge 1$. In the same way, for $q \le N-2R-2$, $Z(q,N)$ agrees with ζ_q on $(-\infty, N-1]$ and with λ_N on $[q+2,\infty)$, so $gZ(q,N)$ agrees with ζ_{q+w} on $(-\infty, N-1-R]$ and with λ_{N+u} on $[q+R+2,\infty)$, and as before $gZ(q,N) = Z(q+w, N+u)$. Now take $t \ge \max(N,N') + 2R + 2$, so that $q = 2N-2-t \le N-2R-2$. By Lemma 5.8, applied to N and to N',

$$\begin{aligned} Z(2N-2-t+w,\, N+u) &= gF^tE(0,N) = F^tgE(0,N) = \sigma^{-v}F^tE(0,N') \\ &= \sigma^{-v}Z(2N'-2-t,\, N') = Z(2N'-2-t+v,\, N+u). \end{aligned}$$

Comparing the positions of the cell $(0,1)$ gives $2N+w = 2N'+v = 2N+2u-v$. □

On $\langle \sigma, F\rangle$ the relation is the identity $-m-n = 2(-m)-(n-m)$ of Proposition 5.7. The proof compares the outgoing kink of the collision before and after applying g (Figure 6). The rules 14 and 42 are related by $\Psi = \kappa \circ \mathrm{rev}$, where $\mathrm{rev}(c)_i = c_{-i}$ and $\kappa(x,y) = (1-x, 1-y)$: reflection takes 14 to 84, $f(a,b,c) \mapsto f(c,b,a)$, and complementation takes 84 to 42, $f(a,b,c) \mapsto f(1-a,1-b,1-c)$, so $\Psi F_{14}\Psi^{-1} = F_{42}$. Also $\Psi\sigma = \sigma^{-1}\Psi$, and a direct check of the cells gives

$$\Psi\eta_p = \rho\eta_{-p}, \qquad \Psi\zeta_p = \eta_{-p-1}, \qquad \Psi\lambda_p = \lambda_{1-p} \qquad (p \in \mathbb{Z}), \tag{7}$$

where on the right η_p and $\rho\eta_p$ are the kinks of Lemma 3.1, which by Lemma 5.1(1) are the fixed points of $\sigma^{\pm1}F_{42}$ other than the constants.

Theorem 5.10. *Lemma K holds for F_{42} at every radius, and every $g \in \mathrm{Aut}(X_{F_{14}})$ equals $\sigma^{-o_\lambda(g)}F_{14}^{o_\eta(g)-o_\lambda(g)}$. Hence*

$$\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2 \qquad \text{for } f \in \{14, 42, 84, 112\}.$$

Proof. By Corollary 5.5, (H_{R+1}) holds for F_{42} for every R, and by Lemma 5.1(3) it implies Lemma K at radius R. Let $g \in \mathrm{Aut}(X_{F_{14}})$ have offsets $(o_\lambda, o_\eta, o_\zeta)(g) = (u, v, w)$, and put $T = \sigma^{-u}F_{14}^{v-u}$. By Proposition 5.7, T has offsets u, v and $2u - v$, and $2u - v = w$ by Proposition 5.9. So $h = T^{-1}g$ has $o_\eta(h) = o_\zeta(h) = 0$: it fixes every η_p and every ζ_p. By (7), $\tilde{h} = \Psi h \Psi^{-1}$ is a block map commuting with F_{42} that fixes $\Psi\zeta_{-1} = \eta_0$ and $\Psi\eta_0 = \rho\eta_0$, so $\tilde{h} = \mathrm{id}$ by Lemma K for F_{42}, at the radius of $\tilde{h}$. Hence $g = T$, and $\langle \sigma, F_{14} \rangle \cong \mathbb{Z}^2$ by Proposition 5.7. For the other members of the class, reflection and complementation conjugate F_{14} to F_f and map $\langle \sigma, F_{14} \rangle$ onto $\langle \sigma, F_f \rangle$. □

The normalisation by T needs neither the parity step of Proposition 3.4 nor a larger radius: T has radius at most $|o_\lambda| + |o_\eta - o_\lambda| = \max(|o_\zeta|, |o_\eta|)$, which is at most the radius of g. For 42 alone, Proposition 3.4 would also do, by Lemma 5.1(1).

5.5. **Rule 46.** For 46 the moves are only $01 \leftrightarrow 10$, and the orbits are described exactly by a conserved bit on each pair of adjacent cells. For a configuration c and $i \in \mathbb{Z}$ put

$$\chi_i(c) = [c_i = (0,0)] \oplus [c_{i+1} = (1,1)].$$

That F_{46} conserves a quantity on every pair of adjacent cells is due to Takesue: it is the local conservation law of 46R of [28, (4.4)–(4.6)], where it is obtained from the vanishing of the flux (4.6). In the form χ_i used here, the bond energy (4.5), with its last factor read as $1 - 2\sigma_{i+1}\hat{\sigma}_{i+1}$ in the notation there, where $\sigma_i = y_i$ and $\hat{\sigma}_i = x_i$, is $2(1 - \chi_i)$. The next lemma is proved in full, since it is needed in this form and its statements on ρ and on the edge moves are not in [28].

Lemma 5.11 (bond bits; cf. [28, (4.4)–(4.6)])**.** *Let $f = 46$.*

1. *$\chi_i(Fc) = \chi_i(c)$ and $\chi_i(\rho c) = \chi_i(c)$ for every configuration c and every i.*
2. *If b is 0-tailed with $e(b) = r$, then $\chi_r(b) = 0$, and $\chi_i(M_m b) = \chi_i(\bar{M}_m b) = \chi_i(b)$ for all i and m.*

Proof. (1) Write $c_j = (x_j, y_j)$. By (1), $F(c)_i = (0,0)$ if and only if $y_i = 0$ and $x_i = f(y_{i-1}, 0, y_{i+1}) = y_{i+1}$, and $F(c)_{i+1} = (1,1)$ if and only if $y_{i+1} = 1$ and $x_{i+1} \oplus f(y_i, 1, y_{i+2}) = x_{i+1} \oplus \neg y_i = 1$, that is $x_{i+1} = y_i$. So $\chi_i(Fc) = [y_i = 0,\, x_i = y_{i+1}] \oplus [y_{i+1} = 1,\, x_{i+1} = y_i]$, to be compared with $\chi_i(c) = [x_i = y_i = 0] \oplus [x_{i+1} = y_{i+1} = 1]$. For $(y_i, y_{i+1}) = (0,0)$ both are $[x_i = 0]$; for $(1,0)$ both are 0; for $(1,1)$ both are $[x_{i+1} = 1]$; and for $(0,1)$ they are $[x_i = 1] \oplus [x_{i+1} = 0]$ and $[x_i = 0] \oplus [x_{i+1} = 1]$, which are equal. Since ρ exchanges the two bits of every cell, it fixes the conditions $c_i = (0,0)$ and $c_{i+1} = (1,1)$.

(2) $b_r \neq (0,0)$ and $b_{r+1} = (0,0)$, so $\chi_r(b) = 0$. By Lemma 5.2(b), $F^m M_m b$ and $F^m \bar{M}_m b$ differ from $F^m b$ at most in the edge cell, which is 01 or 10 before and after a change; so χ_{r-1} and χ_r of the state at time m do not change, and no other χ_i reads the edge cell. By (1), $\chi_i(F^m b) = \chi_i(b)$. □

For 46 the clean reading needs only one target cell, because $f(a, 1, c) = \neg a$ does not read c.

Lemma 5.12 (clean reading for 46)**.** *Let $f = 46$ and $k \geq 1$, and assume (H_k). Let $h \in \mathcal{H}$, let b be 0-tailed with $e(b) = r$, and let O be a finite composite of moves M_m, $\bar{M}_m$ with $|m| \leq k - 1$. If $Y_{Ob}(0, r - k + 1) = 1$, then $Y_{hb}(0, r - k) = Y_b(0, r - k)$. Consequently, if*

(CTRL'_k) for every 0-tailed c, with $r = e(c)$, there is a finite composite O of moves M_m, $\bar{M}_m$ with $|m| \le k-1$ such that $Y_{Oc}(0, r-k+1) = 1$,

then (H_k) *implies* (H_{k+1}).

Proof. As in the proof of Lemma 4.6, with $Y = Y_{Ob}$ and $G' = Y_{hOb} \oplus Y$, the moves leave row 0 unchanged on the columns $\le r-k$, so $G'(0, r-k) = Y_{hb}(0, r-k) \oplus Y_b(0, r-k)$, and G' vanishes on the columns $\ge r-k+1$ at all times, by (H_k) and, for $k = 1$, because both diagrams vanish on column $r+1$. The relation (5) for G' at $(0, r-k+1)$ reads $0 = f(Y(0, r-k) \oplus G'(0, r-k), 1, \cdot) \oplus f(Y(0, r-k), 1, \cdot)$, with the same third argument on both sides, and $f(a, 1, c) = \neg a$ gives $G'(0, r-k) = 0$. The second statement follows as Proposition 4.7 follows from Lemma 4.6. □

Theorem 5.13. *Let $f = 46$, $k \ge 1$, and let c be 0-tailed with $e(c) = r$. Then $\mathcal{O}_k(c)$ is the set of configurations c' with*

(i) *$c'_i = c_i$ for every $i \notin [r-k+1, r]$, and $c'_r \ne (0,0)$; and*
(ii) *$\chi_i(c') = \chi_i(c)$ for $r-k \le i \le r-1$.*

Call this set $\mathcal{T}_k(c)$. Given the exterior cell, condition (ii) determines, cell by cell from the left, whether each of the cells $r-k+1, \dots, r$ is $(1,1)$, given whether the cell before it is $(0,0)$. So every choice of the cells $r-k+1, \dots, j$ that satisfies the conditions $\chi_i(c') = \chi_i(c)$ for $r-k \le i < j$ (and $c'_r \ne (0,0)$ if $j = r$) extends to an element of $\mathcal{T}_k(c)$: for $i = j, \dots, r-1$ in turn, take $c'_{i+1} = (1,1)$ if $\chi_i(c) \oplus [c'_i = (0,0)] = 1$ and $c'_{i+1} = (0,1)$ otherwise. We call this the *filling*. In particular there are 2^k sets $\mathcal{T}_k(c)$ for each exterior cell.

Proof. By Lemma 4.8(a) and Lemma 5.11, $\mathcal{O}_k(c) \subseteq \mathcal{T}_k(c)$. We prove the converse by induction on k. For $k = 1$, $\mathcal{T}_1(c)$ consists of the configurations that differ from c at most in the edge cell, with $[c'_r = (1,1)] = [c_r = (1,1)]$; this is $\{c\}$ if $c_r = (1,1)$ and the two configurations with edge cell 01, 10 otherwise, and the moves at time 0 connect these.

Let $k \ge 2$ and assume the theorem at level $k-1$. For $b \in \mathcal{T}_k(c)$ write $b_{r-k+j} = (u_j, v_j)$ for $j = 0, 1, 2$, where $(u_0, v_0) = c_{r-k}$, and let v_3 be the y-bit of b_{r-k+3} ($v_3 = 0$ if $k = 2$).

Step 1. If $b, b' \in \mathcal{T}_k(c)$ have the same tip, then $b' \in \mathcal{O}_k(b)$. Indeed b' agrees with b outside $[r-k+2, r]$ and has the same χ_i for $r-k+1 \le i \le r-1$, so $b' \in \mathcal{O}_{k-1}(b) \subseteq \mathcal{O}_k(b)$ by the theorem at level $k-1$.

Step 2. If $b \in \mathcal{T}_k(c)$ has tip $(u_1, 0)$, then $\mathcal{O}_k(b)$ contains an element of $\mathcal{T}_k(c)$ with tip $(\neg u_1, 0)$. First, if $(u_2, v_2) = (0,1)$, replace the cell $r-k+2$ by $(1,0)$; this changes no χ_i, since neither cell is $(0,0)$ or $(1,1)$, so by Step 1 we may assume $(u_2, v_2) \ne (0,1)$. The cells $r-k+1$ and $r-k+2$ of Fb are $(v_1, \alpha) = (0, u_1 \oplus v_2)$ and (v_2, β), where $\alpha = u_1 \oplus f(v_0, 0, v_2) = u_1 \oplus v_2$ and $\beta = u_2 \oplus f(0, v_2, v_3)$; if $v_2 = 1$ then $u_2 = 1$ and $\beta = u_2 \oplus 1 = 0$. So $(v_2, \beta) \ne (1,1)$. Let $\tilde{b}$ agree with Fb outside $[r-k+2, r]$, with $\tilde{b}_{r-k+2} = (\neg v_2, v_2)$, and with the cells $r-k+3, \dots, r$ given by the filling for $\mathcal{T}_{k-1}(Fb)$; this is possible because $[\tilde{b}_{r-k+2} = (1,1)] = 0 = [(v_2, \beta) = (1,1)]$, so that $\chi_{r-k+1}(\tilde{b}) = \chi_{r-k+1}(Fb)$. By the theorem at level $k-1$, applied to the 0-tailed configuration Fb, $\tilde{b} = O'Fb$ for a composite O' of moves at the times $|m| \le k-2$, so $b' = F^{-1}\tilde{b} = F^{-1}O'Fb$ is the image of b under the moves at the times $m+1$, and $b' \in \mathcal{O}_k(b)$. Its tip (u'_1, v'_1) is computed from $Y_{b'}(0, \cdot)$, the x-bits of $\tilde{b}$, and $Y_{b'}(-1, i) = Y_{b'}(1, i) \oplus f(Y_{b'}(0, i-1), Y_{b'}(0, i), Y_{b'}(0, i+1))$: the x-bit of $\tilde{b}_{r-k+1}$ gives $v'_1 = 0$, and $u'_1 = \alpha \oplus f(v_0, 0, \neg v_2) = u_1 \oplus v_2 \oplus \neg v_2 = \neg u_1$.

Step 3. If $b \in \mathcal{T}_k(c)$ has tip $(0, v_1)$, then $\mathcal{O}_k(b)$ contains an element of $\mathcal{T}_k(c)$ with tip $(0, \neg v_1)$. Apply Step 2 to ρb, which has tip $(v_1, 0)$, and then ρ, using $\rho\, \mathcal{O}_k(\rho b) = \mathcal{O}_k(b)$ and Lemma 5.11(1).

Finally, $\chi_{r-k} = [c_{r-k} = (0,0)] \oplus [c'_{r-k+1} = (1,1)]$ is fixed on $\mathcal{T}_k(c)$, so either every element of $\mathcal{T}_k(c)$ has tip $(1,1)$, and Step 1 gives $\mathcal{T}_k(c) = \mathcal{O}_k(c)$, or every tip lies in $\{00, 01, 10\}$. In the second case Steps 2 and 3 connect the tips 00 and 10, and 00 and 01, and Step 1 gives $\mathcal{T}_k(c) \subseteq \mathcal{O}_k(c)$. □

Corollary 5.14. *For $f = 46$, (CTRL'_k) holds for every $k \geq 1$, and (H_k) holds for every $k \geq 1$.*

Proof. By the filling, $\mathcal{T}_k(c)$ contains a configuration whose tip is $(1,1)$ or $(0,1)$, which has y-bit 1; by Theorem 5.13 it is Oc for a composite O of moves at the times $|m| \leq k-1$. This is (CTRL'_k). (H_1) is Proposition 4.2, and Lemma 5.12 gives (H_{k+1}) from (H_k) for every $k \geq 1$. □

So for 46 the step (H_2) of Proposition 5.3 is not needed separately.

Theorem 5.15. *Lemma K holds for F_{46} at every radius, every automorphism of $X_{F_{46}}$ with $e^+ = e^- = 0$ is the identity, and*

$$\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2 \qquad \text{for } f \in \{46, 116\}.$$

Proof. By Corollary 5.14, (H_{R+1}) holds for every R, and it implies Lemma K at radius R (Lemma 5.1(3)). An automorphism with $e^+ = e^- = 0$ fixes η_0 and $\rho\eta_0$ by Proposition 3.2, which holds for F_{46} by Lemma 5.1(1), so it is the identity. Proposition 3.4, which holds for F_{46} by the same lemma, gives $\mathrm{Aut}(X_{F_{46}}) = \langle \sigma, F_{46} \rangle \cong \mathbb{Z}^2$, every automorphism having finite radius by Curtis–Hedlund–Lyndon. Reflection, and also complementation, conjugate F_{46} to F_{116}. □

For 46 no computer is used. The bond bits are a conserved field, the local conservation law of Takesue: F and ρ preserve each χ_i (Lemma 5.11). In [28, p. 389] the rules 24R and 46R are singled out as having a local conservation law but no walls. By Theorem 5.15 this conserved structure yields no automorphism outside $\langle \sigma, F \rangle$.

6. The classes of 24 and 189

The method also decides the classes $\{24, 66\}$ and $\{189, 231\}$. Here the automorphism group is larger than $\langle \sigma, F_f \rangle$ by exactly one element of order 2, the complement $\kappa(x, y) = (1-x, 1-y)$ of Section 2.3. We work with the rule 66 throughout and obtain 189 from it, because the two automata agree after four steps. The proof for 66 is reduced to the rule 46 by a factor map. This map takes the configurations behind a stuck edge to those of Section 5.5, takes the kinks of 66 to the kinks of 46, and takes the edge moves of 66 to the edge moves of 46; so the controllability theorem for 46 (Theorem 5.13) can be used without change. Since F_{189} commutes with F_{66}^4 but not with F_{66}, Lemma K is proved for block maps that commute with a power F^N only.

6.1. **The rules.** In this section $f = 66$, $F = F_{66}$ and $F' = F_{189}$, and $\rho(x, y) = (y, x)$ is the time reversal, so that $\rho F \rho = F^{-1}$. The values of 66 on $000, \ldots, 111$ are $0, 1, 0, 0, 0, 0, 1, 0$, and

$$f(a, b, c) = [a = b]\,(b \oplus c).$$

We use the following consequences:

(8)

$$f(a,b,b) = 0, \qquad f(b,b,c) = b \oplus c, \qquad f(a,b,c) = 0 \ \text{ if } a \neq b, \qquad f(a,b,c) = [a = b] \ \text{ if } b \neq c,$$

and $f(1-a, 1-b, 1-c) = f(a, b, c)$. Rule 24 is the reflection $f(c, b, a) = [b = c](a \oplus b)$, rule 189 is $1 \oplus f$, and rule 231 is $1 \oplus f(c, b, a)$, the reflection of 189. The four rules are invariant under $f \mapsto f(1-a, 1-b, 1-c)$, so the classes under reflection and complementation are $\{24, 66\}$ and $\{189, 231\}$. None of the four rules is permutive in any variable: for 66, $f(a, 0, 0) = 0$ for both values of a, $f(0, b, 0) = 0$ for both values of b, and $f(1, 0, c) = 0$ for both values of c; complementing the output and reflecting do not change permutivity.

Lemma 6.1 (the rules)**.** (a) *$\kappa F = F\kappa$ and $\kappa F' = F'\kappa$, so $\kappa \in \mathrm{Aut}(X_F)$ and $\kappa \in \mathrm{Aut}(X_{F'})$.*

(b) *$F' = \nu F$, where $\nu(x, y) = (x, 1-y)$ cellwise. Consequently $F'^2 = \kappa F^2$ and $F'^4 = F^4$.*

(c) *On the constant configurations F acts by $(x, y)^{\mathbb{Z}} \mapsto (y, x)^{\mathbb{Z}}$: it fixes $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$ and exchanges $(0,1)^{\mathbb{Z}}$ and $(1,0)^{\mathbb{Z}}$. F' acts by $(x, y)^{\mathbb{Z}} \mapsto (y, 1-x)^{\mathbb{Z}}$, which is the 4-cycle $(0,0)^{\mathbb{Z}} \mapsto (0,1)^{\mathbb{Z}} \mapsto (1,1)^{\mathbb{Z}} \mapsto (1,0)^{\mathbb{Z}} \mapsto (0,0)^{\mathbb{Z}}$.*

(d) *The allowed triples of* 66 *(Section 2.4) are all triples except* 011 *and* 100*, and those of* 189 *are* 011 *and* 100*. Consequently* $\Phi_{189} = \emptyset$*, so* $\sigma F'$ *and* $\sigma^{-1}F'$ *have no fixed points.*

Proof. (a) By (1), $\kappa F_g \kappa = F_{g'}$ with $g'(a,b,c) = g(1-a, 1-b, 1-c)$ (Section 2.3), and $g' = g$ for $g \in \{66, 189\}$. (b) By (1), $F'(c)_i = (y_i, x_i \oplus 1 \oplus f(y_{i-1}, y_i, y_{i+1})) = \nu F(c)_i$. Since f is invariant under complementing its arguments, $F(\nu c)_i = (1 - y_i,\ x_i \oplus f(y_{i-1}, y_i, y_{i+1}))$, that is $F\nu = \nu' F$ with $\nu'(x,y) = (1-x, y)$. So $F'^2 = \nu F \nu F = \nu\nu' F^2 = \kappa F^2$, and $F'^4 = \kappa F^2 \kappa F^2 = F^4$ by (a). (c) $f(y,y,y) = 0$, so $F((x,y)^{\mathbb{Z}}) = (y,x)^{\mathbb{Z}}$ and $F'((x,y)^{\mathbb{Z}}) = (y, x \oplus 1)^{\mathbb{Z}}$. (d) A triple (a_1, a_2, a_3) is allowed for 66 if $a_1 \oplus a_3 = f(a_1, a_2, a_3)$: if $a_1 = a_2$ both sides are $a_2 \oplus a_3$; if $a_1 \neq a_2$ the right side is 0, so the triple is allowed exactly when $a_3 = a_1$. This excludes 011 and 100 only. The allowed triples of 189 are the complementary set. A sequence all of whose triples are 011 or 100 would contain a triple beginning with 11 or with 00. □

By (d), for 66 a sequence in Φ_{66} alternates from its first change on: if $\varphi(k) \neq \varphi(k+1)$, then $\varphi(k+2) = \varphi(k)$. For $\beta \in \{0,1\}$ and $p \in \mathbb{Z}$ let

$$\varphi_{p,\beta}(i) = \begin{cases} \beta, & i \leq p, \\ \beta \oplus \big((i-p) \bmod 2\big), & i \geq p, \end{cases}$$

and let $\alpha(i) = i \bmod 2$. Then Φ_{66} consists of $0^{\mathbb{Z}}$, $1^{\mathbb{Z}}$, α, its shift, and the $\varphi_{p,\beta}$: a sequence that changes somewhere either has a leftmost change, after a last position p of a constant run, or changes arbitrarily far to the left and so alternates everywhere; and all these sequences use only allowed triples.

Definition 6.2. For $p \in \mathbb{Z}$ and $\beta \in \{0,1\}$ let $K_{p,\beta}$ be the configuration with cells $(K_{p,\beta})_i = (\varphi_{p,\beta}(i+1), \varphi_{p,\beta}(i))$, and $L_{p,\beta} = \rho K_{p,\beta}$. Let A be the configuration with cells $A_i = (\alpha(i+1), \alpha(i))$. Explicitly, with $\neg\beta = 1 - \beta$,

$$K_{p,\beta} = (\beta,\beta)^{(-\infty,p)}\,(\neg\beta,\beta)_p\,(\beta,\neg\beta)_{p+1}\,(\neg\beta,\beta)_{p+2}\cdots,$$

$$L_{p,\beta} = (\beta,\beta)^{(-\infty,p)}\,(\beta,\neg\beta)_p\,(\neg\beta,\beta)_{p+1}\cdots,$$

continued with period 2, and A has the cell $(1,0)$ at even and $(0,1)$ at odd sites.

Lemma 6.3 (kinks)**.** (a) *The fixed points of* σF *are* $(0,0)^{\mathbb{Z}}$*,* $(1,1)^{\mathbb{Z}}$*,* A*,* σA *and the* $K_{p,\beta}$*; the fixed points of* $\sigma^{-1}F$ *are* $(0,0)^{\mathbb{Z}}$*,* $(1,1)^{\mathbb{Z}}$*,* A*,* σA *and the* $L_{p,\beta}$*.*

(b) $\sigma K_{p,\beta} = K_{p-1,\beta}$*,* $FK_{p,\beta} = K_{p+1,\beta}$*,* $\sigma L_{p,\beta} = L_{p-1,\beta}$*,* $FL_{p,\beta} = L_{p-1,\beta}$*,* $\kappa K_{p,\beta} = K_{p,\neg\beta}$ *and* $\kappa L_{p,\beta} = L_{p,\neg\beta}$*. Also* $\sigma^2 A = A \neq \sigma A$ *and* $\rho A = \sigma A$*.*

(c) $K_{p,0}$ *agrees with* $\sigma^{-p}A$ *on* (p,∞)*, and* $L_{p,0}$ *agrees with* $\sigma^{1-p}A$ *on* (p,∞)*.*

Proof. (a) Apply Lemma 2.3 to the list of Φ_{66}. Under $c_i = (\varphi(i+1), \varphi(i))$ the constants give $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, α and its shift give A and σA, and $\varphi_{p,\beta}$ gives $K_{p,\beta}$. Under $d_i = (\varphi(i-1), \varphi(i))$ an alternating sequence gives the same configuration as before, since $\varphi(i-1) = \varphi(i+1)$, and $\varphi_{p,\beta}$ gives the configuration that is (β,β) on $(-\infty, p]$ and has the cell $(\beta,\neg\beta)$ at $p+1$, which is $L_{p+1,\beta}$. (b) $\sigma^{-1}K_{p,\beta}$ is coded by $i \mapsto \varphi_{p,\beta}(i-1) = \varphi_{p+1,\beta}(i)$, and $FK_{p,\beta} = \sigma^{-1}K_{p,\beta}$ by (a). For L use $\rho\sigma = \sigma\rho$ and $F\rho = \rho F^{-1}$, so $FL_{p,\beta} = \rho F^{-1} K_{p,\beta} = \rho K_{p-1,\beta}$. Complementing $\varphi_{p,\beta}$ gives $\varphi_{p,\neg\beta}$. The cells $A_0 = (1,0)$ and $(\sigma A)_0 = A_1 = (0,1)$ differ, and $\rho A_i = (\alpha(i), \alpha(i+1)) = A_{i+1}$. (c) For $i > p$, $(K_{p,0})_i = (\alpha(i+1-p), \alpha(i-p)) = A_{i-p}$, and $(L_{p,0})_i = (\alpha(i-p), \alpha(i+1-p)) = A_{i+1-p}$. □

The second normalisation needs the fixed points of $\sigma^n F^n$ for $n = 4$. They are found by hand, for every $n \geq 1$.

Lemma 6.4 (fixed points of $\sigma^{\pm n}F^n$)**.** *Let* $n \geq 1$*. The fixed points of* $\sigma^n F^n$ *that agree with* $(0,0)^{\mathbb{Z}}$ *on a left half-line are* $(0,0)^{\mathbb{Z}}$ *and the kinks* $K_{p,0}$*,* $p \in \mathbb{Z}$*. Consequently those that agree with* $(1,1)^{\mathbb{Z}}$ *on a left half-line are* $(1,1)^{\mathbb{Z}}$ *and the* $K_{p,1}$*; and the fixed points of* $\sigma^{-n}F^n$ *that agree*

with $(0,0)^{\mathbb{Z}}$ on a left half-line are $(0,0)^{\mathbb{Z}}$ and the $L_{p,0}$. Since $F'^4 = F^4$, the case $n = 4$ describes these fixed points of $\sigma^{\pm 4}F'^4$ as well.

Proof. The listed configurations are fixed by σF, or by $\sigma^{-1}F$, by Lemma 6.3, hence by $\sigma^{\pm n}F^n$. Conversely, let $c \neq (0,0)^{\mathbb{Z}}$ be fixed by $\sigma^n F^n$, with $c_i = (0,0)$ for $i < i_0$, and let $Y = Y_c$. Since $Y_{F^n c}(t,i) = Y(t+n,i)$ and $Y_{\sigma^{-n}c}(t,i) = Y(t,i-n)$, the relation $F^n c = \sigma^{-n} c$ says

$$Y(t+n,i+n) = Y(t,i) \qquad \text{for all } t,i. \tag{9}$$

As $F^{\pm 1}$ have radius 1 and fix $(0,0)^{\mathbb{Z}}$, row t of Y vanishes on $(-\infty, i_0 - |t|)$. When row t is not zero, let $\lambda(t)$ be the least i with $Y(t,i) = 1$. We use two instances of (2), which hold at every (t,i) where row t vanishes on $[i-1, i+1]$, respectively at $i = \lambda(t) - 1$:

$$Y(t+1,i) = Y(t-1,i) \quad (i \leq \lambda(t) - 2), \qquad Y(t+1,\lambda(t)-1) \oplus Y(t-1,\lambda(t)-1) = 1, \tag{10}$$

since $f(0,0,0) = 0$ and $f(0,0,1) = 1$; the first also holds for all i if row t is zero.

No row is zero. Suppose row t is zero. Row $t+1$ is not zero, since two consecutive rows determine the diagram; put $a = \lambda(t+1)$. We show $\lambda(t+1+j) = a - j$ for all $j \geq 0$. Assume this for j; row $t+j$ vanishes on $(-\infty, a-j]$ (it is row t if $j = 0$, and $\lambda(t+j) = a - j + 1$ otherwise). By (10) at time $t+1+j$, row $t+2+j$ equals row $t+j$, hence 0, on $(-\infty, a-j-2]$, and $Y(t+2+j, a-j-1) = 1 \oplus Y(t+j, a-j-1) = 1$; so $\lambda(t+2+j) = a-j-1$. By (9), row $t+1+n$ is row $t+1$ shifted, so $\lambda(t+1+n) = a+n$, while we have shown $\lambda(t+1+n) = a-n$. This contradicts $n \geq 1$.

The first ones move at speed 1. So λ is defined everywhere, and $\lambda(t+n) = \lambda(t) + n$ by (9). Put $d(t) = \lambda(t+1) - \lambda(t)$, so that $d(t) + \cdots + d(t+n-1) = n$ for every t. Fix t and let $a = \lambda(t)$. By (10) rows $t \pm 1$ agree on $(-\infty, a-2]$, and exactly one of them is 1 at $a-1$. If one of $\lambda(t \pm 1)$ is at most $a-2$, both are, and they are equal. Otherwise one of them is $a-1$ and the other is at least a. So one of the following holds:

(A) $d(t-1) \geq 2$, $d(t) = -d(t-1)$; (B) $d(t-1) \leq 0$, $d(t) = -1$; (C) $d(t-1) = 1$, $d(t) \geq 0$.

These conditions determine $d(t)$ from $d(t-1)$ as follows: if $d(t-1) \geq 2$, only (A) is possible, and $d(t) \leq -2$; if $d(t-1) \leq 0$, only (B) is possible, and $d(t) = -1$; if $d(t-1) = 1$, only (C) is possible, and $d(t) \geq 0$. Hence if $d(t_1) \neq 1$ for some t_1, then $d(t_1+1) \leq 0$ if $d(t_1) \geq 2$, and in both cases $d(t) = -1$ for all $t \geq t_1 + 2$, contradicting $d(t) + \cdots + d(t+n-1) = n$. So $d(t) = 1$ for every t. With $p = \lambda(0) - 1$ we get $\lambda(t) = p + 1 + t$: row t vanishes on $(-\infty, p+t]$ and is 1 at $p+t+1$.

The diagram, diagonal by diagonal. For $j \geq 0$ let $D_j(t) = Y(t, p+t+j)$. Then $D_0 = 0$ and $D_1 = 1$. The relation (2) at $(t, p+t+j)$, $j \geq 1$, reads

$$D_{j-1}(t+1) \oplus D_{j+1}(t-1) = f\big(D_{j-1}(t), D_j(t), D_{j+1}(t)\big).$$

If D_{j-1} is constant equal to b and D_j is constant equal to $\neg b$, then the right side is 0 by (8), so $D_{j+1}(t-1) = b$ for every t. By induction on j, $D_j = j \bmod 2$ for all $j \geq 0$. So $Y(t,i) = [i > p+t]\,\big((i-p-t) \bmod 2\big)$, which is the diagram of $K_{p,0}$, as $F^t K_{p,0} = K_{p+t,0}$ has the y-row $\varphi_{p+t,0}$. Hence $c = K_{p,0}$.

For the consequences: κ commutes with σ and F, exchanges $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, and maps $K_{p,0}$ to $K_{p,1}$. And $\rho\sigma^{-n}F^n\rho = \sigma^{-n}F^{-n}$ is the inverse of $\sigma^n F^n$, so ρ maps the fixed points of $\sigma^n F^n$ onto those of $\sigma^{-n}F^n$; it fixes $(0,0)^{\mathbb{Z}}$, preserves agreement with it on a left half-line, and maps $K_{p,0}$ to $L_{p,0}$. □

6.2. **The factor map to 46.** Let $\delta\colon \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ be the block map

$$\delta(c)_i = (x_i \oplus x_{i+1},\ y_i \oplus y_{i+1}), \qquad c_i = (x_i, y_i).$$

Write F_{46} for the automaton of rule 46 and η_p for its kinks (Lemma 3.1).

Lemma 6.5 (the factor map). (a) *$f(a,b,c)\oplus f(b,c,d) = f_{46}(a\oplus b, b\oplus c, c\oplus d)$ for all a,b,c,d. Consequently $\delta F = F_{46}\delta$ and $\delta F' = F_{46}\delta$.*

(b) *$\delta\sigma = \sigma\delta$, $\delta\rho = \rho\delta$, $\delta\kappa = \delta$, and $\delta(c) = \delta(c')$ if and only if $c' = c \oplus s^{\mathbb{Z}}$ for a cell s. The diagram of δc is $Y_{\delta c}(t,i) = Y_c(t,i) \oplus Y_c(t,i+1)$.*

(c) *$\delta K_{p,\beta} = \eta_{p-1}$ and $\delta L_{p,\beta} = \rho\eta_{p-1}$.*

Proof. (a) Recall $f_{46}(a,0,c) = c$ and $f_{46}(a,1,c) = \neg a$ (Section 5). If $b = c$, the left side is $0 \oplus (b \oplus d) = c \oplus d$, by (8), and the right side is $f_{46}(a \oplus b, 0, c \oplus d) = c \oplus d$. If $b \neq c$, the left side is $[a = b] \oplus 0$, and the right side is $\neg(a \oplus b) = [a = b]$. Now let $\delta(c) = (w,z)$, so $w_i = x_i \oplus x_{i+1}$ and $z_i = y_i \oplus y_{i+1}$. By (1), $\delta(Fc)_i = \big(z_i,\ w_i \oplus f(y_{i-1}, y_i, y_{i+1}) \oplus f(y_i, y_{i+1}, y_{i+2})\big) = \big(z_i,\ w_i \oplus f_{46}(z_{i-1}, z_i, z_{i+1})\big) = F_{46}(\delta c)_i$. For F' the two added constants 1 cancel. (b) The first three identities are clear. If $\delta c = \delta c'$, then $c_i \oplus c'_i = c_{i+1} \oplus c'_{i+1}$ for all i. The y-row of $F^t\delta c = \delta F^t c$ is $Y_c(t,i) \oplus Y_c(t,i+1)$. (c) $(K_{p,\beta})_i \oplus (K_{p,\beta})_{i+1}$ is $(0,0)$ for $i < p-1$, $(\beta,\beta) \oplus (\neg\beta,\beta) = (1,0)$ for $i = p-1$, and $(1,1)$ for $i \geq p$, since the cells alternate there. Apply ρ. □

Two parts of this structure are in the literature. The relations $F'^2 = \kappa F^2$ and $F'^4 = F^4$ of Lemma 6.1(b) are the correspondence between the orbits of a rule that is invariant under complementing its arguments and those of its complement, due to Takesue [28, (A.1)–(A.4)], where the classes of 24 and 189 are listed as such a pair. And 24R has a local conservation law, which sorts the pairs of adjacent cells into two classes of eight pairs each [28, (4.9), (4.10)]; for the reflected rule 66 the two classes are the mirror images, and the bond bit $\chi_i(\delta c)$ of Lemma 5.11 equals 1 exactly when the pairs (c_i, c_{i+1}) and (c_{i+1}, c_{i+2}) lie in the same class, as one checks on the 64 words of three cells. So under δ the conserved bond bits of 46 correspond to this conservation law. The factor map δ and its use below are not in [28].

6.3. **Tailed configurations, gluing and the tail flip.** For $\beta \in \{0,1\}$ call c *β-tailed* if $c \neq (\beta,\beta)^{\mathbb{Z}}$ and $c_i = (\beta,\beta)$ for all large i. Then β is the *colour* of c, and the *edge* $e(c)$ is the largest i with $c_i \neq (\beta,\beta)$. A configuration is *tailed* if it is β-tailed for some β. So the 0-tailed configurations are those of Section 4.1, and κ maps β-tailed configurations to $\neg\beta$-tailed ones with the same edge. Call a cell *mixed* if it is $(0,1)$ or $(1,0)$. If c is β-tailed with $e(c) \leq r$, or $c = (\beta,\beta)^{\mathbb{Z}}$, and Λ is a configuration with $\Lambda_i = (\beta,\beta)$ for $i \leq r+1$, write $c \sqcup \Lambda$ for the configuration equal to c on $(-\infty, r]$ and to Λ on (r,∞), *glued at* r; it equals c on $(-\infty, r+1]$. For $p \geq r+2$, $K_{p,\beta}$ and $L_{p,\beta}$ are (β,β) on $(-\infty, r+1]$.

Lemma 6.6 (tailed configurations). *Let c be β-tailed with $e(c) = e$.*

(a) *The columns of Y_c to the right of e are constant equal to β, column e is never β in two consecutive rows, and $F^t c$ is β-tailed with $e(F^t c) = e$ for all $t \in \mathbb{Z}$.*

(b) *δc is 0-tailed with $e(\delta c) = e$. The map $c \mapsto (\delta c, \beta)$ is a bijection from the tailed configurations with edge e onto the pairs consisting of a 0-tailed configuration with edge e and a colour.*

Proof. (a) If rows $t-1$ and t are β on the columns $i > e$, then so is row $t+1$, since $Y(t+1,i) = \beta \oplus f(Y(t,i-1), \beta, \beta) = \beta$ by (8), and likewise row $t-2$. By induction in both directions of time, these columns are constant. If $Y(t-1,e) = Y(t,e) = \beta$, the same computation at column e gives $Y(t+1,e) = Y(t-2,e) = \beta$, so column e is constant and $c_e = (\beta,\beta)$, a contradiction. So the cell $(Y(t-1,e), Y(t,e))$ of $F^t c$ is never (β,β). (b) $(\delta c)_i = (0,0)$ for $i > e$ and $(\delta c)_e = c_e \oplus (\beta,\beta) \neq (0,0)$. Conversely, given a 0-tailed w with edge e and a colour β, the configuration $c_i = (\beta,\beta) \oplus \bigoplus_{j \geq i} w_j$ (a finite sum) is the unique β-tailed configuration with $\delta c = w$, by Lemma 6.5(b). □

Lemma 6.7 (gluing)**.** (a) *Let u agree with $K_{p,\beta}$ on $[q,\infty)$, where $p \geq q+1$. Then Fu agrees with $K_{p+1,\beta}$ on $[q,\infty)$. Let u agree with $L_{p,\beta}$ on $[q,\infty)$, where $p \geq q+2$. Then Fu agrees with $L_{p-1,\beta}$ on $[q,\infty)$. No assumption is made on u to the left of q.*

(b) *Let d be β-tailed with $e(d) \leq r$, or $d = (\beta,\beta)^{\mathbb{Z}}$. Then $F(d \sqcup K_{p,\beta}) = Fd \sqcup K_{p+1,\beta}$ for $p \geq r+2$, and $F(d \sqcup L_{p,\beta}) = Fd \sqcup L_{p-1,\beta}$ for $p \geq r+3$, all glued at r.*

(c) *Let h be a block map of radius R with $h((\beta,\beta)^{\mathbb{Z}}) = (\beta,\beta)^{\mathbb{Z}}$, let d be β-tailed with $e(d) \leq r$, or $d = (\beta,\beta)^{\mathbb{Z}}$, let $\Lambda \in \{K_{p,\beta}, L_{p,\beta}\}$ with $p \geq r+2R+1$, and $u = d \sqcup \Lambda$. Then hu agrees with hd on $(-\infty, p-1-R]$ and with $h\Lambda$ on $[r+R+1,\infty)$, and $(hd)_i = (\beta,\beta)$ for $i > r+R$. If moreover $(hd)_i = (\beta,\beta)$ for $i > r$ and $h\Lambda = \Lambda$, then $hu = hd \sqcup \Lambda$, glued at r.*

Proof. (a) For $i \geq q+1$ the cell $(Fu)_i$ reads only cells of u in $[q,\infty)$, so it equals $(FK_{p,\beta})_i = (K_{p+1,\beta})_i$. The cell $(Fu)_q = (y_q,\, x_q \oplus f(y_{q-1}, y_q, y_{q+1}))$ has $x_q = y_q = y_{q+1} = \beta$, read off $K_{p,\beta}$ as $q+1 \leq p$; by $f(a,\beta,\beta) = 0$ it is (β,β), whatever y_{q-1} is, which is the cell of $K_{p+1,\beta}$ at q. For $L_{p,\beta}$ the same argument applies, with $x_q = y_q = y_{q+1} = \beta$ because $q+2 \leq p$, and $(L_{p-1,\beta})_q = (\beta,\beta)$ because $q < p-1$. (b) $u = d \sqcup \Lambda$ agrees with d on $(-\infty, r+1]$, so Fu agrees with Fd on $(-\infty, r]$; by (a) with $q = r+1$, Fu agrees with $F\Lambda$ on $[r+1,\infty)$; and Fd is β-tailed with edge $e(d) \leq r$, by Lemma 6.6, or $Fd = d = (\beta,\beta)^{\mathbb{Z}}$. (c) The local rule of h maps $(\beta,\beta)^{2R+1}$ to (β,β), which gives the statement about hd. The configuration u equals d on $(-\infty, p-1]$, because both are (β,β) on $(r, p-1]$, and equals Λ on (r,∞); take the windows of radius R. The two half-lines cover $\mathbb{Z}$, as $p-1-R \geq r+R$, and on $(r, p-1-R]$ both hd and Λ are (β,β) under the additional hypotheses. □

Definition 6.8 (tail flip)**.** Let b be β-tailed with $e(b) = r$, and let $s \in \mathbb{Z}$ be such that the edge cell $(F^s b)_r$ is mixed. Let $T_s(b)$ be the configuration with $F^s T_s(b)$ equal to $F^s b$ on $(-\infty, r]$ and to $(\neg\beta, \neg\beta)$ on (r,∞).

So $T_s(b)$ is $\neg\beta$-tailed with edge r: it is obtained by complementing the tail of the state at time s. Clearly $\kappa T_s(b) = T_s(\kappa b)$, and $\rho T_s(b) = T_{-s}(\rho b)$, since $F^{-s}\rho b = \rho F^s b$.

6.4. **The bounce.**

Lemma 6.9 (bounce)**.** *Let d be β-tailed with $e(d) = e$, let $p \geq e+2$, $c = d \sqcup L_{p,\beta}$ glued at e, $s_0 = p-e-2$, and*

$$\varepsilon = [\, Y_d(s_0, e) = \beta \,], \qquad U = 2e - p + 3 - \varepsilon.$$

Put $D = d$ if $\varepsilon = 0$, and $D = T_{s_0+1}(d)$ if $\varepsilon = 1$; in the second case the edge cell of $F^{s_0+1}d$ is $(\beta, \neg\beta)$, so D is defined. Then D is $(\beta \oplus \varepsilon)$-tailed with $e(D) = e$, and

$$F^t c = F^t d \sqcup L_{p-t,\beta} \quad (0 \leq t \leq s_0), \qquad F^t c = F^t D \sqcup K_{t+U,\,\beta\oplus\varepsilon} \quad (t \geq s_0 + 1 + \varepsilon),$$

all glued at e. In particular $F^t c$ agrees with $K_{t+U,\beta\oplus\varepsilon}$ on $[e+1,\infty)$ for $t \geq s_0 + 1 + \varepsilon$.

So the kink $L_{p,\beta}$, glued to the right of the edge, moves to the left, bounces off the edge and leaves as a kink K moving to the right. If the edge value it meets is β, the kink comes out with the other colour and one step later, and the tail of the configuration left behind has been complemented.

Proof. Since κ commutes with F, δ and T_s, maps $L_{p,\beta}$, $K_{p,\beta}$ to $L_{p,\neg\beta}$, $K_{p,\neg\beta}$, and complements diagrams, which leaves ε unchanged, we may assume $\beta = 0$. Write $Y = Y_c$. By Lemma 6.7(b), applied s_0 times (the step from t to $t+1$ needs $p - t \geq e+3$), $F^t c = F^t d \sqcup L_{p-t,0}$ for $0 \leq t \leq s_0$. So the state at time s_0 is $F^{s_0} d \sqcup L_{e+2,0}$. Let (x_i, y_i), $i \leq e$, be the cells of $F^{s_0} d$; thus $x_i = Y(s_0 - 1, i)$, $y_i = Y(s_0, i)$, $(x_e, y_e) \neq (0,0)$ and $\varepsilon = \neg y_e$. On the columns $i > e$, rows $s_0 - 1$ and s_0 are those of $L_{e+2,0}$:

$$Y(s_0 - 1, i) = [\, i \geq e+3,\ i - e \text{ odd} \,], \qquad Y(s_0, i) = [\, i \geq e+2,\ i - e \text{ even} \,] \qquad (i > e).$$

In particular both rows vanish at $e+1$, as do the rows of Y_d. The relation (2) at (s_0,i), $i \le e$, reads row s_0 on $[i-1,i+1] \subseteq (-\infty,e+1]$ and row s_0-1 at i, so row s_0+1 of Y agrees with that of Y_d on $(-\infty,e]$. On $[e+3,\infty)$ it is the y-row of $FL_{e+2,0} = L_{e+1,0}$, which is $[i-e \text{ odd}]$, because these entries read only columns $\ge e+2$. At $e+2$ it is $0 \oplus f(0,1,0) = 0$, and at $e+1$ it is $0 \oplus f(y_e,0,1) = \neg y_e$.

Case $\varepsilon = 0$, that is $y_e = 1$. On $[e+1,\infty)$ the state at time s_0+1 has the cells $(Y(s_0,i), Y(s_0+1,i))$, namely $(0,0)$ at $e+1$ and $([i-e \text{ even}],[i-e \text{ odd}])$ for $i \ge e+2$. These are the cells of $K_{e+2,0}$. So $F^{s_0+1}c = F^{s_0+1}d \sqcup K_{e+2,0}$, and Lemma 6.7(b) gives $F^t c = F^t d \sqcup K_{t+e+1-s_0,0}$ for $t \ge s_0+1$. Here $e+1-s_0 = 2e-p+3 = U$.

Case $\varepsilon = 1$, that is $y_e = 0$. Then $x_e = 1$, and $Y(s_0+1,e) = x_e \oplus f(y_{e-1},0,0) = 1$; so the edge cell of $F^{s_0+1}d$ is $(0,1)$. On $[e+1,\infty)$ row s_0+1 is $[i-e \text{ odd}]$, and the state at time s_0+1 has the cells $(0,1)$ at $e+1$ and $([i-e \text{ even}],[i-e \text{ odd}])$ for $i \ge e+2$, which are those of $L_{e+1,0}$. Row s_0+2 is therefore the y-row of $FL_{e+1,0} = L_{e,0}$, that is $[i-e \text{ even}]$, on $[e+2,\infty)$. At $e+1$ it is $Y(s_0,e+1) \oplus f(1,1,0) = 1$, and at e it is $Y(s_0,e) \oplus f(Y(s_0+1,e-1),1,1) = 0$. So the state at time s_0+2 has the cells $(1,0)$ at e, $(1,1)$ at $e+1$, and $([i-e \text{ odd}],[i-e \text{ even}])$ for $i \ge e+2$. On $[e+1,\infty)$ these are the cells of $K_{e+2,1}$. Let D' equal $F^{s_0+2}c$ on $(-\infty,e]$ and $(1,1)$ beyond. It is 1-tailed with edge e, as its cell e is $(1,0)$, and $F^{s_0+2}c = D' \sqcup K_{e+2,1}$. Put $D = F^{-s_0-2}D'$. By Lemma 6.6, $F^{s_0+1}D = F^{-1}D'$ is 1-tailed with edge e. Its y-row is the x-row of D', which on $(-\infty,e]$ is row s_0+1 of Y, hence of Y_d. Its x-row is row s_0 of Y_D. By (2) at (s_0+1,i), solved for the earlier row, this row is $Y(s_0,i) = Y_d(s_0,i)$ for $i \le e-1$, as only columns $\le e$ of rows s_0+1 and s_0+2 are read. At e it is $Y_D(s_0+2,e) \oplus f(Y_D(s_0+1,e-1),1,1) = 0 = y_e$. So $F^{s_0+1}D$ agrees with $F^{s_0+1}d$ on $(-\infty,e]$, that is, $D = T_{s_0+1}(d)$. Lemma 6.7(b) gives $F^t c = F^t D \sqcup K_{t+e-s_0,1}$ for $t \ge s_0+2$. Here $e-s_0 = 2e-p+2 = U$. □

Since a kink $K_{q,\beta'}$ with $q > j$ is determined by its restriction to $[j,\infty)$, the integer U and the colour $\beta \oplus \varepsilon$ are determined by c: they are read off $F^t c$ on $[e+1,\infty)$ for any large t.

6.5. **Lemma K for block maps commuting with a power of F.** For $N \ge 1$ let $\mathcal{H}_N$ be the set of block maps h, of any radius and not necessarily invertible, that commute with F^N and fix $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$, $K_{0,0}$ and $L_{0,0}$. Such an h commutes with σ, so it fixes every $K_{p,0}$ and $L_{p,0}$. We prove:

Lemma 6.10 (Lemma K′)**.** *Let $N \ge 1$ and $h \in \mathcal{H}_N$. Then $h = \mathrm{id}$.*

Only $N = 1$ and $N = 4$ are used below. The plan is that of Section 4, with three changes. The kinks of colour 1 are not assumed to be fixed: Proposition 6.12 shows that they are. A move does not change the edge cell; it complements the tail. And since h commutes only with F^N, the statements about h at time t are reduced to time 0 by conjugating h with F^t, which preserves $\mathcal{H}_N$.

Lemma 6.11. *Let $h \in \mathcal{H}_N$ have radius R. Then h maps $(\beta,\beta)^{2R+1}$ to (β,β) for $\beta \in \{0,1\}$; $F^t h F^{-t} \in \mathcal{H}_N$ for every $t \in \mathbb{Z}$; and $\rho h \rho \in \mathcal{H}_N$.*

Proof. The first statement holds because h fixes $(\beta,\beta)^{\mathbb{Z}}$. $F^t h F^{-t}$ is a block map commuting with F^N; it fixes $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, which F fixes, and $F^t h F^{-t} K_{p,0} = F^t h K_{p-t,0} = K_{p,0}$, $F^t h F^{-t} L_{p,0} = F^t h L_{p+t,0} = L_{p,0}$, by Lemma 6.3(b). $\rho h \rho$ commutes with $\rho F^N \rho = F^{-N}$, fixes the two constants, and exchanges the roles of $K_{p,0}$ and $L_{p,0} = \rho K_{p,0}$. □

Proposition 6.12 ((H_1') and the kinks of colour 1)**.** *Let $h \in \mathcal{H}_N$.*

(a) *h fixes $K_{p,1}$ and $L_{p,1}$ for every p, and $\kappa h \kappa \in \mathcal{H}_N$.*
(b) *For every tailed d, the configuration hd is tailed, with the colour and the edge of d, and $F^t hd$ agrees with $F^t d$ on $[e(d),\infty)$ for every $t \in \mathbb{Z}$; that is, $Y_{hd}(t,i) = Y_d(t,i)$ for all t and all $i \ge e(d)$.*

Proof. Let R be the radius of h, let d be 0-tailed with $e(d) = e$, and let $p \geq e + 2R + 2$. Let $c = d \sqcup L_{p,0}$ be glued at e, and let s_0, ε, U be as in Lemma 6.9, so that $s_0 \geq 2R$. By Lemma 6.7(c), hd is $(0,0)^{\mathbb{Z}}$ or 0-tailed with $e' = e(hd) \leq e + R$. Moreover hc agrees with hd on $(-\infty, p-1-R]$ and with $L_{p,0}$ on $[e+R+1, \infty)$. Since hd and $L_{p,0}$ are $(0,0)$ on $(e+R, p-1]$, it follows that $hc = hd \sqcup L_{p,0}$, glued at $e+R$; this is $L_{p,0}$ if $hd = (0,0)^{\mathbb{Z}}$, and the same configuration glued at e' otherwise. In the second case Lemma 6.9 applies to hc, with e' in place of e, and gives $\varepsilon' = [Y_{hd}(p-e'-2, e') = 0]$ and $U' = 2e' - p + 3 - \varepsilon'$. In both cases, for $t \in N\mathbb{Z}$ large, $F^t hc = hF^t c$. By Lemma 6.9 for c, $F^t c$ agrees with $K_{t+U,\varepsilon}$ on $[e+1, \infty)$, so $hF^t c$ agrees with $hK_{t+U,\varepsilon}$ on $[e+R+1, \infty)$. Hence

$$(11)\quad hK_{t+U,\varepsilon} \text{ agrees with } \begin{cases} L_{p-t,0} & \text{if } hd = (0,0)^{\mathbb{Z}}, \\ K_{t+U',\varepsilon'} & \text{otherwise,} \end{cases} \quad \text{on } [e+R+1, \infty), \text{ for all large } t \in N\mathbb{Z}.$$

Step 1: $hd \neq (0,0)^{\mathbb{Z}}$ and $e(hd) = e$. Column e of Y_d is never 0 in two consecutive rows (Lemma 6.6), so we may choose $p \geq e + 2R + 2$ with $Y_d(s_0, e) = 1$, that is $\varepsilon = 0$. Then $hK_{t+U,0} = K_{t+U,0}$. For large t this kink is $(0,0)$ on $[e+R+1, e+R+2]$, while $L_{p-t,0}$ has mixed cells there. So $hd \neq (0,0)^{\mathbb{Z}}$, and $K_{t+U,0}$ agrees with $K_{t+U',\varepsilon'}$ on $[e+R+1, \infty)$ for large t. Hence $\varepsilon' = 0$ and $U' = U$, that is $e' = e$.

Step 2: the edge column at large times. Now let $p \geq e + 2R + 2$ be arbitrary. By Step 1, the second case of (11) holds, with $e' = e$. If $\varepsilon = 0$, the argument of Step 1 gives $\varepsilon' = 0$. If $\varepsilon = 1$, then $hK_{t+U,1} = \sigma^{-t-U} hK_{0,1}$, so by (11) $hK_{0,1}$ agrees with $\sigma^{t+U} K_{t+U',\varepsilon'} = K_{U'-U,\varepsilon'}$ on $[e+R+1-t-U, \infty)$, for arbitrarily large t; hence $hK_{0,1} = K_{U'-U,\varepsilon'}$. Since $K_{0,1}$ is $(1,1)$ on $(-\infty, 0)$ and h fixes $(1,1)^{\mathbb{Z}}$, $hK_{0,1}$ is $(1,1)$ on $(-\infty, -R)$, so $\varepsilon' = 1$ and $U' - U = 2(e' - e) = 0$. In both cases $\varepsilon' = \varepsilon$, that is, $Y_{hd}(s, e) = Y_d(s, e)$ for every $s = p - e - 2 \geq 2R$; and if $\varepsilon = 1$ occurs for some d and p, then $hK_{0,1} = K_{0,1}$.

Step 3: (a). Let b_0 have the cell $(0,1)$ at 0 and $(0,0)$ elsewhere, and let $d_0 = F^{-2R-1} b_0$, which is 0-tailed with edge 0 (Lemma 6.6) and has $Y_{d_0}(2R, 0) = Y_{b_0}(-1, 0) = 0$. With $d = d_0$ and $p = 2R + 2$ we have $\varepsilon = 1$, so h fixes $K_{0,1}$, and hence every $K_{p,1}$. The same applies to $\rho h \rho \in \mathcal{H}_N$ (Lemma 6.11), so h fixes $\rho K_{p,1} = L_{p,1}$. Now $\kappa h \kappa$ commutes with F^N, as κ commutes with F, and it fixes $(0,0)^{\mathbb{Z}} = \kappa((1,1)^{\mathbb{Z}})$, $(1,1)^{\mathbb{Z}}$, $K_{0,0} = \kappa K_{0,1}$ and $L_{0,0} = \kappa L_{0,1}$; so $\kappa h \kappa \in \mathcal{H}_N$.

Step 4: (b). For 0-tailed d, Steps 1 and 2 give $e(hd) = e$ and $Y_{hd}(s, e) = Y_d(s, e)$ for $s \geq 2R$. For $\ell \in N\mathbb{Z}$, $F^{-\ell} d$ is 0-tailed with edge e and $hF^{-\ell} d = F^{-\ell} hd$. Applying the same to $F^{-\ell} d$ gives $Y_{hd}(s-\ell, e) = Y_d(s-\ell, e)$ for all $s \geq 2R$ and $\ell \in N\mathbb{Z}$, that is, at all times. The columns to the right of e are 0 in both diagrams, by Lemma 6.6, and the cells of $F^t hd$ are $(Y_{hd}(t-1, i), Y_{hd}(t, i))$. If d is 1-tailed, apply this to $\kappa h \kappa \in \mathcal{H}_N$ and the 0-tailed configuration κd, and use $\kappa(\kappa h \kappa)\kappa d = hd$. □

Probes and moves. For a tailed b with colour β and $e(b) = r$, and $n \in \mathbb{Z}$, let the *probe* be

$$P_n(b) = F^{-n}\big(F^n b \sqcup L_{r+3,\beta}\big),$$

glued at r. Write (x_r, y_r) for the edge cell of $F^m b$; it is $(\beta, \neg\beta)$, $(\neg\beta, \beta)$ or $(\neg\beta, \neg\beta)$. Define the *moves*

$$M_m(b) = \begin{cases} T_m(b), & (x_r, y_r) = (\beta, \neg\beta), \\ b, & \text{otherwise,} \end{cases} \qquad \bar{M}_m(b) = \begin{cases} T_m(b), & (x_r, y_r) = (\neg\beta, \beta), \\ b, & \text{otherwise.} \end{cases}$$

So the moves at time m complement the tail of the state at time m whenever its edge cell is mixed, and they never change a configuration whose edge cell at time m is $(\neg\beta, \neg\beta)$. A move maps a tailed configuration with edge r to a tailed configuration with edge r. It is described by its effect on the state at time m alone, so $F^j M_m = M_{m-j} F^j$ and $F^j \bar{M}_m = \bar{M}_{m-j} F^j$, and by the remark after Definition 6.8, $\bar{M}_m = \rho M_{-m} \rho$.

Lemma 6.13 (probes and moves)**.** *Let $h \in \mathcal{H}_N$, let b be tailed with colour β and $e(b) = r$, and let $m \in \mathbb{Z}$.*

(a) *Let $\varepsilon = [Y_b(m-1, r) = \beta]$. Then $F^s P_{m-2}(b) = F^s M_m(b) \sqcup K_{s-m+r+2-\varepsilon,\, \beta\oplus\varepsilon}$, glued at r, for every $s \geq m + \varepsilon$.*
(b) *$hP_n(b) = P_n(hb)$ for every n.*
(c) *$hM_m(b) = M_m(hb)$ and $h\bar{M}_m(b) = \bar{M}_m(hb)$.*
(d) (Light cone.) *$M_m(b)$ and $\bar{M}_m(b)$ agree with b on $(-\infty, r - |m|]$.*

Proof. (a) Apply Lemma 6.9 to $d = F^{m-2}b$, which is β-tailed with edge r, and $p = r + 3$. Then $s_0 = 1$, $\varepsilon = [Y_d(1, r) = \beta] = [Y_b(m-1, r) = \beta]$, $U = r - \varepsilon$, and $D = d$ or $D = T_2(d) = F^{m-2}T_m(b)$. The condition $Y_b(m-1, r) = x_r = \beta$ says that the edge cell of $F^m b$ is $(\beta, \neg\beta)$, so $D = F^{m-2}M_m(b)$ in both cases. Replace t by $s - m + 2$.

(b) By Lemma 6.7(b), $F^n P_n(b) = F^j(F^{n-j}b \sqcup L_{r+3+j,\beta})$ for $j \geq 0$. Choose $j \geq 2R$ with $j \equiv n \pmod N$, where R is the radius of h. By Proposition 6.12, $hF^{n-j}b$ is β-tailed with edge r, and h fixes $L_{r+3+j,\beta}$, so Lemma 6.7(c) gives $h(F^{n-j}b \sqcup L_{r+3+j,\beta}) = hF^{n-j}b \sqcup L_{r+3+j,\beta} = F^{n-j}hb \sqcup L_{r+3+j,\beta}$, since $n - j \in N\mathbb{Z}$. Apply F^{j-n}, which commutes with h.

(c) Let ε be as in (a) and $\beta' = \beta \oplus \varepsilon$. By Proposition 6.12, hb is β-tailed with edge r and $Y_{hb}(m-1, r) = Y_b(m-1, r)$, so (a) holds for hb with the same ε and the same kink. Take $s \in N\mathbb{Z}$ with $s \geq m + 1$ and $s - m + r + 2 - \varepsilon \geq r + 2R + 1$. By (a), (b) and $hF^s = F^s h$,

$$F^s M_m(hb) \sqcup K = F^s P_{m-2}(hb) = hF^s P_{m-2}(b) = h\big(F^s M_m(b) \sqcup K\big),$$

where $K = K_{s-m+r+2-\varepsilon,\beta'}$. By Proposition 6.12, $hF^s M_m(b)$ is β'-tailed with edge r, and h fixes K. So, by Lemma 6.7(c), the right side is $hF^s M_m(b) \sqcup K = F^s hM_m(b) \sqcup K$. Comparing on $(-\infty, r]$, the β'-tailed configurations $F^s M_m(hb)$ and $F^s hM_m(b)$ are equal, and so $M_m(hb) = hM_m(b)$. For $\bar{M}_m = \rho M_{-m}\rho$ use $\rho h\rho \in \mathcal{H}_N$: $h\bar{M}_m(b) = \rho(\rho h\rho)M_{-m}(\rho b) = \rho M_{-m}(\rho h\rho\, \rho b) = \bar{M}_m(hb)$.

(d) $F^m T_m(b)$ differs from $F^m b$ only on (r, ∞), and F^{-m} has radius $|m|$. □

By (a), M_m is the passage from a probe at time $m - 2$ to the configuration it leaves behind, as in Lemma 4.4; by (c) the moves commute with every $h \in \mathcal{H}_N$, for every m. As in Section 4, this is an identity of operators defined by F alone. For 46 write M_m^{46} and $\bar{M}_m^{46}$ for the moves of Section 5.5. By Lemma 5.2(b) they are given by their effect on the state at time m: M_m^{46} changes the edge cell 01 to 10 and fixes b if the edge cell has x-bit 1, and $\bar{M}_m^{46}$ changes 10 to 01 and fixes b if the edge cell has y-bit 1; nothing else changes.

Lemma 6.14 (moves of 66 and of 46)**.** *Let b be tailed with $e(b) = r$. Then $\delta M_m(b) = M_m^{46}(\delta b)$ and $\delta\bar{M}_m(b) = \bar{M}_m^{46}(\delta b)$ for every m. Consequently, if O is a finite composite of moves M_m, $\bar{M}_m$ and $\tilde{O}$ is the same composite of the moves M_m^{46}, $\bar{M}_m^{46}$, then $\delta(Ob) = \tilde{O}(\delta b)$.*

Proof. Let β be the colour of b and (x_r, y_r) the edge cell of $F^m b$. By Lemma 6.5, $F_{46}^m \delta b = \delta F^m b$, whose edge cell, at r, is $(x_r \oplus \beta, y_r \oplus \beta)$, by Lemma 6.6(b). If $(x_r, y_r) = (\beta, \neg\beta)$, this edge cell is $(0, 1)$, and $M_m^{46}(\delta b)$ is obtained by changing it to $(1, 0)$ at time m. On the other hand $F^m M_m(b) = F^m T_m(b)$ has the same cells as $F^m b$ on $(-\infty, r]$ and the tail $(\neg\beta, \neg\beta)$. So $\delta F^m T_m(b)$ agrees with $\delta F^m b$ at the cells $i < r$, its cell r is $(\beta, \neg\beta) \oplus (\neg\beta, \neg\beta) = (1, 0)$, and it vanishes beyond. Hence $F_{46}^m \delta M_m(b) = F_{46}^m M_m^{46}(\delta b)$. Otherwise the edge cell of $\delta F^m b$ has x-bit 1, and both sides are unchanged. The case of $\bar{M}_m$ is the same with $(\neg\beta, \beta)$ and $(1, 0) \mapsto (0, 1)$. The last statement follows by induction on the length of O, since every move maps tailed configurations with edge r to tailed configurations with edge r. □

So, under δ, the moves of 66 are exactly the moves $01 \leftrightarrow 10$ of 46. The colour, which δ forgets, changes with every nontrivial move.

The inductive statement. For $k \geq 1$ consider

(H'_k) *for every $h \in \mathcal{H}_N$ and every tailed c, $(hc)_i = c_i$ for all $i \geq e(c) - k + 1$.*

This is a statement about the state at time 0. It is the analogue of (H_k) of Section 4.1, stated for block maps commuting with F^N. (H'_1) is Proposition 6.12(b).

Lemma 6.15 (all times)**.** *Assume* (H'_k)*. Then for every $h \in \mathcal{H}_N$, every tailed c and every $t \in \mathbb{Z}$, $F^t hc$ agrees with $F^t c$ on $[e(c) - k + 1, \infty)$; that is, $Y_{hc}(t, i) = Y_c(t, i)$ for all t and all $i \geq e(c) - k + 1$.*

Proof. $h_t = F^t h F^{-t}$ lies in $\mathcal{H}_N$ (Lemma 6.11), and $F^t c$ is tailed with edge $e(c)$ (Lemma 6.6). (H'_k) for h_t and $F^t c$ is the claim, since $h_t F^t c = F^t hc$. □

Conjugating h by F^t amounts to conjugating the moves: $F^{-t} M_m F^t = M_{m+t}$, and h_t commutes with the moves because h does.

Proposition 6.16 (the inductive step)**.** *For every $k \geq 1$,* (H'_k) *implies* (H'_{k+1}).

Proof. Reading. Let $h \in \mathcal{H}_N$ and let b be tailed with $e(b) = r$. We show $Y_{hb}(0, r-k) = Y_b(0, r-k)$. By Lemma 6.6(b), δb is 0-tailed with edge r. By (CTRL'_k) for 46 (Lemma 5.12 and Corollary 5.14), there is a finite composite $\tilde{O}$ of moves M_m^{46}, $\bar{M}_m^{46}$ with $|m| \leq k-1$ such that $Y_{\tilde{O}\delta b}(0, r-k+1) = 1$. Let O be the same composite of the moves of 66. By Lemmas 6.14 and 6.5(b),

$$Y_{Ob}(0, r-k+1) \oplus Y_{Ob}(0, r-k+2) = Y_{\delta Ob}(0, r-k+1) = 1.$$

Put $Y = Y_{Ob}$ and $Y' = Y_{hOb}$. As Ob is tailed with edge r, Lemma 6.15 gives $Y' = Y$ on the columns $\geq r-k+1$ at all times. So the relation (2) at $(0, r-k+1)$, in the two diagrams, gives

$$f\big(Y'(0, r-k), \beta_1, \beta_2\big) = f\big(Y(0, r-k), \beta_1, \beta_2\big), \qquad \beta_1 = Y(0, r-k+1) \neq \beta_2 = Y(0, r-k+2).$$

By (8), $f(a, \beta_1, \beta_2) = [a = \beta_1]$, so $Y'(0, r-k) = Y(0, r-k)$. Now $hOb = Ohb$ by Lemma 6.13(c), and hb is tailed with edge r by Proposition 6.12. Every factor of O is a move at a time $|m| \leq k-1$, so by Lemma 6.13(d), applied factor by factor, Ob agrees with b, and Ohb with hb, on $(-\infty, r-k+1]$. Hence $Y_{hb}(0, r-k) = Y'(0, r-k) = Y(0, r-k) = Y_b(0, r-k)$.

Both rows. Let $h \in \mathcal{H}_N$ and let c be tailed with $e(c) = r$. Reading with $b = c$ gives $Y_{hc}(0, r-k) = Y_c(0, r-k)$. Reading with $F^{-1} h F \in \mathcal{H}_N$ and $b = F^{-1} c$, which is tailed with edge r, gives $Y_{hc}(-1, r-k) = Y_{F^{-1}hc}(0, r-k) = Y_{F^{-1}c}(0, r-k) = Y_c(-1, r-k)$. So $(hc)_{r-k} = c_{r-k}$, and with (H'_k) this is (H'_{k+1}). □

The reading target is the y-bit of the cell $r-k+1$ of $\delta(Ob)$. This is exactly the target of (CTRL'_k) for 46, so no new controllability theorem is needed. The rule 66 reads its left argument when the two other arguments differ, as 46 reads its left argument when the middle one is 1.

Proof of Lemma 6.10. By Proposition 6.12 and Proposition 6.16, (H'_k) holds for every $k \geq 1$. Let $h \in \mathcal{H}_N$ have radius R, and let $w \in \mathcal{A}^{2R+1}$. If $w = (0,0)^{2R+1}$, then $h(w) = (0,0)$ because h fixes $(0,0)^{\mathbb{Z}}$. Otherwise let c be w on $[-R, R]$ and $(0,0)$ elsewhere. It is 0-tailed with $e(c) \leq R$, so (H'_{R+1}) gives $h(w) = (hc)_0 = c_0 = w_0$. So $h = \mathrm{id}$. □

6.6. **Normalisation and the automorphism groups.**

Lemma 6.17 (normalisation)**.** *Let $N \geq 1$ and let g be an invertible block map that commutes with F^N and fixes $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$. Then $g = \sigma^a F^b$ for some $a, b \in \mathbb{Z}$.*

Proof. Let R be the radius of g. Since g and g^{-1} commute with $\sigma^N F^N$, g permutes its fixed points. The point $gK_{0,0}$ agrees with $g((0,0)^{\mathbb{Z}}) = (0,0)^{\mathbb{Z}}$ on $(-\infty, -R)$, and it is not $(0,0)^{\mathbb{Z}}$ because g is injective. By Lemma 6.4 with $n = N$, $gK_{0,0} = K_{e^+,0}$ for some $e^+ \in \mathbb{Z}$, and then $gK_{p,0} = g\sigma^{-p} K_{0,0} = K_{p+e^+,0}$ for every p. In the same way, with the fixed points of $\sigma^{-N} F^N$, $gL_{p,0} = L_{p+e^-,0}$ for some e^-.

Parity. By Lemma 6.3(c), $gK_{p,0}$ agrees with $g\sigma^{-p}A = \sigma^{-p}gA$ on $(p+R,\infty)$, and $K_{p+e^+,0}$ agrees with $\sigma^{-p-e^+}A$ on $(p+e^+,\infty)$. Both gA and A are fixed by σ^2, and two configurations of period 2 that agree on a half-line are equal; so $gA = \sigma^{-e^+}A$. In the same way, from $L_{p,0}$, $gA = \sigma^{-e^-}A$. Since $\sigma A \neq A = \sigma^2 A$, $e^+ \equiv e^- \pmod 2$.

So $a = -\frac{1}{2}(e^+ + e^-)$ and $b = \frac{1}{2}(e^+ - e^-)$ are integers. By Lemma 6.3(b), $\sigma^a F^b$ maps $K_{p,0}$ to $K_{p+b-a,0} = K_{p+e^+,0}$ and $L_{p,0}$ to $L_{p-a-b,0} = L_{p+e^-,0}$. Put $h = F^{-b}\sigma^{-a}g$. It is a block map commuting with F^N; it fixes $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$, which F and σ fix; and it fixes $K_{0,0}$ and $L_{0,0}$. So $h \in \mathcal{H}_N$, and $h = \mathrm{id}$ by Lemma 6.10. □

Theorem 6.18. *For $f \in \{24, 66, 189, 231\}$,*

$$\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \times \langle \kappa \rangle \cong \mathbb{Z}^2 \times \mathbb{Z}/2,$$

and every automorphism is $\sigma^a F_f^b \kappa^j$ for unique $a, b \in \mathbb{Z}$ and $j \in \{0,1\}$.

Proof. The rule 66. By Lemma 6.1(a), $\kappa \in \mathrm{Aut}(X_F)$; it commutes with σ and F, and $\kappa^2 = \mathrm{id}$. Let $g \in \mathrm{Aut}(X_F)$. It commutes with σ, so it permutes the fixed points of σ, which are the four constant configurations. It commutes with F, so it permutes the constants fixed by F, which are $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$ by Lemma 6.1(c). Replacing g by κg if necessary, g fixes both, and Lemma 6.17 with $N = 1$ gives $g = \sigma^a F^b$. So $\mathrm{Aut}(X_F) = \langle \sigma, F \rangle \langle \kappa \rangle$. For uniqueness, suppose $\sigma^a F^b \kappa^j = \mathrm{id}$. Then $j = 0$, since $\sigma^a F^b$ fixes $(0,0)^{\mathbb{Z}}$ and κ does not; and comparing $\sigma^a F^b K_{0,0} = K_{b-a,0}$ and $\sigma^a F^b L_{0,0} = L_{-a-b,0}$ with $K_{0,0}$ and $L_{0,0}$ gives $a = b = 0$.

The rule 189. Let $g \in \mathrm{Aut}(X_{F'})$. It permutes the four constant configurations by a permutation π that commutes with the 4-cycle γ by which F' acts (Lemma 6.1(c)). Then $\pi\gamma^k(s) = \gamma^k\pi(s)$ for a fixed constant s and all k, so π is determined by $\pi(s)$, and $\pi = \gamma^j$ for the j with $\gamma^j(s) = \pi(s)$. So $g_1 = F'^{-j}g$ fixes all four constants. It commutes with $F'^4 = F^4$ (Lemma 6.1(b)), so Lemma 6.17 with $N = 4$ gives $g_1 = \sigma^a F^b$. Now F^b exchanges $(0,1)^{\mathbb{Z}}$ and $(1,0)^{\mathbb{Z}}$ if b is odd (Lemma 6.1(c)), while g_1 fixes them; so b is even, and $F^b = (F^2)^{b/2} = (\kappa F'^2)^{b/2} = \kappa^{b/2} F'^b$. Hence $g = F'^j \sigma^a \kappa^{b/2} F'^b \in \langle \sigma, F' \rangle \langle \kappa \rangle$, and $\kappa \in \mathrm{Aut}(X_{F'})$ by Lemma 6.1(a). For uniqueness, suppose $\sigma^a F'^b \kappa^j = \mathrm{id}$. On the constants, κ acts as γ^2, so $\gamma^{b+2j} = \mathrm{id}$ and $b \equiv 2j \pmod 4$. Then b is even, $F'^b \kappa^j = \kappa^{b/2+j} F^b$, and $b/2 \equiv j \pmod 2$, so $\sigma^a F^b = \mathrm{id}$, hence $a = b = 0$ by the case of 66, and then $j = 0$ since $2j \equiv 0 \pmod 4$.

Reflection. The reflection rev conjugates F_{66} to F_{24} and F_{189} to F_{231}, σ to σ^{-1}, and κ to itself (Section 2.3). So it carries the statement for 66 and 189 to the statement for 24 and 231. □

For the class of 189 the normalisation cannot use kinks of $\sigma^{\pm 1}F'$, since there are none (Lemma 6.1(d)). It uses the kinks of 66, which are fixed by $\sigma^{\pm 4}F'^4$ (Lemma 6.4). The element F_{66} does not commute with F': on $(0,0)^{\mathbb{Z}}$, FF' gives $(1,0)^{\mathbb{Z}}$ and $F'F$ gives $(0,1)^{\mathbb{Z}}$. Its square $F_{66}^2 = \kappa F'^2$ does commute with F', and this is why b is even. No computer is used in this section.

7. The class of 79

In this section $f = 93$ and $F = F_{93}$. The rule is $f(a,b,c) = \neg c \vee (\neg a \wedge b)$, with values $1,0,1,1,1,0,1,0$ on $000, \ldots, 111$. Reflection $f(a,b,c) \mapsto f(c,b,a)$ takes 93 to 79, complementation $f(a,b,c) \mapsto f(1-a,1-b,1-c)$ takes 93 to 186 and 79 to 242, so by Section 2.3 it suffices to treat 93. The rule does not satisfy (4), since $f(0,0,0) = 1$, and the constant configurations are not a quiet background; their role is taken by a configuration of period 2 in space and 6 in time, the *vacuum*. We use the following values of f:

$$f(a,b,0) = 1, \qquad f(a,b,1) = \neg a \wedge b, \qquad f(a,1,1) = \neg a, \qquad f(0,1,c) = 1, \tag{12}$$

and $f(0,b,c) = f(1,b,c)$ unless $(b,c) = (1,1)$. On the constant configurations, F acts by $(0,0)^{\mathbb{Z}} \mapsto (0,1)^{\mathbb{Z}} \mapsto (1,0)^{\mathbb{Z}} \mapsto (0,0)^{\mathbb{Z}}$ and fixes $(1,1)^{\mathbb{Z}}$, since $f(0,0,0) = 1$ and $f(1,1,1) = 0$.

The proof follows the plan of Section 4, with three changes. The edge is the last cell at which a configuration differs from a vacuum, and a move changes the phase of the vacuum to the right of the edge rather than the edge cell. The second and third columns are read off by moves at a few times next to each time, using the period-6 structure of the edge column. And the controllability theorem starts at $k = 4$, from a statement at level 3 that holds up to five exceptional configurations and is proved from a table of the moves at the times $|m| \le 3$.

7.1. **The vacuum and the edge.** Let $V\colon \mathbb{Z}^2 \to \{0,1\}$ be given by

$$V(t,i) = \begin{cases} [\,i \text{ odd}\,] & t \equiv 0,1 \pmod 6, \\ 0 & t \equiv 2,5 \pmod 6, \\ [\,i \text{ even}\,] & t \equiv 3,4 \pmod 6, \end{cases}$$

and for $\omega \in \mathbb{Z}/6$ let V_ω be the configuration with cells $(V_\omega)_i = (V(\omega-1,i), V(\omega,i))$, the *vacuum of phase* ω. Its cells at an even and an odd site are $(0,0),(0,1)$ for $\omega = 0$; $(0,0),(1,1)$ for $\omega = 1$; $(0,0),(1,0)$ for $\omega = 2$; $(0,1),(0,0)$ for $\omega = 3$; $(1,1),(0,0)$ for $\omega = 4$; and $(1,0),(0,0)$ for $\omega = 5$.

Lemma 7.1 (vacuum)**.** (a) *V satisfies the spacetime relation (2), so V is the diagram of V_0 and $F^t V_0 = V_t$. Moreover $\sigma V_\omega = V_{\omega+3}$, $\sigma^2 V_\omega = V_\omega$, $F^3 V_\omega = \sigma V_\omega$ and $\rho V_\omega = V_{2-\omega}$.*

(b) *Two vacua that agree at two adjacent sites are equal, and V_ω and $V_{\omega+3}$ differ at every site.*

(c) (Left-blindness.) *$(V(t,i), V(t,i+1)) \ne (1,1)$ for all t,i. Consequently, if d_i and d_{i+1} agree with V_ω, then $F(d)_i$ and $F^{-1}(d)_i$ do not depend on d_{i-1}.*

Proof. (a) Row t of V is [i odd], [i odd], 0, [i even], [i even], 0 for $t \equiv 0,\dots,5$. On a row [i odd] or [i even] the triple at i is $(0,1,0)$ or $(1,0,1)$, and $f(0,1,0) = 1$, $f(1,0,1) = 0$, so the right side of (2) is the row itself; on a zero row it is $f(0,0,0) = 1$. In each of the six cases the left side $V(t+1,i) \oplus V(t-1,i)$ is the same. The row of V at time $t+3$ is that at time t shifted by one site, which gives $\sigma V_\omega = V_{\omega+3}$ and $F^3 V_\omega = V_{\omega+3}$; the period 2 is clear. By (3), ρV_ω has diagram $V(\omega-1-t,i)$, and $V(1-t,i) = V(t,i)$ for all t, so $V(\omega-1-t,i) = V(t+2-\omega,i)$. (b) is read off the list of cells above. (c) No row of V has two adjacent ones, and $f(a,b,c)$ depends on a only if $(b,c) = (1,1)$. The y-row and the x-row of V_ω are rows of V, and $F(d)_i$ reads $a = y_{i-1}$ with $(b,c) = (y_i, y_{i+1})$, while $F^{-1}(d)_i$ reads x_{i-1} with (x_i, x_{i+1}). □

For $r \in \mathbb{Z}$ let Ω_r be the set of configurations c such that, for some ω, $c_i = (V_\omega)_i$ for all $i > r$ and $c_r \ne (V_\omega)_r$; by Lemma 7.1(b), ω is unique, and we call it the *phase* $\omega(c)$ and r the *edge* of c. Put $\vartheta(c) = \omega(c) + 3r \in \mathbb{Z}/6$; by Lemma 7.1(a), $\sigma^r c \in \Omega_0$ has phase $\vartheta(c)$. For $c \in \Omega_r$ write $z_t = Y_c(t,r)$ for the edge column. If c agrees with V_ω on (r,∞) and Λ agrees with V_ω on $(-\infty,p)$ with $p \ge r+1$, let $c \sqcup_r \Lambda$ be the configuration equal to c on $(-\infty,r]$ and to Λ on (r,∞); it equals $c \sqcup_{r'} \Lambda$ for every $r' \in [r,p-1]$. The same notation, with the same meaning, is used when c agrees with a vacuum V_ω on (r,∞) and Λ is a vacuum $V_{\omega'}$ of a possibly different phase, as in Definition 7.9; Lemmas 7.2(a) and 7.8 assume a common phase.

Lemma 7.2 (gluing and the edge)**.** (a) *Let c agree with V_ω on (r,∞) and Λ with V_ω on $(-\infty,p)$, where $p \ge r+3$. Then $F^{\pm1}(c \sqcup_r \Lambda) = F^{\pm1} c \sqcup_r F^{\pm1}\Lambda$, where $F^{\pm1}c$ agrees with $V_{\omega\pm1}$ on (r,∞) and $F^{\pm1}\Lambda$ agrees with $V_{\omega\pm1}$ on $(-\infty,p-1)$.*

(b) *If $c \in \Omega_r$, then $F^t c \in \Omega_r$ with phase $\omega(c)+t$, for every $t \in \mathbb{Z}$. The columns of Y_c to the right of r are those of V shifted by $\omega(c)$ in time.*

(c) *For $c \in \Omega_r$ and $t \in \mathbb{Z}$, with $s_t = V(t+\omega(c), r+1)$,*

$$z_{t+1} = z_{t-1} \oplus f\big(Y_x(t,r-1), z_t, s_t\big),$$

and $s_t = 1$ if and only if $t + \vartheta(c) \equiv 0, 1 \pmod 6$. So $z_{t+1} = \neg z_{t-1}$ if $t + \vartheta(c) \equiv 2, 3, 4, 5$, and $z_{t+1} = z_{t-1} \oplus (z_t \wedge \neg Y_c(t, r-1))$ otherwise. The cell of V_ω at r at time t has y-bit 1 if and only if $t + \omega + 3r \equiv 3, 4 \pmod 6$.

Proof. (a) Put $\hat{c} = c \sqcup_r \Lambda$, which equals c on $(-\infty, p-1]$ and Λ on (r, ∞). For $i \le p-2$ the cell $F^{\pm1}(\hat{c})_i$ reads $\hat{c}$ on $[i-1, i+1] \subseteq (-\infty, p-1]$, so it is $F^{\pm1}(c)_i$; for $i \ge r+2$ it reads only cells of Λ; and for $i = r+1$ it reads $\hat{c}_{r+1} = \Lambda_{r+1}$ and $\hat{c}_{r+2} = \Lambda_{r+2}$, which agree with V_ω, so by Lemma 7.1(c) it does not depend on $\hat{c}_r$ and equals $F^{\pm1}(\Lambda)_{r+1}$. As $p - 2 \ge r + 1$, this is the identity. Applying the same argument to c glued to V_ω, and to V_ω glued to Λ, gives the statements about $F^{\pm1}c$ and $F^{\pm1}\Lambda$, since $F^{\pm1}V_\omega = V_{\omega\pm1}$.

(b) By (a), $F^{\pm1}c$ agrees with $V_{\omega\pm1}$ on (r, ∞). If $F(c)_r$ were $(V_{\omega+1})_r$, then $F^{-1}(Fc)_r$ would read the cells $r, r+1$ of Fc, which agree with $V_{\omega+1}$, so by Lemma 7.1(c) it would equal $(V_\omega)_r$, contradicting $c \in \Omega_r$. The same argument applies to F^{-1}; induct on $|t|$.

(c) This is (2) at (t, r), with $Y_c(t, r+1) = V(t + \omega, r+1)$ by (b), and (12). If r is even, then $\vartheta \equiv \omega$, and $V(t + \omega, r+1) = 1$ exactly when $t + \omega \equiv 0, 1$; if r is odd, then $\vartheta \equiv \omega + 3$, and $V(t + \omega, r+1) = 1$ exactly when $t + \omega \equiv 3, 4$. The last statement is the same computation at column r. □

7.2. **Kinks and normalisation.** For $s = \pm1$ let Fix_s be the set of fixed points of $\sigma^{-s}F^3$. As in Lemma 2.3, these are coded by sequences. Let Φ be the set of $\varphi \in \{0,1\}^{\mathbb{Z}}$ with

$$\varphi(n-1) \oplus \varphi(n+1) = f\big(\varphi(n-3), \varphi(n), \varphi(n+3)\big) \qquad \text{for all } n \in \mathbb{Z}, \tag{13}$$

and write (P_n) for the instance of (13) at n. Φ is invariant under the shift $n \mapsto n+1$.

Lemma 7.3 (coding of Fix_1)**.** (a) *The map $\varphi \mapsto c$, $c_i = (\varphi(3i-1), \varphi(3i))$, is a bijection from Φ onto Fix_1, and $Y_c(t, i) = \varphi(t + 3i)$. The sequence of $F^j c$ is $n \mapsto \varphi(n + j)$.*

(b) *Let $\varphi, \psi \in \Phi$ code c, d. Then c and d agree on a left (right) half-line if and only if φ and ψ agree on a left (right) half-line.*

(c) *The vacuum V_ω is coded by $v_\omega(n) = v(n + \omega)$, where $v(n) = [\, n \equiv 3, 4 \pmod 6 \,]$; the constant configuration $F^s((0,0)^{\mathbb{Z}})$ ($s \in \mathbb{Z}/3$) by $\kappa_s(n) = [\, n \equiv 1 - s \pmod 3 \,]$; and $(1,1)^{\mathbb{Z}}$ by the constant sequence 1. The six points of $\mathcal{W}$ (defined below) lie in Fix_1 and are coded by the shifts of $w(n) = [\, n \not\equiv 3, 5 \pmod 6 \,]$.*

Proof. (a) $c \in \mathrm{Fix}_1$ means $F^3 c = \sigma c$, that is $Y_c(t+3, i) = Y_c(t, i+1)$ for all t, i, since a configuration is determined by its diagram and $Y_{F^3 c}(t, i) = Y_c(t+3, i)$, $Y_{\sigma c}(t, i) = Y_c(t, i+1)$. Then $\varphi(n) = Y_c(n, 0)$ satisfies $Y_c(t, i) = \varphi(t + 3i)$, and (2) at (t, i) is (P_{t+3i}). Conversely, for $\varphi \in \Phi$ the function $Y(t, i) = \varphi(t + 3i)$ satisfies (2), so it is the diagram of the configuration c formed by its rows -1 and 0, and $Y(t+3, i) = Y(t, i+1)$ gives $c \in \mathrm{Fix}_1$. The last statement is $Y_{F^j c}(t, i) = Y_c(t + j, i)$.

(b) If c and d agree on $(-\infty, a]$, then, as $F^{\pm1}$ have radius 1, $Y_c(t, i) = Y_d(t, i)$ for $t \in \{0, 1, 2\}$ and $i \le a - 2$. Since $\varphi(n) = Y_c(n - 3i, i)$ for every i, taking $i = \lfloor n/3 \rfloor$ gives $\varphi(n) = \psi(n)$ for $n \le 3a - 4$. Conversely, if $\varphi = \psi$ on $(-\infty, M]$, then $c_i = d_i$ for $3i \le M$. Right half-lines are the same, with $t \in \{-2, -1, 0\}$.

(c) By Lemma 7.1(a), $V(t, i) = V(t + 3i, 0) = v(t + 3i)$, and $F^3 V_\omega = \sigma V_\omega$; so $V_\omega \in \mathrm{Fix}_1$, coded by $n \mapsto V(n + \omega, 0) = v(n + \omega)$. The diagram of $(0,0)^{\mathbb{Z}}$ is $Y(t, i) = [\, t \equiv 1 \pmod 3 \,]$, as F acts by $(0,0)^{\mathbb{Z}} \mapsto (0,1)^{\mathbb{Z}} \mapsto (1,0)^{\mathbb{Z}} \mapsto (0,0)^{\mathbb{Z}}$; it is constant in i and of period 3 in t, so it is coded by κ_0, and $F^s((0,0)^{\mathbb{Z}})$ by κ_s. The diagram of $(1,1)^{\mathbb{Z}}$ is 1. For $\mathcal{W}$: the sequence w is $1, 1, 1, 0, 1, 0$ on $n \equiv 0, \dots, 5$, and the configuration it codes has the cells $(w(-1), w(0)) = (0, 1)$ at even sites and $(w(2), w(3)) = (1, 0)$ at odd sites. One checks (P_n) for $n \equiv 0, \dots, 5$ with the values of f below; so this point of $\mathcal{W}$ lies in Fix_1, and its images under F and σ, which are the six points of $\mathcal{W}$ by the list in the proof of Proposition 7.6, are coded by the shifts of w. □

We use $f(a,b,0)=1$ and $f(a,b,1)=\neg a\wedge b$ from (12). So (P_n) says: if $\varphi(n+3)=0$, then $\varphi(n-1)\neq\varphi(n+1)$; if $\varphi(n+3)=1$, then $\varphi(n-1)\oplus\varphi(n+1)=\neg\varphi(n-3)\wedge\varphi(n)$. Read from left to right, this is a rule for $\varphi(n+3)$:

Lemma 7.4. *Let $\varphi\in\Phi$ and $n,p\in\mathbb{Z}$.*

(a) (Forward rule.) *The values $\varphi(n-3),\varphi(n-1),\varphi(n),\varphi(n+1)$ determine $\varphi(n+3)$, unless $\varphi(n-1)\neq\varphi(n+1)$, $\varphi(n-3)=0$ and $\varphi(n)=1$, in which case both values of $\varphi(n+3)$ satisfy (P_n). Precisely: if $\varphi(n-1)=\varphi(n+1)$, then $\varphi(n+3)=1$; if $\varphi(n-1)\neq\varphi(n+1)$ and $(\varphi(n-3),\varphi(n))\neq(0,1)$, then $\varphi(n+3)=0$.*

(b) (Constants absorb.) *If φ agrees with $\kappa\in\{\kappa_0,\kappa_1,\kappa_2,1\}$ on a left half-line, then $\varphi=\kappa$.*

(c) (No new 00.) *If $\varphi(p)=\varphi(p+1)=0$, then $\varphi(q)=\varphi(q+1)=0$ for some $q\in\{p-1,p-3,p-4\}$. Consequently, if $\varphi(q)\varphi(q+1)\neq 00$ for $q=N,\dots,N+3$, then $\varphi(q)\varphi(q+1)\neq 00$ for every $q\geq N$.*

(d) *If $\varphi=1$ on a right half-line, then $\varphi=1$.*

Proof. (a) is the reformulation of (P_n) above: if $\varphi(n-1)=\varphi(n+1)$, the value $\varphi(n+3)=0$ is excluded; if $\varphi(n-1)\neq\varphi(n+1)$, the value $\varphi(n+3)=1$ requires $\neg\varphi(n-3)\wedge\varphi(n)=1$, and when this holds both values are allowed, since $f(0,1,c)=1$.

(b) κ lies in Φ by Lemma 7.3(c), and the exceptional case of (a) never occurs in κ: in κ_s a value 1 at n has the values 0 at $n\pm1$, and in 1 the value at $n-3$ is 1. So if $\varphi=\kappa$ on $(-\infty,M]$, then by (a) at $n=M-2$ the value $\varphi(M+1)$ is the forced value $\kappa(M+1)$; induct.

(c) By (P_{p-2}) and $\varphi(p+1)=0$, $\varphi(p-3)\neq\varphi(p-1)$; by ($\mathrm{P}_{p-3}$) and $\varphi(p)=0$, $\varphi(p-4)\neq\varphi(p-2)$. If $\varphi(p-1)=0$, take $q=p-1$. Otherwise $\varphi(p-1)=1$ and $\varphi(p-3)=0$; if $\varphi(p-2)=0$, take $q=p-3$; otherwise $\varphi(p-2)=1$, so $\varphi(p-4)=0$, and take $q=p-4$. For the consequence, a smallest $p\geq N+4$ with $\varphi(p)\varphi(p+1)=00$ would give such a q with $N\leq q<p$.

(d) If $\varphi(p)=0$ and $\varphi(p+2)=\varphi(p+3)=\varphi(p+4)=\varphi(p+6)=1$, then ($\mathrm{P}_{p+3}$) reads $0=f(0,1,1)=1$. So if $\varphi=1$ on $[M,\infty)$, then $\varphi(M-2)=\varphi(M-1)=1$ (take $p=M-2$ and $p=M-1$); induct. □

Let $\varphi_K\in\{0,1\}^{\mathbb{Z}}$ be given by

$$\varphi_K(n)=\begin{cases}v(n)=[\,n\equiv 3,4 \pmod 6\,] & (n\leq 6),\\ \kappa_0(n)=[\,n\equiv 1 \pmod 3\,] & (n\geq 4);\end{cases}$$

the two formulas agree on $\{4,5,6\}$, and $\varphi_K(n)$ for $n=1,\dots,10$ is $0,0,1,1,0,0,1,0,0,1$. Let K^- be the configuration equal to V_0 on $(-\infty,1]$ and to $(0,0)$ on $[2,\infty)$, and K^+ the configuration equal to V_0 on $(-\infty,0]$ and to $(0,1)$ on $[1,\infty)$. For $j\in\mathbb{Z}$ put $K^-_j=F^jK^-$ and $K^+_j=F^jK^+$. Let $\mathcal{W}$ be the set of the six configurations of period 2 whose cells at an even and at an odd site are two distinct elements of $\{(0,1),(1,0),(1,1)\}$.

Lemma 7.5 (kinks)**.** (a) *$\varphi_K\in\Phi$, and φ_K codes K^-; so $Y_{K^-}(t,i)=\varphi_K(t+3i)$. Moreover $K^+=F^{-2}\rho K^-$, so $Y_{K^+}(t,i)=\varphi_K(1-t+3i)$. Hence $F^3K^-=\sigma K^-$, $F^3K^+=\sigma^{-1}K^+$ and $\rho K^-=F^2K^+$; so $K^-_j\in\mathrm{Fix}_1$, $K^+_j\in\mathrm{Fix}_{-1}$, $\sigma K^-_j=K^-_{j+3}$, $\sigma^{-1}K^+_j=K^+_{j+3}$, $\rho K^-_j=K^+_{2-j}$ and $\rho K^+_j=K^-_{2-j}$.*

(b) *K^-_j agrees with V_j exactly on $(-\infty,p^-(j))$, and K^+_j with V_j exactly on $(-\infty,q^+(j))$, where $p^-(j)=2-\lfloor(j-1)/3\rfloor$ and $q^+(j)=2+\lfloor(j+1)/3\rfloor$. The configuration K^-_j is constant, equal to $F^j((0,0)^{\mathbb{Z}})$, exactly on $[2-\lfloor(j+1)/3\rfloor,\infty)$, and K^+_j is constant, equal to $F^j((0,1)^{\mathbb{Z}})$, on the cells i with $3i\geq j+3$.*

(c) *The points of Fix_1 that agree with a vacuum on a left half-line and with a constant configuration on a right half-line are the K^-_j, $j\in\mathbb{Z}$, and these are pairwise distinct; the*

same holds for Fix_{-1} *and the* K_j^+*. No point of* $\mathrm{Fix}_{\pm1}$ *agrees with a point of* $\mathcal{W}$ *on a left half-line and with a constant configuration on a right half-line, and a point of* $\mathrm{Fix}_{\pm1}$ *that agrees with constant configurations on a left and on a right half-line is constant.*

Proof. (a) (P_n) holds for $n \le 3$, where it involves only values of $v \in \Phi$, and for $n \ge 7$, where it involves only values of $\kappa_0 \in \Phi$. The remaining three instances are (P_4): $\varphi_K(3) \oplus \varphi_K(5) = 1 = f(0,1,1)$; (P_5): $\varphi_K(4) \oplus \varphi_K(6) = 1 = f(0,0,0)$; and (P_6): $\varphi_K(5) \oplus \varphi_K(7) = 1 = f(1,0,0)$. The configuration coded by φ_K has the cells $(\varphi_K(3i-1), \varphi_K(3i))$, which are $(v(3i-1), v(3i)) = (V_0)_i$ for $i \le 1$ (Lemma 7.3(c)) and $(\kappa_0(3i-1), \kappa_0(3i)) = (0,0)$ for $i \ge 2$; this is K^-. By (3), the diagram of $F^{-2}\rho K^-$ is $Y_{K^-}(1-t,i) = \varphi_K(1-t+3i)$, and its cells are $(\varphi_K(2+3i), \varphi_K(1+3i))$. For $i \le 0$ these are $(v(2+3i), v(1+3i)) = (v(3i-1), v(3i)) = (V_0)_i$, by $v(1-n) = v(n)$ and the period 6; for $i \ge 1$ they are $(\kappa_0(2+3i), \kappa_0(1+3i)) = (0,1)$. So $F^{-2}\rho K^- = K^+$. Now $K^- \in \mathrm{Fix}_1$, and $F^3K^+ = F^{-2}\rho F^{-3}K^- = F^{-2}\rho\sigma^{-1}K^- = \sigma^{-1}K^+$; the rest follows, as σ commutes with F and ρ, and $\rho F^j = F^{-j}\rho$.

(b) By (a), $(K_j^-)_i = (\varphi_K(j-1+3i), \varphi_K(j+3i))$ and $(V_j)_i = (v(j-1+3i), v(j+3i))$. As $\varphi_K = v$ on $(-\infty, 6]$ while $\varphi_K(7) = 1 \ne v(7)$ and $\varphi_K(9) = 0 \ne v(9)$, the two cells agree if $j+3i \le 6$ and differ if $j+3i \in \{7,8,9\}$ (one of the two arguments is then 7 or 9). So $p^-(j)$ is the least i with $j+3i \ge 7$, which is $\lceil (7-j)/3 \rceil = 2 - \lfloor (j-1)/3 \rfloor$. In the same way $(K_j^+)_i = (\varphi_K(2-j+3i), \varphi_K(1-j+3i))$, the vacuum cell is the same with v, and $q^+(j)$ is the least i with $2-j+3i \ge 7$, which is $2 + \lfloor (j+1)/3 \rfloor$. If $j-1+3i \ge 4$, then $(K_j^-)_i = (\kappa_0(j-1), \kappa_0(j))$, the cell of $F^j((0,0)^{\mathbb{Z}})$ by Lemma 7.3(c); and if $1-j+3i \ge 4$, then $(K_j^+)_i = (\kappa_0(2-j), \kappa_0(1-j)) = (\kappa_0(j), \kappa_0(j+1))$, the cell of $F^{j+1}((0,0)^{\mathbb{Z}}) = F^j((0,1)^{\mathbb{Z}})$. The least i with $j-1+3i \ge 4$ is $i_0 = 2 - \lfloor (j+1)/3 \rfloor$, and $j-1+3(i_0-1) = e+1$ with $e \in \{0,1,2\}$, $e \equiv j+1 \pmod 3$; the cell $(K_j^-)_{i_0-1} = (\varphi_K(e+1), \varphi_K(e+2))$ is $(0,0)$, $(0,1)$, $(1,1)$ for $e = 0,1,2$, while the constant cell $(\kappa_0(e+1), \kappa_0(e+2))$ is $(1,0)$, $(0,0)$, $(0,1)$. So K_j^- is not constant on $[i_0-1, \infty)$.

(c) By Lemma 7.3, a point $c \in \mathrm{Fix}_1$ that agrees with a vacuum on a left half-line and with a constant configuration on a right half-line is coded by a $\varphi \in \Phi$ that agrees with some v_ω on a left half-line and with some $\kappa \in \{\kappa_0, \kappa_1, \kappa_2, 1\}$ on a right half-line. By Lemma 7.4(d), $\kappa \ne 1$, since $\varphi \ne 1$. So $\varphi \ne v_\omega$, and, replacing φ by a shift, we may assume that $\omega = 0$ and that N is the least integer with $\varphi(N) \ne v(N)$. The values $\varphi(N-6), \varphi(N-4), \varphi(N-3), \varphi(N-2)$ are those of v, and $v(N)$ satisfies (P_{N-3}); by Lemma 7.4(a), $\varphi(N) \ne v(N)$ is possible only if $v(N-3) = 1$, $v(N-6) = 0$ and $v(N-4) \ne v(N-2)$. This means $N \equiv 0$ or $1 \pmod 6$ (then $v(N-6) = v(N) = 0$ and $v(N-4) \ne v(N-2)$ hold), and $\varphi(N) = 1$.

If $N \equiv 0$, then $\varphi(N-5), \dots, \varphi(N)$ are $0,0,1,1,0,1$, and by Lemma 7.4(a) at $n = N-1$, where $\varphi(N-2) = \varphi(N) = 1$, $\varphi(N+2) = 1$. So no 00 starts at $N-2, N-1, N, N+1$, and by Lemma 7.4(c) none starts at any $q \ge N-2$; but κ_s contains 00 in every interval of length 3, a contradiction. Hence $N \equiv 1$, and $\varphi(N-6), \dots, \varphi(N)$ are $0,0,1,1,0,0,1$. Lemma 7.4(a) at $n = N-2$ (values $0,1,0,0$) gives $\varphi(N+1) = 0$, at $n = N-1$ (values $1,0,0,1$) gives $\varphi(N+2) = 0$, and at $n = N$ (values $1,0,1,0$) gives $\varphi(N+3) = 1$. So $\varphi = \kappa$ on $[N-2, N+3]$ for the $\kappa \in \{\kappa_0, \kappa_1, \kappa_2\}$ with $\kappa(N) = 1$, and by the argument of Lemma 7.4(b), applied from $N+3$ on, $\varphi = \kappa$ on $[N-2, \infty)$. Thus $\varphi = v$ on $(-\infty, N-1]$ and $\varphi = \kappa$ on $[N-2, \infty)$, with $N \equiv 1 \pmod 6$; this is the shift $n \mapsto \varphi_K(n+7-N)$ of φ_K, which codes $F^{7-N}K^-$ by Lemma 7.3(a). Undoing the first shift, c is some K_j^-. The K_j^- are pairwise distinct, since they are coded by the shifts of φ_K, which is not periodic.

If $c \in \mathrm{Fix}_1$ agrees with a point of $\mathcal{W}$ on a left half-line, then φ agrees with a shift of w on a left half-line (Lemma 7.3(b),(c)). As w contains no 00, Lemma 7.4(c) shows that φ contains no 00, so it does not agree with any κ_s on a right half-line, and by Lemma 7.4(d) it does not agree with

1 on a right half-line, since $\varphi \neq 1$. If c agrees with a constant configuration on a left half-line, then φ agrees with some κ on a left half-line, and $\varphi = \kappa$ by Lemma 7.4(b), so c is constant.

For Fix_{-1}: since $\rho\sigma^{-1}F^3\rho = \sigma^{-1}F^{-3}$, whose fixed points are those of σF^3, ρ maps Fix_1 bijectively onto Fix_{-1}. It acts cellwise, maps the vacua onto the vacua ($\rho V_\omega = V_{2-\omega}$), the constant configurations onto the constant configurations, $\mathcal{W}$ onto $\mathcal{W}$, and K^-_j to K^+_{2-j} by (a). So the statements for Fix_{-1} are those for Fix_1. □

Proposition 7.6 (normalisation)**.** *Let $g \in \mathrm{Aut}(X_F)$. There are $j \in \{0,1,2\}$ and $a, b \in \mathbb{Z}$ such that $h = F^{-3b}\sigma^{-a}F^{-j}g$ fixes K^- and K^+.*

Proof. The automorphism g permutes the fixed points of σ, which are the four constant configurations, and the induced permutation commutes with that of F, the 3-cycle $(0,0)^{\mathbb{Z}} \mapsto (0,1)^{\mathbb{Z}} \mapsto (1,0)^{\mathbb{Z}}$. The centralizer of a 3-cycle in $\mathrm{Sym}(4)$ is the group it generates, so g acts on the constants as F^j for one $j \in \{0,1,2\}$, and $g' = F^{-j}g$ fixes every constant configuration. It permutes the fixed points of σ^2, which are the configurations of period 2, and preserves the sizes of their F-orbits. By Lemma 7.1(a) the vacua form one F-orbit of size 6, and the only other orbit of size 6 is $\mathcal{W}$: since $f(p,q,p) = \neg p$, F maps the configuration of period 2 with cells (x_e, y_e) at even and (x_o, y_o) at odd sites to the one with cells $(y_e, x_e \oplus \neg y_o)$ and $(y_o, x_o \oplus \neg y_e)$, and its orbits on the sixteen configurations of period 2, written as pairs (even cell, odd cell) in the coding $2x+y$, are $\{(0,0),(1,1),(2,2)\}$, $\{(3,3)\}$, the vacua $(0,1) \mapsto (0,3) \mapsto (0,2) \mapsto (1,0) \mapsto (3,0) \mapsto (2,0) \mapsto (0,1)$, and $\mathcal{W}$: $(1,2) \mapsto (3,1) \mapsto (3,2) \mapsto (2,1) \mapsto (1,3) \mapsto (2,3) \mapsto (1,2)$. Since K^- agrees with V_0 on a left half-line and with $(0,0)^{\mathbb{Z}}$ on a right half-line, and g' is a block map, $g'K^- \in \mathrm{Fix}_1$ agrees with $g'V_0$ on a left half-line and with $(0,0)^{\mathbb{Z}}$ on a right half-line. By Lemma 7.5(c), $g'V_0 \notin \mathcal{W}$, so $g'V_0 = V_{\omega_0}$ for some ω_0, and $g'K^- = K^-_{j'}$ for some j'. Its right constant is $F^{j'}((0,0)^{\mathbb{Z}}) = (0,0)^{\mathbb{Z}}$, so $j' = 3e^-$ for an integer e^-, and $g'K^- = \sigma^{e^-}K^-$. In the same way, $g'K^+ \in \mathrm{Fix}_{-1}$ agrees with $g'V_0$ on a left half-line and with $(0,1)^{\mathbb{Z}}$ on a right half-line; by the statements of Lemma 7.5(c) for Fix_{-1}, $g'V_0 \notin \mathcal{W}$ and $g'K^+ = K^+_{j''}$ for some j'', whose right constant $F^{j''}((0,1)^{\mathbb{Z}}) = (0,1)^{\mathbb{Z}}$ gives $j'' = 3n$ for an integer n, so $g'K^+ = K^+_{3n} = \sigma^{-n}K^+$; put $e^+ = -n$, so $g'K^+ = \sigma^{e^+}K^+$. Moreover, $g'K^-$ agrees on a left half-line with $g'V_0 = V_{\omega_0}$ and also with $\sigma^{e^-}V_0 = V_{3e^-}$; so $\omega_0 \equiv 3e^-$, and likewise $\omega_0 \equiv 3e^+ \pmod 6$. Hence $e^- \equiv e^+ \pmod 2$. Put $a = \frac{1}{2}(e^- + e^+)$ and $b = \frac{1}{2}(e^- - e^+)$. Since F^3 fixes the constants and $F^3K^\mp = \sigma^{\pm 1}K^\mp$, the map $\sigma^a F^{3b}$ sends K^- to $\sigma^{a+b}K^- = \sigma^{e^-}K^-$ and K^+ to $\sigma^{a-b}K^+ = \sigma^{e^+}K^+$, so $h = F^{-3b}\sigma^{-a}g'$ fixes K^- and K^+. □

Lemma 7.7 (Lemma K_{93})**.** *Let h be a block map that commutes with F_{93} and fixes K^- and K^+. Then $h = \mathrm{id}$.*

Lemma K_{93} is proved in Section 7.6. Let $\mathcal{H}$ be the set of block maps h, of any radius, that commute with F and fix K^- and K^+. Such an h fixes every $K^\pm_j$. It fixes every vacuum: hV_0 has period 2 and agrees with V_0 on a left half-line, because $hK^- = K^-$; so $hV_0 = V_0$, and $hV_\omega = V_\omega$ by F. It fixes $(0,0)^{\mathbb{Z}}$, $(0,1)^{\mathbb{Z}}$ and $(1,0)^{\mathbb{Z}}$, the right constants of the $K^\pm_j$. And $\rho h \rho \in \mathcal{H}$, since $\rho F \rho = F^{-1}$ and $\rho h \rho K^\mp = \rho h F^2 K^\pm = \rho F^2 K^\pm = K^\mp$ by Lemma 7.5(a).

Lemma 7.8. *Let h be a block map of radius R that fixes V_ω and Λ, let c agree with V_ω on (r, ∞) and Λ with V_ω on $(-\infty, p)$, where $p \geq r + 2R + 1$. Then hc agrees with V_ω on $(r+R, \infty)$ and $h(c \sqcup_r \Lambda) = hc \sqcup_{r+R} \Lambda$.*

Proof. For $i \geq r + R + 1$ the window of i lies in (r, ∞), where $c = V_\omega$, so $(hc)_i = (hV_\omega)_i = (V_\omega)_i$. Put $\hat{c} = c \sqcup_r \Lambda$. For $i \leq p - 1 - R$ the window of i lies in $(-\infty, p-1]$, where $\hat{c} = c$, so $(h\hat{c})_i = (hc)_i$; for $i \geq r + R + 1$ it lies in (r, ∞), where $\hat{c} = \Lambda$, so $(h\hat{c})_i = (h\Lambda)_i = \Lambda_i$. These ranges cover $\mathbb{Z}$, and on their overlap hc and Λ both agree with V_ω. □

7.3. **Moves.**

Definition 7.9. Let $c \in \Omega_r$ have phase ω, let $m \in \mathbb{Z}$ and $\omega' \in \mathbb{Z}/6$. Let $S_m^{\omega'}(c)$ be the configuration D with $F^m D = F^m c \sqcup_r V_{\omega'+m}$, that is, $F^m D$ agrees with $F^m c$ on $(-\infty, r]$ and with the vacuum of phase $\omega' + m$ on (r, ∞). Put $\vartheta = \vartheta(c)$ and $z_t = Y_c(t, r)$, and define the *move at time* m

$$T_m(x) = \begin{cases} S_m^{\omega+2}(x) & \text{if } m + \vartheta \equiv 2 \text{ and } z_{m-1} = 0, \\ S_m^{\omega-1}(x) & \text{if } m + \vartheta \equiv 2 \text{ and } z_{m-1} = 1, \\ S_m^{\omega-2}(x) & \text{if } m + \vartheta \equiv 0 \text{ and } z_m = 0, \\ S_m^{\omega+1}(x) & \text{if } m + \vartheta \equiv 0 \text{ and } z_m = 1, \\ x & \text{otherwise,} \end{cases}$$

congruences modulo 6. We call T_m a *forward move* in the first two cases and a *backward move* in the next two, and write M_m, $\bar{M}_m$ for T_m restricted to the configurations on which it is forward, respectively backward.

So a move keeps the state at time m on the columns $\le r$ and switches the phase of the vacuum to its right, by $+2$ or -1 (forward) or by -2 or $+1$ (backward), according to one bit of the edge column.

Lemma 7.10. *Let $c \in \Omega_r$ and $m \in \mathbb{Z}$.*

(a) *$T_m c \in \Omega_r$.*
(b) *$T_m c$ agrees with c on $(-\infty, r - |m|]$; and $Y_{T_m c}(t, r) = z_t$ for $t \in \{m-1, m\}$.*
(c) *$T_m(c) = F^{-\ell} T_{m-\ell}(F^\ell c)$ for every $\ell \in \mathbb{Z}$, and $\sigma T_m = T_m \sigma$ (with edges r and $r-1$). Moreover $\rho c \in \Omega_r$, $\vartheta(\rho c) = 2 - \vartheta(c)$, the edge column of ρc is $t \mapsto z_{-1-t}$, and $\bar{M}_m = \rho M_{-m} \rho$. So under ρ the time t of a diagram corresponds to the time $-1-t$, while the move at time m corresponds to the move at time $-m$.*

Proof. (a) By Lemma 7.2(b) it suffices that the cell r of $F^m c$, which is (z_{m-1}, z_m), differs from the cell of $V_{\omega'+m}$ at r, which by Lemma 7.2(c) has bits $[m - 1 + \vartheta' \equiv 3, 4]$ and $[m + \vartheta' \equiv 3, 4]$, where $\vartheta' = \omega' + 3r$. In the four cases of the definition, $m + \vartheta' \equiv 4, 1, 4, 1$, so this cell is $(1,1)$, $(0,0)$, $(1,1)$, $(0,0)$, while (z_{m-1}, z_m) is $(0, \cdot)$, $(1, \cdot)$, $(\cdot, 0)$, $(\cdot, 1)$.

(b) $F^m T_m c$ and $F^m c$ agree on $(-\infty, r]$, and F^{-m} has radius $|m|$. The second statement holds because rows $m-1$ and m form the state at time m.

(c) $F^\ell c \in \Omega_r$ has phase $\omega + \ell$, so $\vartheta(F^\ell c) = \vartheta + \ell$, its edge column is $t \mapsto z_{t+\ell}$, and the case of $T_{m-\ell}$ at $F^\ell c$ is the case of T_m at c; and $F^{m-\ell}(F^\ell S_m^{\omega'} c) = F^m c \sqcup_r V_{\omega'+m}$, which says $F^\ell S_m^{\omega'}(c) = S_{m-\ell}^{\omega'+\ell}(F^\ell c)$. The shift: $\sigma c \in \Omega_{r-1}$ has phase $\omega + 3$ and the same ϑ and edge column. For ρ: by Lemma 7.1(a) and (3), $\rho c \in \Omega_r$ has phase $2 - \omega$, $\vartheta(\rho c) = 2 - \vartheta$, and edge column $t \mapsto z_{-1-t}$; and $\rho S_{m'}^{\omega'} \rho = S_{-m'}^{2-\omega'}$, since $F^{-m'} \rho = \rho F^{m'}$ exchanges the two rows of each cell. The forward move M_{-m} at ρc applies when $-m + 2 - \vartheta \equiv 2$, that is $m + \vartheta \equiv 0$, reads $Y_{\rho c}(-m-1, r) = z_m$, and gives the phase $2 - \omega + 2$ or $2 - \omega - 1$; conjugating by ρ gives $S_m^{\omega-2}$ or $S_m^{\omega+1}$, which is $\bar{M}_m$. □

A move is not undone by a move at the same time: after T_m the class of $m + \vartheta$ is 4 or 1, and T_m acts as the identity. Orbits of moves are therefore defined below by directed reachability.

The moves are realised by gluing a kink. If $c \in \Omega_r$ has phase ω and $j \equiv \omega \pmod 6$, $j \le -3r$, then K_j^- agrees with $V_j = V_\omega$ on $(-\infty, r+2]$ by Lemma 7.5(b), and $c \sqcup_r K_j^-$ is defined. We say that a configuration $\hat{c}$ has *outgoing index* $J \in \mathbb{Z}$ if there are $q \in \mathbb{Z}$ and s_0 such that $F^s \hat{c}$ agrees with K_{s+J}^+ on $[q, \infty)$ for every $s \ge s_0$. The index is unique if it exists: for $J \not\equiv J' \pmod 3$ the right constants of K_{s+J}^+ and $K_{s+J'}^+$ differ; for $J' = J + 3d$ with d odd, $K_{s+J'}^+ = \sigma^{-d} K_{s+J}^+$ and the vacua V_{s+J}, $V_{s+J'}$ to the left of the two kinks differ at every site; and for $d \ne 0$ even the two

vacua coincide while the first sites $q^+(s+J) \neq q^+(s+J')$ at which the configurations leave them both tend to ∞. For the same reason, K^-_j has no outgoing index: $F^s K^-_j = K^-_{j+s}$ is constant on $[q,\infty)$ for large s, while K^+_{s+J} is not, as no two adjacent cells of a vacuum are equal.

Lemma 7.11 (moves from kinks)**.** *Let $c \in \Omega_r$ have phase ω, let $j \equiv \omega \pmod 6$ with $j \le -3r$, and put $\hat{c} = c \sqcup_r K^-_j$, $m = 2-j-3r$, $z = Y_c(m-1,r)$, $D = M_m c$ and*

$$J = j + 6r + 3z - 4.$$

Then $m \ge 2$, $m + \vartheta(c) \equiv 2 \pmod 6$, and for every $s \ge m$ the configuration $F^s\hat{c}$ agrees with $F^s D$ on $(-\infty, r]$ and with K^+_{s+J} on (r,∞):

$$F^s\hat{c} = F^s D \sqcup_r K^+_{s+J} \qquad (s \ge m).$$

In particular $\hat{c}$ has outgoing index J.

Proof. $m+\vartheta \equiv 2-j-3r+\omega+3r \equiv 2$, and $m \ge 2$ as $j \le -3r$. The phase of D is $\omega' = \omega+2-3z \equiv J \pmod 6$. We write down the diagram of $\hat{c}$. Put $\tau = t-m$, $k = i-r$, and let

$$A(t,i) = Y_{K^-_j}(t,i) = \varphi_K(t+j+3i) = \varphi_K(\tau+3k+2),$$
$$B(t,i) = Y_{K^+_J}(t,i) = \varphi_K(1-t-J+3i) = \varphi_K(3k-\tau+3-3z),$$

by Lemma 7.5(a). Both are functions of (t,i); written in τ and k, neither depends on r, ω or j, and below we say that a statement holds at (τ,k) when it holds at $(t,i) = (m+\tau, r+k)$. Define $Z\colon \mathbb{Z}^2 \to \{0,1\}$ by

$$Z(t,i) = \begin{cases} Y_c(t,i) & i \le r,\ t \le m, \\ Y_D(t,i) & i \le r,\ t \ge m-1, \\ A(t,i) & i > r,\ t \le m-1, \\ B(t,i) & i > r,\ t \ge m. \end{cases}$$

The first two lines are consistent, since $F^m D$ agrees with $F^m c$ on $(-\infty, r]$, and the rows $m-1$, m form the state at time m. Recall $\varphi_K(n) = v(n)$ for $n \le 6$, $\varphi_K(n) = \kappa_0(n) = [n \equiv 1\ (3)]$ for $n \ge 4$, and $\varphi_K(1),\dots,\varphi_K(9) = 0,0,1,1,0,0,1,0,0$. We use four facts.

(i) $A(t,i) = V(t+\omega,i)$ whenever $\tau+3k+2 \le 6$, in particular for $k=1$, $\tau \le 1$ and for $k=2$, $\tau \le -2$; and $B(t,i) = V(t+J,i) = V(t+\omega',i)$ whenever $3k-\tau+3-3z \le 6$, in particular for $k=1$, $\tau \ge -3z$ and for $k=2$, $\tau \ge 3-3z$. (By Lemma 7.3(c), $V(t,i) = v(t+3i)$, and $v(1-n) = v(n)$; and $j \equiv \omega$, $J \equiv \omega'$.)

(ii) $A(t,i) = B(t,i)$ for $\tau = -1$, $k \ge 1$, and for $\tau = 0$, $k \ge 2$. (For $\tau = -1$ the arguments $3k+1$ and $3k+4-3z$ are ≥ 4 and $\equiv 1 \pmod 3$, so both values are 1; for $\tau = 0$ the arguments $3k+2 \ge 8$ and $3k+3-3z \ge 6$ are $\not\equiv 1 \pmod 3$, so both values are 0.)

(iii) For $k=1$ and $\tau \ge 0$, $B(t,r+1) = B(t,r+2) = 1$ only if $(\tau,z) = (2,0)$; for $k=1$ and $\tau \le -2$, $A(t,r+1) = A(t,r+2) = 1$ never holds. (In the ranges of (i) these are two values of a row of V at adjacent sites, which are never both 1 by Lemma 7.1(c); the remaining values are $B(m,r+1) = \varphi_K(6-3z)$ and $B(m,r+2) = \varphi_K(9-3z) = 0$, $B(m+1,r+1) = \varphi_K(5-3z) = 0$, and, for $z=0$, $B(m+2,r+1) = \varphi_K(4) = 1$ and $B(m+2,r+2) = \varphi_K(7) = 1$.)

(iv) If $z = 0$, then $Y_D(m+2,r) = 0$. (The cell (z_{m-1}, z_m) of $F^m c$ at r differs from the cell of $V_{\omega+m}$ at r, which by Lemma 7.2(c) is $(0,0)$, as $m+\vartheta \equiv 2$; so $z_m = 1$. D has $\vartheta(D) = \vartheta+2$, so $(m+1)+\vartheta(D) \equiv 5$, and Lemma 7.2(c) for D gives $Y_D(m+2,r) = \neg Y_D(m,r) = \neg z_m = 0$.)

We check (2) for Z at every (t,i); it involves Z at $(t\pm1,i)$ and $(t,i-1),(t,i),(t,i+1)$.

Columns $i \le r-1$. For $t \le m-1$ all values are those of Y_c, for $t \ge m$ all are those of Y_D.

Column r. For $t \le m-1$ the values on the columns $\le r$ are those of Y_c, and $Z(t,r+1) = A(t,r+1) = V(t+\omega, r+1)$ by (i), since $\tau \le -1$; this is (2) for Y_c, whose column $r+1$ is $V(t+\omega, r+1)$ (Lemma 7.2(b)). For $t \ge m$ they are those of Y_D, and $Z(t,r+1) = B(t,r+1) = V(t+\omega', r+1)$ by (i), since $\tau \ge 0 \ge -3z$; this is (2) for Y_D.

Columns $i \ge r+2$. For $t \le m-2$ all values are values of A, and for $t \ge m+1$ all are values of B. For $t = m-1$ they are values of A, except $Z(m,i) = B(m,i)$, which equals $A(m,i)$ by (ii); for $t = m$ they are values of B, except $Z(m-1,i) = A(m-1,i) = B(m-1,i)$ by (ii). In each case (2) is that of the diagram A or B.

Column $r+1$. Here $f(Z(t,r), Z(t,r+1), Z(t,r+2))$ depends on $Z(t,r)$ only if $Z(t,r+1) = Z(t,r+2) = 1$. For $t \le m-2$ all other values are values of A, and by (iii) the relation is that of A. For $t \ge m$ all other values are values of B (at $t = m$, $Z(m-1,r+1) = B(m-1,r+1)$ by (ii)), and by (iii) the relation is that of B, except at $(\tau, z) = (2,0)$. There it reads $B(m+3,r+1) \oplus B(m+1,r+1) = \varphi_K(3) \oplus \varphi_K(5) = 1 = f(Z(m+2,r),1,1) = \neg Y_D(m+2,r)$, which is (iv). At $t = m-1$ the values are $Z(m-1,r) = z$, $Z(m-1,r+1) = A(m-1,r+1) = \varphi_K(4) = 1$, $Z(m-1,r+2) = \varphi_K(7) = 1$, $Z(m-2,r+1) = \varphi_K(3) = 1$ and $Z(m,r+1) = B(m,r+1) = \varphi_K(6-3z) = z$, and the relation reads $z \oplus 1 = f(z,1,1) = \neg z$.

So Z satisfies (2) everywhere. Its rows -1 and 0 are those of c on $(-\infty, r]$ and of K_j^- on (r, ∞), as $m \ge 2$; so $Z = Y_{\hat{c}}$. For $s \ge m$, rows $s-1$ and s of Z are those of Y_D on the columns $\le r$, and those of B on the columns $> r$ (for $s = m$, by (ii) at $\tau = -1$); this is the claim. Finally $\hat{c}$ has outgoing index J, with $q = r+1$ and $s_0 = m$. □

The lemma holds, with the same proof, for every $j \equiv \omega$ with $j \le 1 - 3r$ (then $m \ge 1$). In the coordinates (τ, k), the incoming kink leaves the vacuum of phase ω on the line $\tau + 3k + 2 = 7$, the outgoing kink leaves the vacuum of phase ω' on the line $3k - \tau + 3 - 3z = 7$, and the edge bit $z = Y_c(m-1, r)$ is read at the single point $(\tau, k) = (-1, 1)$, through $f(z,1,1) = \neg z$.

By Lemma 7.10(c), the backward moves are realised in the same way by ρ: $\bar{M}_m = \rho M_{-m} \rho$, and gluing K_j^- to ρc is gluing $\rho K_j^- = K_{2-j}^+$ to c in reversed time.

7.4. **The first columns.** For $k \ge 1$ consider

(H_k) *for every $h \in \mathcal{H}$, every r and every $c \in \Omega_r$, $hc \in \Omega_r$ has the phase of c, and $Y_{hc}(t,i) = Y_c(t,i)$ for all $t \in \mathbb{Z}$ and $i \ge r - k + 1$.*

Proposition 7.12 ((H_1))**.** (H_1) *holds.*

Proof. Let $h \in \mathcal{H}$ have radius R, let $c \in \Omega_r$ have phase ω, and write $z_t = Y_c(t,r)$, $\vartheta = \vartheta(c)$. By Lemma 7.8 with $\Lambda = V_\omega$, hc agrees with V_ω on $(r+R, \infty)$. Let $j \equiv \omega \pmod 6$ with $j \le -3(r+2R)$, so that $p^-(j) \ge r + 2R + 3$, and $\hat{c} = c \sqcup_r K_j^-$. By Lemma 7.8, $h\hat{c} = hc \sqcup_{r+R} K_j^-$. If $hc = V_\omega$, then $h\hat{c} = K_j^-$, which has no outgoing index; but $F^s h\hat{c} = hF^s\hat{c}$ agrees with $hK_{s+J}^+ = K_{s+J}^+$ on $[q+R, \infty)$ whenever $F^s\hat{c}$ agrees with K_{s+J}^+ on $[q, \infty)$, so $h\hat{c}$ has the outgoing index J of $\hat{c}$, which exists by Lemma 7.11. Hence $hc \in \Omega_{r'}$ with phase ω, for some $r' \le r + R$, and $h\hat{c} = hc \sqcup_{r'} K_j^-$ with $j \le -3r'$. Lemma 7.11 for c and for hc, with $m = 2 - j - 3r$ and $m' = 2 - j - 3r'$, gives for the common outgoing index J of $\hat{c}$ and $h\hat{c}$

$$j + 6r' + 3Y_{hc}(m'-1, r') - 4 = J = j + 6r + 3z_{m-1} - 4.$$

So $6(r'-r) \in \{-3, 0, 3\}$, hence $r' = r$, $m' = m$ and $Y_{hc}(m-1,r) = z_{m-1}$. As j ranges over $j \equiv \omega$, $j \le -3(r+2R)$, the time $t = m-1$ ranges over all $t \equiv 1 - \vartheta \pmod 6$ with $t \ge t_0$, for some t_0. Applying this to $F^{-6i}c$, which lies in Ω_r with phase ω and satisfies $hF^{-6i}c = F^{-6i}hc$, gives $Y_{hc}(t,r) = z_t$ for every $t \equiv 1 - \vartheta$. Next apply the same to $\rho h \rho \in \mathcal{H}$ and $\rho c \in \Omega_r$, which has $\vartheta(\rho c) = 2 - \vartheta$ and edge column $t \mapsto z_{-1-t}$ (Lemma 7.10(c)): $Y_{\rho hc}(t,r) = Y_{\rho c}(t,r)$ for $t \equiv \vartheta - 1$, that is $Y_{hc}(t,r) = z_t$ for $t \equiv -\vartheta$. Both c and hc lie in Ω_r with phase ω, so by Lemma 7.2(c) both

edge columns satisfy $z_{t+1} = \neg z_{t-1}$ whenever $t + \vartheta \equiv 2, 3, 4, 5$. Writing $t + \vartheta \equiv n$, the values with $n = 0, 1$ are known; $n = 3$ follows from $n = 1$, $n = 4$ from $n = 0$ (at $t + \vartheta \equiv 5$), $n = 2$ from $n = 4$, and $n = 5$ from $n = 3$. So the edge columns agree at all times, and to the right of r both diagrams are the vacuum of phase ω. □

Lemma 7.13 (moves commute with $\mathcal{H}$)**.** *For every $h \in \mathcal{H}$, $r \in \mathbb{Z}$, $c \in \Omega_r$ and $m \in \mathbb{Z}$, $hT_m c = T_m hc$.*

Proof. By (H_1), $hc \in \Omega_r$ has the phase and the edge column of c, so T_m is in the same case at c and at hc. Let h have radius R, and let T_m be forward at c, with $m \geq 2 + 6R$. Put $j = 2 - m - 3r \leq -3(r + 2R)$, $\hat{c} = c \sqcup_r K_j^-$, and $D = M_m c$, $D' = M_m hc$. By Lemma 7.8 and (H_1), $h\hat{c} = hc \sqcup_r K_j^-$. By Lemma 7.11 for c and hc, which have the same index J, $F^s\hat{c} = F^s D \sqcup_r K_{s+J}^+$ and $F^s h\hat{c} = F^s D' \sqcup_r K_{s+J}^+$ for large s. For s so large that $q^+(s + J) \geq r + 2R + 1$, Lemma 7.8 gives $F^s h\hat{c} = hF^s\hat{c} = hF^s D \sqcup_{r+R} K_{s+J}^+$. By ($H_1$), $hF^s D \in \Omega_r$ has the phase of $F^s D$, which is that of $F^s D'$. So $F^s D'$ and $hF^s D = F^s hD$ agree on $(-\infty, r]$ and lie in Ω_r with the same phase; hence they are equal, and $hD = D'$. For arbitrary m, Lemma 7.10(c) gives $T_m = F^{6i} T_{m+6i} F^{-6i}$, and h commutes with F. For backward moves use $\bar{M}_m = \rho M_{-m} \rho$ and $\rho h \rho \in \mathcal{H}$. □

Blocks. For $c \in \Omega_r$ and $t \in \mathbb{Z}$ write $n(t) = t + \vartheta(c) \bmod 6$ and

$$z_t = Y_c(t, r), \qquad y_t = Y_c(t, r - 1), \qquad w_t = Y_c(t, r - 2).$$

A *block* of c is a set $\{t_1, t_1 + 1, \ldots, t_1 + 5\}$ of times with $n(t_1) = 1$; the time $t_j = t_1 + j - 1$ is at *position* j, so that position j is $n \equiv j \pmod 6$, and (z_{t_1}, z_{t_2}) is the *type* of the block. We also write $t_0 = t_1 - 1$ and $t_7 = t_6 + 1$, the last time of the previous block and the first of the next. For a configuration D obtained from c by moves, blocks, types and positions refer to $\vartheta(D)$, and z^D, y^D, w^D are the columns $r, r - 1, r - 2$ of Y_D.

Lemma 7.14 (blocks)**.** *Let $c \in \Omega_r$ and let (p, q) be the type of a block of c.*

(a) $(z_{t_1}, \ldots, z_{t_6}) = (p, q, \neg p, \neg q, p, q)$.
(b) *The next block has type (p', q') with $p' = p \oplus (q \wedge \neg y_{t_6})$ and $q' = q \oplus (p' \wedge \neg y_{t_7})$. In particular $p' = 1$ if $(p, q) = (1, 0)$, and the q-value of the previous block is q if $p = 0$.*
(c) $(p, q) \neq (0, 0)$.
(d) *The time t is at position j of a block of type (p, q) of c if and only if $-1 - t$ is at position $7 - j$ of a block of type (q, p) of ρc.*

Proof. (a) By Lemma 7.2(c), $z_{t+1} = \neg z_{t-1}$ at the positions $2, \ldots, 5$. (b) Lemma 7.2(c) at t_6 and t_7, where $n \equiv 0, 1$, gives $z_{t_7} = z_{t_5} \oplus (z_{t_6} \wedge \neg y_{t_6})$ and $z_{t_8} = z_{t_6} \oplus (z_{t_7} \wedge \neg y_{t_7})$, and $z_{t_5} = p$, $z_{t_6} = q$ by (a). For the last statement apply this to the previous block (p_-, q_-): $q = q_- \oplus (p \wedge \neg y_{t_1}) = q_-$ if $p = 0$. (c) If $(p, q) = (0, 0)$, then (b) gives $(p', q') = (0, 0)$, and for the previous block $q_- = q \oplus (p \wedge \neg y_{t_1}) = 0$ and then $p_- = p \oplus (q_- \wedge \neg y_{t_0}) = 0$. So every block has type $(0, 0)$, and by (a) $z_t = 1$ exactly when $n(t) \in \{3, 4\}$. By the last statement of Lemma 7.2(c), this is the y-bit at r of $F^t V_\omega$, ω the phase of c; at $t = -1, 0$ it says $c_r = (V_\omega)_r$, which contradicts $c \in \Omega_r$. (d) By Lemma 7.10(c) and (3), $\vartheta(\rho c) = 2 - \vartheta(c)$ and $Y_{\rho c}(t, i) = Y_c(-1 - t, i)$; so the value of n for ρc at $-1 - t$ is $1 - n(t)$, and the block is traversed in reverse order. □

Lemma 7.15 (one move)**.** *Let $c \in \Omega_r$, let $n(m) \in \{0, 2\}$ and $D = T_m c$.*

(a) *For $j \geq 0$ the rows $m - 1 - j$ and $m + j$ of Y_D and Y_c agree on the columns $\leq r - j$; in particular $z_t^D = z_t$ for $t \in \{m - 1, m\}$, $y_t^D = y_t$ for $m - 2 \leq t \leq m + 1$, and $w_t^D = w_t$ for $m - 3 \leq t \leq m + 2$. For every $h \in \mathcal{H}$ and $j \geq 0$, $Y_{hD}(t, r - j) \oplus Y_D(t, r - j) = Y_{hc}(t, r - j) \oplus Y_c(t, r - j)$ for $m - 1 - j \leq t \leq m + j$.*
(b) *Exactly one of the following holds.*

F$^-$: $n(m)=2$ *and* $z_{m-1}=1$. *Then* $\vartheta(D)=\vartheta(c)-1$, *and* m *is at position* 1 *of a block of* D *of type* $\big(z_m,\ 1\oplus(z_m\wedge\neg y_m)\big)$.

F$^+$: $n(m)=2$ *and* $z_{m-1}=0$. *Then* $\vartheta(D)=\vartheta(c)+2$, *and* m *is at position* 4 *of a block of* D *of type* $(1,\neg z_m)$.

B$^-$: $n(m)=0$ *and* $z_m=0$. *Then* $\vartheta(D)=\vartheta(c)-2$, *and* m *is at position* 4 *of a block of* D *of type* $(\neg z_{m-1},1)$.

B$^+$: $n(m)=0$ *and* $z_m=1$. *Then* $\vartheta(D)=\vartheta(c)+1$, m *is at position* 1 *of a block of* D *of type* $(1,z_{m+1})$, *and* $m-1$ *is at position* 6 *of a block of* D *of type* $\big(1\oplus(z_{m-1}\wedge\neg y_{m-1}),\ z_{m-1}\big)$.

Proof. (a) By Definition 7.9, the rows $m-1$ and m of Y_D and Y_c agree on the columns $\le r$; as $F^{\pm1}$ have radius 1, the rows $m-1-j$ and $m+j$ agree on the columns $\le r-j$. The same holds for hD and hc, since $hD=T_mhc$ by Lemma 7.13. (b) The four cases are those of Definition 7.9, and $\vartheta(D)-\vartheta(c)$ is the change of phase. So the value of n for D at m is $1,4,4,1$ in the four cases. In F$^-$, $z^D_m=z_m$, and Lemma 7.2(c) for D at m gives $z^D_{m+1}=z^D_{m-1}\oplus(z^D_m\wedge\neg y^D_m)=1\oplus(z_m\wedge\neg y_m)$. In F$^+$ and B$^-$, $m-1$ and m are at the positions 3 and 4, where by Lemma 7.14(a) the values are the negations of the type; they are $z_{m-1}=0$ and z_m, respectively z_{m-1} and $z_m=0$. In B$^+$, $z^D_m=z_m=1$, and Lemma 7.2(c) at m, for D (where $n\equiv1$) and for c (where $n\equiv0$), gives $z^D_{m+1}=z_{m-1}\oplus\neg y_m=z_{m+1}$. Finally $m-1$ is at position 6 for D, with $z^D_{m-1}=z_{m-1}$, and Lemma 7.2(c) for D at $m-1$ gives $z^D_{m-2}=z^D_m\oplus(z^D_{m-1}\wedge\neg y^D_{m-1})=1\oplus(z_{m-1}\wedge\neg y_{m-1})$. □

Proposition 7.16 ((H$_2$))**.** (H$_2$) *holds.*

Proof. For $h\in\mathcal{H}$ and $c\in\Omega_r$ put $g_t=g^c_t=Y_{hc}(t,r-1)\oplus Y_c(t,r-1)$. By (H$_1$) it suffices to show $g^c=0$, and Y_{hc} and Y_c agree on the columns $\ge r$ for every c. We use three rules, valid for every $h\in\mathcal{H}$ and $c\in\Omega_r$.

(G1) If $z_t=1$ and $n(t)\in\{0,1\}$, then $g_t=0$: by Lemma 7.2(c) for c and hc, which have the same phase and edge column, $z_{t+1}\oplus z_{t-1}=f(Y_c(t,r-1),1,1)=f(Y_{hc}(t,r-1),1,1)$, and $f(a,1,1)=\neg a$.

(G2) If $z_t=0$, then $g_{t-1}=g_{t+1}$: by (2) at $(t,r-1)$ and $f(a,b,0)=1$, $Y(t+1,r-1)=\neg Y(t-1,r-1)$ for both diagrams.

(G3) If $D=T_mc$, then $g^D_t=g^c_t$ for $m-2\le t\le m+1$, by Lemma 7.15(a).

By Lemma 7.14(c) every block of c has type $(1,1)$, $(1,0)$ or $(0,1)$. We show that g vanishes at every position of every block; t_j denote the times of the block considered.

Step 1. Type $(1,1)$. At t_1 and t_6 by (G1). The move at t_2 is F$^-$, as $z_{t_1}=1$, and t_2 is at position 1 of a block of $D=T_{t_2}c$ of type $(1,\cdot)$; so $g^D_{t_2}=0$ by (G1) for D, and $g_{t_2}=0$ by (G3). The move at t_6 is B$^+$, as $z_{t_6}=1$, and t_5 is at position 6 of a block of $D=T_{t_6}c$ whose q-value is $z_{t_5}=1$; (G1) for D and (G3) give $g_{t_5}=0$. Finally (G2) at t_3 and t_4, where $z=0$, gives $g_{t_4}=g_{t_2}$ and $g_{t_3}=g_{t_5}$.

Step 2. Type $(0,1)$, *positions* 6, 4, 1. At t_6 by (G1), and $g_{t_4}=g_{t_6}$ by (G2) at t_5. The move at t_2 is F$^+$, and t_2 is at position 4, so t_1 at position 3, of a block of $D=T_{t_2}c$ of type $(1,0)$. In D, (G1) at position 1 and (G2) at position 2 give $g^D=0$ at position 3; so $g_{t_1}=0$ by (G3).

Step 3. Type $(1,0)$, *positions* 1, 3, 6. At t_1 by (G1), and $g_{t_3}=g_{t_1}$ by (G2) at t_2. The move at t_6 is B$^-$, and t_6 is at position 4 of a block of $D=T_{t_6}c$ of type $(\neg z_{t_5},1)=(0,1)$; by Step 2 for D and (G3), $g_{t_6}=0$.

Step 4. Type $(1,0)$, *positions* 2, 4, 5. The move at t_2 is F$^-$, and t_2 is at position 1 of a block of $D=T_{t_2}c$ of type $(z_{t_2},1)=(0,1)$; by Step 2 for D and (G3), $g_{t_2}=0$. Then $g_{t_4}=g_{t_2}$ by (G2) at t_3. By (G2) at t_6, $g_{t_5}=g_{t_7}$, and t_7 is at position 1 of the next block, whose p-value is 1 by Lemma 7.14(b); so $g_{t_7}=0$ by (G1).

Step 5. Type $(0,1)$, *positions* 2, 3, 5. Steps 1–4 hold for every $h\in\mathcal{H}$ and every configuration. Apply Steps 3 and 4 to $\rho h\rho\in\mathcal{H}$ and ρc: by Lemma 7.14(d) the block becomes a block of type

$(1,0)$ of ρc, and by (3) the quantity g for $\rho h \rho$ and ρc at $-1-t$ is g^c_t. Steps 3 and 4 are applied to ρc itself, with its own moves; translated back to c, the time t' of ρc is the time $-1-t'$ of c, while a move of ρc at time m' is the move of c at time $-m'$ (Lemma 7.10(c)), and the window $[m'-2, m'+1]$ of (G3) becomes $[-m'-2, -m'+1]$, the window of the move at $-m'$. □

Proposition 7.17 ((H$_3$))**.** (H$_3$) *holds.*

Proof. For $h \in \mathcal{H}$ and $c \in \Omega_r$ put $g_t = g^c_t = Y_{hc}(t, r-2) \oplus Y_c(t, r-2)$. By (H$_2$) it suffices to show $g^c = 0$, and Y_{hc} and Y_c agree on the columns $\geq r-1$ for every c. The rules are now, for every h and c:

(K1) if $y_t = z_t = 1$, then $g_t = 0$, by (2) at $(t, r-1)$ and $f(a,1,1) = \neg a$;
(K2) if $y_t = 0$, then $g_{t-1} = g_{t+1}$, by (2) at $(t, r-2)$ and $f(a,b,0) = 1$;
(K3) if $D = T_m c$, then $g^D_t = g^c_t$ for $m-3 \leq t \leq m+2$, by Lemma 7.15(a).

Moreover, by (2) at (t,r), $(t,r-1)$, $(t,r-2)$ and $f(a,0,c) = \neg c$, $f(0,1,c) = 1$, $f(a,b,0) = 1$, if $y_t = 0$ then

$$\text{(Y)}\ \ y_{t+1} = y_{t-1} \oplus \neg z_t, \qquad \text{(Z)}\ \ z_{t+1} = \neg z_{t-1} \ \text{ if } z_t = 1, \qquad \text{(W)}\ \ w_{t+1} = \neg w_{t-1}.$$

Each of the following steps holds for every $h \in \mathcal{H}$, $c \in \Omega_r$ and t; by Lemma 7.14(d) and (3), applying a step to $\rho h \rho$ and ρc at $-1-t$ gives the mirror statement, in which position j becomes $7-j$ with the type reversed. In it the times are reflected by $t \mapsto -1-t$ and the move times by $m \mapsto -m$ (Lemma 7.10(c)): the time $t+s$ becomes $t-s$, and a move at time $t+s$ becomes a move at time $t+1-s$; the windows of (K3) correspond, since $t \in [m-3, m+2]$ if and only if $-1-t \in [-m-3, -m+2]$. This is what "by ρ" means below.

Step 1. If $y_t = 1$, then $g_t = 0$. A move at a time m with $t-1 \leq m \leq t+2$ does not change y_t and g_t (Lemma 7.15(a)). We show that at most four such moves lead to D with $z^D_t = 1$; then $g_t = g^D_t = 0$ by (K3) and (K1) for D. Let $z_t = 0$. By Lemma 7.14(a), t is at position 3 or 4 of a block of type $(1,1)$, at position 2, 3 or 6 of type $(1,0)$, or at position 1, 4 or 5 of type $(0,1)$. By ρ it suffices to treat the positions 3 of $(1,1)$ and $(1,0)$, 2 of $(1,0)$ and 1 of $(0,1)$. *Position* 3*, type* $(1,q)$. The move at $t-1$ is F$^-$, and t is at position 2 of a block of D of type $\big(q, 1 \oplus (q \wedge \neg y_{t-1})\big)$. So $z^D_t = 1$, unless $q = 1$ and $y_{t-1} = 0$, and then t is at position 2 of a block of D of type $(1,0)$. *Position* 2*, type* $(1,0)$. The move at t is F$^-$, and t is at position 1 of a block of D of type $(0,1)$. *Position* 1*, type* $(0,1)$. The move at $t+1$ is F$^+$, and $t+1$ is at position 4 of a block of D of type $(1,0)$; so t is at position 3 of a block of type $(1,0)$, which is settled by one more move.

Step 2. If $y_t = 0$ and $z_t = 0$, then $g_t = 0$. By (Y), $y_{t+1} \neq y_{t-1}$; by ρ we may assume $y_{t+1} = 0$. Then $g_t = g_{t+2}$ by (K2) at $t+1$, and $y_{t+2} = \neg z_{t+1}$ by (Y) at $t+1$. If $z_{t+1} = 0$, then $g_{t+2} = 0$ by Step 1. If $z_{t+1} = 1$, then (Z) and (Y) at $t+1$, $t+2$, $t+3$ give $z_{t+2} = 1$, $y_{t+2} = y_{t+3} = 0$, $z_{t+3} = 0$ and $y_{t+4} = 1$; so $g_{t+2} = g_{t+4}$ by (K2) at $t+3$, and $g_{t+4} = 0$ by Step 1.

Step 3. If $y_t = 0$, $z_t = 1$ and $y_{t-1}y_{t+1} = 0$, then $g_t = 0$. By (Y), $y_{t-1} = y_{t+1} = 0$, and by (Z), $z_{t-1} \neq z_{t+1}$; by ρ we may assume $z_{t+1} = 0$. Then $g_t = g_{t+2}$ by (K2) at $t+1$, and $y_{t+2} = 1$ by (Y) at $t+1$; so $g_{t+2} = 0$ by Step 1.

Step 4. If $y_t = 0$, $z_t = 1$ and $y_{t-1} = y_{t+1} = 1$, then $g_t = 0$. We find D, obtained from c by one or two moves at times m with $m-3 \leq t \leq m+2$, with $z^D_t = 0$ or $y^D_t = 1$; then $g^D_t = 0$ by Steps 1–3 for D, and $g_t = g^D_t$ by (K3). By (Z) and (W), $z_{t-1} \neq z_{t+1}$ and $w_{t-1} \neq w_{t+1}$; by ρ we may assume $z_{t-1} = 0$ and $z_{t+1} = 1$. By Lemma 7.14(a), t is then at position 1 of a block of type $(1,1)$ (and $z_{t-1} = 0$ is the q-value of the previous block), at position 5 of type $(1,1)$, at position 2 of type $(0,1)$, at position 6 of type $(0,1)$ (and $z_{t+1} = 1$ is the p-value of the next block), or at position 4 of type $(1,0)$. *Position* 1*, type* $(1,1)$. The previous block has type $(1,0)$, by Lemma 7.14(c), so $z_{t-2} = 1$. The move at $t-1$ is B$^-$, and $t-1$ is at position 4 of a block of D of type $(\neg z_{t-2}, 1) = (0,1)$; t is at position 5, so $z^D_t = 0$. *Position* 5*, type* $(1,1)$. By Lemma 7.14(b) the next block has p-value $1 \oplus \neg y_{t+1} = 1$, so the move at $t+3$ is F$^-$. Then $t+3$ is at position 1

of a block of D, so $t+2$ is at position 6 of a block of D whose q-value is $z^D_{t+2} = z_{t+2} = 1$, and t, at position 4 of that block, has $z^D_t = 0$. *Position* 2, *type* $(0,1)$. By Lemma 7.14(b) the previous block has q-value 1, so the move at $t-2$ is B^+, and $t-2$ is at position 1 of a block of D of type $(1, z_{t-1}) = (1,0)$; t is at position 3, so $z^D_t = 0$. *Position* 6, *type* $(0,1)$. The move at t is B^+, and $t-1$ is at position 6 of a block of D of type $\big(1 \oplus (z_{t-1} \wedge \neg y_{t-1}), z_{t-1}\big) = (1,0)$. In D the move at $t-1$ is B^-, and $t-1$ is at position 4 of a block of $D' = T_{t-1}D$ of type $(\neg z^D_{t-2}, 1) = (0,1)$; t is at position 5, so $z^{D'}_t = 0$. *Position* 4, *type* $(1,0)$. The move at $t-2$ is F^-, and $t-2$ is at position 1 of a block of D of type $(z_{t-2}, 1) = (0,1)$; so $z^D_{t-1} = z^D_t = 1$ and $z^D_{t+1} = 0$. By (2) at $(t-1, r-1)$ for D and for c, with $y^D_{t-2} = y_{t-2}$, $y^D_{t-1} = y_{t-1} = 1$ and $w^D_{t-1} = w_{t-1}$,

$$y^D_t = y_{t-2} \oplus f(w_{t-1}, 1, 1) = y_{t-2} \oplus \neg w_{t-1}, \qquad 0 = y_t = y_{t-2} \oplus f(w_{t-1}, 1, 0) = y_{t-2} \oplus 1,$$

so $y^D_t = w_{t-1}$. If $w_{t-1} = 1$, then $y^D_t = 1$. If $w_{t-1} = 0$, then $y^D_t = 0$, $z^D_t = 1$, $y^D_{t-1} = 1$, and $w^D_{t+1} = \neg w^D_{t-1} = 1$ by (W) for D: so (D,t) is in the situation of Step 4 with $z^D_{t-1} = 1$, $z^D_{t+1} = 0$, at position 3 of a block of type $(0,1)$, and with $w^D_{t+1} = 1$. This is the mirror of the case just treated with $w_{t-1} = 1$; so one more move, at $t+3$, gives D' with $y^{D'}_t = 1$.

By Steps 1–4, $g_t = 0$ for every t. □

7.5. **Controllability.** For $k \geq 1$ and $c \in \Omega_r$ let $\mathcal{O}_k(c)$ be the set of configurations $T_{m_n} \cdots T_{m_1} c$ with $n \geq 0$ and $|m_1|, \dots, |m_n| \leq k$. As in Section 4.5, c_{r-k+1} is the tip and c_{r-k} the exterior cell of c at level k.

The orbits at level 3 are almost those of Theorem 7.20. We determine them in Proposition 7.19, from an explicit description of the moves at the times $|m| \leq 3$.

Notation at level 3. For $c \in \Omega_r$ write $a_t = Y_c(t, r-3)$, in addition to z_t, y_t, w_t; so

$$c_{r-3} = (a_{-1}, a_0), \qquad c_{r-2} = (w_{-1}, w_0), \qquad c_{r-1} = (y_{-1}, y_0), \qquad c_r = (z_{-1}, z_0).$$

A cell (x,y) is also written as the string xy, and c is abbreviated to

$$\vartheta \,|\, \tau\, \mu\, \varepsilon \qquad \text{with} \qquad \vartheta = \vartheta(c), \quad \tau = c_{r-2}, \quad \mu = c_{r-1}, \quad \varepsilon = c_r.$$

This determines c when its cells on $(-\infty, r-3]$ are given, and by Lemma 7.10(b) no move at a time $|m| \leq 3$ changes these cells. By the last statement of Lemma 7.2(c), the cell at r of the vacuum of c is 00, 00, 00, 01, 11, 10 for $\vartheta = 0, \dots, 5$, and ε can be any other cell; so there are eighteen pairs (ϑ, ε). For bits u, v the bracket $[uv \neq 01]$ is 0 if $(u,v) = (0,1)$ and 1 otherwise; by (12),

$$f(u,v,0) \oplus f(u,v,1) = [uv \neq 01], \qquad f(u,v,1) = \neg u \wedge v. \tag{14}$$

For $c, D \in \Omega_r$ we write $c \xrightarrow{m} D$ if $D = T_m c \neq c$, and we say that c *reaches* D if $D \in \mathcal{O}_3(c)$; this relation is transitive.

Mirror images. By Lemma 7.10(c), ρ maps Ω_r onto itself, it exchanges the two bits of every cell, $\vartheta(\rho c) = 2 - \vartheta(c)$, and $\bar{M}_m = \rho M_{-m} \rho$, hence also $M_m = \rho \bar{M}_{-m} \rho$. Moreover $-m + \vartheta(\rho c) = 2 - (m + \vartheta(c))$, so T_{-m} is backward, forward or the identity at ρc exactly when T_m is forward, backward or the identity at c. So $T_{-m} = \rho T_m \rho$ for every m, and $c \xrightarrow{m} D$ if and only if $\rho c \xrightarrow{-m} \rho D$. Therefore every statement below that holds for all $c \in \Omega_r$ has a *mirror image*, which also holds for all $c \in \Omega_r$: in it ϑ is replaced by $2 - \vartheta$, the two bits of every cell are exchanged, including those of c_{r-3}, and T_m is replaced by T_{-m}. We write $\bar{\tau}$ for the cell τ with its two bits exchanged.

Lemma 7.18 (the moves at the times $0, \dots, 3$)**.** *Let $c \in \Omega_r$, let $m \in \{0,1,2,3\}$ with $n(m) \in \{0,2\}$, and let $D = T_m c$. Write ϑ', z'_t, y'_t, w'_t for the quantities ϑ, z_t, y_t, w_t of D, and put*

$$y_1 = y_{-1} \oplus f(w_0, y_0, z_0), \qquad w_1 = w_{-1} \oplus f(a_0, w_0, y_0),$$

which are $Y_c(1, r-1)$ *and* $Y_c(1, r-2)$ *by* (2).

(a) *The value* ϑ'*, the edge cell* $D_r = (z'_{-1}, z'_0)$ *and, for* $m = 3$*, the value* z'_1 *are given by the following table. In it* $z_1 = z_{-1} \oplus (z_0 \wedge \neg y_0)$ *if* $\vartheta = 0$*, and* $z_1 = \neg z_{-1}$ *if* $\vartheta \in \{3, 5\}$*;* $z_2 = z_0 \oplus (z_1 \wedge \neg y_1)$ *if* $\vartheta = 5$*; and* $\pi = 1 \oplus (\neg z_0 \wedge y_0)$*.*

m	ϑ	*case*	ϑ'	D_r	
0	2	$z_{-1} = 0$	4	c_r	
0	2	$z_{-1} = 1$	1	c_r	
0	0	$z_0 = 0$	4	c_r	
0	0	$z_0 = 1$	1	c_r	
1	1	$z_0 = 0$	3	00	
1	1	$z_0 = 1$	0	c_r	
1	5	$z_{-1} = 1$	3	c_r	
1	5	$z_{-1} = 0$	0	$\big(1 \oplus (z_0 \wedge \neg y_0),\, z_0\big)$	
2	0	$z_1 = 0$	2	10	
2	0	$z_1 = 1$	5	$(0, z_0)$	
2	4	$z_0 = 1$	2	c_r	
2	4	$z_0 = 0$	5	$\big(z_{-1},\, 1 \oplus (\neg z_{-1} \wedge y_{-1})\big)$	
3	5	$z_2 = 0$	1	$(\neg y'_0, 1)$	$z'_1 = 0,\ z_1 = 1$
3	5	$z_2 = 1$	4	$(z_{-1}, 0)$	$z'_1 = z_1$
3	3	$z_{-1} = 0$	1	10	$z'_1 = z_1$
3	3	$z_{-1} = 1$	4	$(\neg\pi, z_0)$	$z'_1 = \pi,\ z_1 = 0$

(b) D *agrees with* c *on* $(-\infty, r-m]$*. If* $m = 2$*, or if* $m = 3$ *and* $z'_1 = z_1$*, then* $D_{r-2} = c_{r-2}$*,* $y'_0 = y_0$ *and*

$$y'_{-1} = y_{-1} \oplus \big([w_0 y_0 \neq 01] \wedge (z_0 \oplus z'_0)\big).$$

If $m = 3$ *and* $z'_1 \neq z_1$*, then, with* $\delta = [w_1 y_1 \neq 01]$*,*

$$y'_0 = y_0 \oplus \delta, \qquad w'_0 = w_0, \qquad w'_{-1} = w_{-1} \oplus \big(\delta \wedge [a_0 w_0 \neq 01]\big), \qquad y'_{-1} = y_1 \oplus f(w_0, y'_0, z'_0).$$

Proof. (b) The first statement is Lemma 7.10(b). By Lemma 7.15(a), for $j \geq 0$ the row $m-1-j$ of Y_D agrees with that of Y_c on the columns $\leq r-j$, and the row m on the columns $\leq r$. Let $m = 2$. Then $y'_0 = y_0$, $y'_1 = y_1$ and $w'_0 = w_0$, and (2) at $(0, r-1)$, for D and for c, gives $y'_{-1} \oplus y_{-1} = f(w_0, y_0, z'_0) \oplus f(w_0, y_0, z_0)$, which is $[w_0 y_0 \neq 01] \wedge (z_0 \oplus z'_0)$ by (14). Let $m = 3$. Then $y'_1 = y_1$, $y'_2 = y_2$, $w'_0 = w_0$, $w'_1 = w_1$ and $Y_D(0, r-3) = a_0$. By (2) at $(1, r-1)$, $(0, r-2)$ and $(0, r-1)$, for D and for c,

$$y'_0 \oplus y_0 = [w_1 y_1 \neq 01] \wedge (z_1 \oplus z'_1), \quad w'_{-1} \oplus w_{-1} = [a_0 w_0 \neq 01] \wedge (y_0 \oplus y'_0), \quad y'_{-1} = y_1 \oplus f(w_0, y'_0, z'_0).$$

If $z'_1 = z_1$, then $y'_0 = y_0$ and $w'_{-1} = w_{-1}$, and $y'_{-1} \oplus y_{-1}$ is found as for $m = 2$. If $z'_1 \neq z_1$, these are the formulas of (b).

(a) For $m = 0$ this is Definition 7.9. Let $m \geq 1$. The value of ϑ' is given by Lemma 7.15(b), and the case is decided by z_{m-1} if $n(m) = 2$ and by z_m if $n(m) = 0$. By Lemma 7.14(a), at the positions $1, \ldots, 6$ of a block of type (p, q) the edge column is $p, q, \neg p, \neg q, p, q$, for c and for D. For c this gives: if $\vartheta = 5$, the times $-1, 0, 1$ are at the positions $4, 5, 6$, so $z_1 = \neg z_{-1}$; if $\vartheta = 4$, the times $-1, \ldots, 2$ are at the positions $3, \ldots, 6$, so $z_1 = \neg z_{-1}$ and $z_2 = \neg z_0$; if $\vartheta = 3$, the times $-1, \ldots, 3$ are at the positions $2, \ldots, 6$, so $z_1 = \neg z_{-1}$, $z_2 = \neg z_0$ and $z_3 = z_{-1}$. By Lemma 7.2(c), $z_1 = z_{-1} \oplus (z_0 \wedge \neg y_0)$ if $\vartheta = 0$, and $z_2 = z_0 \oplus (z_1 \wedge \neg y_1)$ if $\vartheta = 5$. So the cases of the table are those of Lemma 7.15(b), in the order F^+, F^- for $(m, \vartheta) = (1, 1), (2, 0), (3, 5)$ and B^-, B^+ for $(m, \vartheta) = (1, 5), (2, 4), (3, 3)$. We determine z'_t for $m - 4 \leq t \leq m - 1$ in the four cases; recall $z'_{m-1} = z_{m-1}$ and $z'_m = z_m$.

F^+. Here $z_{m-1}=0$, so $z_m=1$ by Lemma 7.14(c), and m is at position 4 of a block of D of type $(1,0)$; so $(z'_{m-3},\dots,z'_m)=(1,0,0,1)$. This gives $D_r=00$ for $m=1$ and $D_r=10$ for $m=2$. For $m=3$ it gives $z'_0=1$ and $z'_1=0$, and Lemma 7.2(c) for D at $t=0$, where the value of n for D is $\vartheta'=1$, gives $z'_1=z'_{-1}\oplus(z'_0\wedge\neg y'_0)$, so $z'_{-1}=\neg y'_0$. Moreover $z_1=1$: the block of c that begins at time 2 has type $(0,1)$, so by Lemma 7.14(b) the previous block, which ends at time 1, has q-value 1.

F^-. Here $z_{m-1}=1$, and m is at position 1 of a block of D, so $m-1$ is at position 6 of the previous block of D, whose q-value is $z'_{m-1}=1$. Lemma 7.2(c) at $t=m-1$, for D (where $n=0$) and for c (where $n=1$), gives $z'_m=z'_{m-2}\oplus(z'_{m-1}\wedge\neg y'_{m-1})$ and the same relation without primes; as $y'_{m-1}=y_{m-1}$, it follows that $z'_{m-2}=z_{m-2}$. So that block of D has type $(z_{m-2},1)$, and $(z'_{m-4},z'_{m-3},z'_{m-2})=(\neg z_{m-2},0,z_{m-2})$. This gives $D_r=c_r$ for $m=1$, $D_r=(0,z_0)$ for $m=2$, and $D_r=(\neg z_1,0)=(z_{-1},0)$, $z'_1=z_1$ for $m=3$.

B^-. Here $z_m=0$, so $z_{m-1}=1$ by Lemma 7.14(c), and m is at position 4 of a block of D of type $(0,1)$; so $(z'_{m-3},\dots,z'_m)=(0,1,1,0)$, and $z'_{m-4}=1$, the q-value of the previous block of D (Lemma 7.14(b)). For c, m is at position 6 of a block of type $(1,0)$, so $(z_{m-4},\dots,z_m)=(0,0,1,1,0)$. Hence $D_r=c_r$ for $m=1,2$, and $D_r=10$, $z'_1=z_1$ for $m=3$.

B^+. Here $z_m=1$, and $m-1$ is at position 6 of a block of D of type (π_m,z_{m-1}) with $\pi_m=1\oplus(z_{m-1}\wedge\neg y_{m-1})$; so $(z'_{m-4},\dots,z'_{m-1})=(\neg\pi_m,\neg z_{m-1},\pi_m,z_{m-1})$. For $m=1$, $D_r=(\pi_1,z_0)$. For $m=2$, $D_r=(\neg z_1,\pi_2)=(z_{-1},\pi_2)$; here $z_0=\neg z_2=0$, so $y_1=y_{-1}\oplus f(w_0,y_0,0)=\neg y_{-1}$ and $\pi_2=1\oplus(\neg z_{-1}\wedge y_{-1})$. For $m=3$, $D_r=(\neg\pi_3,\neg z_2)=(\neg\pi_3,z_0)$ and $z'_1=\pi_3$; here $z_1=\neg z_{-1}=\neg z_3=0$, so $y_2=y_0\oplus f(w_1,y_1,0)=\neg y_0$ and $\pi_3=1\oplus(\neg z_0\wedge y_0)=\pi$. □

Let $\mathcal{E}$ be the set of configurations $b\in\Omega_r$, for any r, with

$$b_{r-2}=(0,0),\qquad b_{r-1}=(1,1),\qquad (\vartheta(b),b_r)\in\big\{(0,(1,1)),\,(1,(1,1)),\,(2,(1,1)),\,(3,(1,0)),\,(5,(0,1))\big\}.$$

Proposition 7.19 ((F^*_3))**.** *Let $c\in\Omega_r$. Then $\mathcal{O}_3(c)$ contains every $c'\in\Omega_r$, of any phase, with $c'\notin\mathcal{E}$, $c'_i=c_i$ for $i\le r-3$, and, if $c_{r-3}=(0,0)$, $c'_{r-2}=(1,1)$ if and only if $c_{r-2}=(1,1)$.*

Proof. The proof has four steps. Twelve of the eighteen pairs (ϑ,ε) occur in sixteen *sectors* of twelve configurations each, and any two configurations of a sector reach each other (Step 1). A configuration with one of the other six pairs reaches a sector (Step 2) and, unless it lies in $\mathcal{E}$, is reached from one (Step 3). Moves from sector to sector, mainly the moves at the times ±3 applied to the configurations with $\vartheta\in\{3,5\}$ and edge cell 11, join the sectors as far as the condition on $(1,1)$ allows (Step 4).

All configurations in this proof lie in Ω_r and agree with c on $(-\infty,r-3]$. Every arrow stated below is proved for arbitrary cells on $(-\infty,r-3]$; so its mirror image holds as well, for arbitrary cells on $(-\infty,r-3]$. In a configuration $\vartheta\,|\,\tau\,\nu\,\varepsilon$ we always write $\tau=(w_{-1},w_0)$, $\nu=(y_{-1},y_0)$, $\varepsilon=(z_{-1},z_0)$, and $(a_{-1},a_0)=c_{r-3}$; the middle cell is called ν or μ. For cells $\tau=(w_{-1},w_0)$ and $\nu=(y_{-1},y_0)$ put

$$\nu^x=\big(y_{-1}\oplus[w_0y_0\ne01],\ y_0\big),\qquad \nu^y=\big(y_{-1},\ y_0\oplus[w_{-1}y_{-1}\ne01]\big).$$

These cells depend on τ, which is always the cell written to the left of ν. Since ν^x has the y-bit of ν and ν^y has its x-bit, $(\nu^x)^x=\nu$ and $(\nu^y)^y=\nu$. The mirror image of the operation $\nu\mapsto\nu^x$ is $\nu\mapsto\nu^y$.

The arrows of Steps 1–3. Table 1 lists the arrows that are used in Steps 1–3. Each of them holds for all cells τ, ν that satisfy the condition in its third column, and is read off Lemma 7.18: the fourth column names the row of the table of the lemma, by (m,ϑ) and the case, and the last column evaluates this row. An arrow at a time $m<0$ is the mirror image of the arrow at the time $-m$ that is displayed in the last column, and the fourth column names the row for the displayed arrow. The cells c_{r-2} and c_{r-1} are found from Lemma 7.18(b). If $m\le1$, they do not

change. If $m = 2$, or if $m = 3$ and $z_1' = z_1$, then τ does not change, and ν is replaced by ν^x if z_0 changes and does not change otherwise. In the two arrows with $m = 3$ and $z_1' \neq z_1$, which are (25) and (26), the last column evaluates the second set of formulas of Lemma 7.18(b).

Step 1: sectors. For cells τ, ν let

$$\begin{aligned}
\mathsf{H}(\tau,\nu) &= \{3\,|\,\tau\,\nu\,11,\ 5\,|\,\tau\,\nu\,11\},\\
\mathsf{A}(\tau,\nu) &= \{4\,|\,\tau\,\nu\,10,\ 0\,|\,\tau\,\nu\,10,\ 5\,|\,\tau\,\nu\,00,\ 1\,|\,\tau\,\nu\,01,\ 0\,|\,\tau\,\nu\,01\},\\
\mathsf{D}(\tau,\nu) &= \{4\,|\,\tau\,\nu\,01,\ 2\,|\,\tau\,\nu\,01,\ 3\,|\,\tau\,\nu\,00,\ 1\,|\,\tau\,\nu\,10,\ 2\,|\,\tau\,\nu\,10\}.
\end{aligned}$$

The mirror image of $\mathsf{A}(\tau,\nu)$ is $\mathsf{D}(\bar\tau,\bar\nu)$, and that of $\mathsf{H}(\tau,\nu)$ is $\mathsf{H}(\bar\tau,\bar\nu)$.

The arrows of (15) form a closed chain through the five configurations of $\mathsf{A}(\tau,\nu)$; so these reach each other, and by the mirror image so do those of $\mathsf{D}(\tau,\nu)$. By (16), applied to $\nu = \mu^x$ and to $\nu = \mu$, the configurations of $\mathsf{H}(\tau,\mu)$ and of $\mathsf{A}(\tau,\mu^x)$ reach each other, and by the mirror image so do those of $\mathsf{H}(\tau,\mu)$ and of $\mathsf{D}(\tau,\mu^y)$. Hence any two of the twelve configurations of the *sector*

$$\Sigma(\tau,\mu) = \mathsf{H}(\tau,\mu) \cup \mathsf{A}(\tau,\mu^x) \cup \mathsf{D}(\tau,\mu^y)$$

reach each other. The mirror image of $\Sigma(\tau,\mu)$ is $\Sigma(\bar\tau,\bar\mu)$. Twelve of the eighteen pairs (ϑ,ε) occur in H, A, D, and a configuration $\vartheta\,|\,\tau\,\nu\,\varepsilon$ with one of them lies in a sector: $\mathsf{H}(\tau,\nu) \subseteq \Sigma(\tau,\nu)$, $\mathsf{A}(\tau,\nu) \subseteq \Sigma(\tau,\nu^x)$ and $\mathsf{D}(\tau,\nu) \subseteq \Sigma(\tau,\nu^y)$. The other six pairs are $(0,11)$, $(1,11)$, $(2,11)$, $(5,01)$, $(3,10)$ and $(4,00)$; we call a configuration with one of them *irregular*, and put $\mathsf{C}(\tau,\nu) = \{0\,|\,\tau\,\nu\,11, 1\,|\,\tau\,\nu\,11, 2\,|\,\tau\,\nu\,11\}$. Thus $\mathcal{E}$ consists of $\mathsf{C}(00,11)$, $5\,|\,00\,11\,01$ and $3\,|\,00\,11\,10$.

Step 2: every irregular configuration reaches a configuration in a sector. Let an irregular configuration be given.

- $4\,|\,\tau\,\nu\,00$. If $y_{-1} = 1$, (21) leads to $5\,|\,\tau\,\nu\,00 \in \mathsf{A}(\tau,\nu)$. If $y_0 = 1$, the mirror image of (21) leads to $3\,|\,\tau\,\nu\,00 \in \mathsf{D}(\tau,\nu)$. If $\nu = 00$, then $\nu^x = 10$, and (21) and (19) lead to $5\,|\,\tau\,10\,01$ and to $0\,|\,\tau\,10\,01 \in \mathsf{A}(\tau,10)$.
- $5\,|\,\tau\,\nu\,01$. If $y_0 = 0$, (19) leads to $0\,|\,\tau\,\nu\,01 \in \mathsf{A}(\tau,\nu)$. If $y_{-1} = 1$, the mirror image of (20) leads to $4\,|\,\tau\,\nu\,01 \in \mathsf{D}(\tau,\nu)$. If $\nu = 01$, (19) leads to $0\,|\,\tau\,\nu\,11 \in \mathsf{C}(\tau,01)$, treated below. The configurations $3\,|\,\tau\,\nu\,10$ are the mirror images of these.
- $\mathsf{C}(\tau,\nu)$. By (17) its three configurations reach each other. If $y_0 = 0$, (18) leads to $2\,|\,\tau\,\nu^x\,10 \in \mathsf{D}(\tau,\nu^x)$. If $y_{-1} = 0$, the mirror image of (18) leads from $2\,|\,\tau\,\nu\,11$ to $0\,|\,\tau\,\nu^y\,01 \in \mathsf{A}(\tau,\nu^y)$. If $\nu = 11$, (18) and the mirror image of (20) lead to $5\,|\,\tau\,\nu\,01$ and to $4\,|\,\tau\,\nu\,01 \in \mathsf{D}(\tau,\nu)$.

Step 3: every irregular configuration outside $\mathcal{E}$ is reached from a configuration in a sector. Call a configuration *good* if it is reached from a configuration in a sector; configurations in sectors are good, and a configuration reached from a good one is good.

- $4\,|\,\tau\,\nu\,00$ is good. If $y_{-1} = 1$, by (22) from $5\,|\,\tau\,\nu\,00 \in \mathsf{A}(\tau,\nu)$; if $y_0 = 1$, by the mirror image; and if $\nu = 00$, by (24) and (23) from $0\,|\,\tau\,10\,01 \in \mathsf{A}(\tau,10)$, through $5\,|\,\tau\,10\,01$.
- $5\,|\,\tau\,\nu\,01$ with $y_0 = 0$ is good, by (24); and so is its mirror image, $3\,|\,\tau\,\nu\,10$ with $y_{-1} = 0$.
- If $y_0 = 1$, then $0\,|\,\tau\,\nu\,11 \xrightarrow{2} 5\,|\,\tau\,\nu\,01 \xrightarrow{1} 0\,|\,\tau\,\nu\,11$ by (18) and (19); with (17), the four configurations of $\mathsf{C}(\tau,\nu) \cup \{5\,|\,\tau\,\nu\,01\}$ reach each other. By the mirror image, if $y_{-1} = 1$, the four configurations of $\mathsf{C}(\tau,\nu) \cup \{3\,|\,\tau\,\nu\,10\}$ reach each other. So it remains to show that $\mathsf{C}(\tau,\nu)$ contains a good configuration for every $(\tau,\nu) \neq (00,11)$; the remaining irregular configurations, those of $\mathsf{C}(00,11)$, $5\,|\,00\,11\,01$ and $3\,|\,00\,11\,10$, form $\mathcal{E}$.
- $\nu = 11$ and $w_0 = 1$, or $\nu = 01$ and $w_0 = 0$. The configuration $4\,|\,\tau\,01\,00$ is good, and by (21) and (19) it reaches $5\,|\,\tau\,\nu\,01$ and $0\,|\,\tau\,\nu\,11$ with $\nu = (01)^x$, which is 11 if $w_0 = 1$ and 01 if $w_0 = 0$.
- $\nu = 10$ and $w_{-1} = 1$. By (25), from $5\,|\,\tau\,00\,00 \in \mathsf{A}(\tau,00)$.

	arrow	condition	row	evaluation
(15)	$4\,\vert\,\tau\,\nu\,10 \xrightarrow{-2} 0\,\vert\,\tau\,\nu\,10$		$(2,4)$, $z_0=1$	mirror image of $4\,\vert\,\bar\tau\,\bar\nu\,01 \xrightarrow{2} 2\,\vert\,\bar\tau\,\bar\nu\,01$, where $D_r=c_r$
	$0\,\vert\,\tau\,\nu\,10 \xrightarrow{2} 5\,\vert\,\tau\,\nu\,00$		$(2,0)$, $z_1=1$	$z_1=z_{-1}\oplus(z_0\wedge\neg y_0)=1$; $D_r=(0,z_0)=00$, and z_0 does not change
	$5\,\vert\,\tau\,\nu\,00 \xrightarrow{-3} 1\,\vert\,\tau\,\nu\,01$		$(3,3)$, $z_{-1}=0$	mirror image of $3\,\vert\,\bar\tau\,\bar\nu\,00 \xrightarrow{3} 1\,\vert\,\bar\tau\,\bar\nu\,10$, where $D_r=10$, $z_1'=z_1$, and z_0 does not change
	$1\,\vert\,\tau\,\nu\,01 \xrightarrow{1} 0\,\vert\,\tau\,\nu\,01$		$(1,1)$, $z_0=1$	$D_r=c_r$
	$0\,\vert\,\tau\,\nu\,01 \xrightarrow{0} 1\,\vert\,\tau\,\nu\,01$		$(0,0)$, $z_0=1$	$D_r=c_r$
	$1\,\vert\,\tau\,\nu\,01 \xrightarrow{-1} 5\,\vert\,\tau\,\nu\,00$		$(1,1)$, $z_0=0$	mirror image of $1\,\vert\,\bar\tau\,\bar\nu\,10 \xrightarrow{1} 3\,\vert\,\bar\tau\,\bar\nu\,00$, where $D_r=00$
	$5\,\vert\,\tau\,\nu\,00 \xrightarrow{1} 0\,\vert\,\tau\,\nu\,10$		$(1,5)$, $z_{-1}=0$	$D_r=(1\oplus(0\wedge\neg y_0),0)=10$
	$0\,\vert\,\tau\,\nu\,10 \xrightarrow{0} 4\,\vert\,\tau\,\nu\,10$		$(0,0)$, $z_0=0$	$D_r=c_r$
(16)	$4\,\vert\,\tau\,\nu\,10 \xrightarrow{2} 5\,\vert\,\tau\,\nu^x\,11$		$(2,4)$, $z_0=0$	$D_r=(1,1\oplus(\neg1\wedge y_{-1}))=11$, and z_0 changes
	$5\,\vert\,\tau\,\nu\,11 \xrightarrow{3} 4\,\vert\,\tau\,\nu^x\,10$		$(3,5)$, $z_2=1$	$z_1=\neg z_{-1}=0$ and $z_2=z_0\oplus(z_1\wedge\neg y_1)=1$; $D_r=(z_{-1},0)=10$, $z_1'=z_1$, and z_0 changes
	$5\,\vert\,\tau\,\nu\,11 \xrightarrow{1} 3\,\vert\,\tau\,\nu\,11$		$(1,5)$, $z_{-1}=1$	$D_r=c_r$
	$3\,\vert\,\tau\,\nu\,11 \xrightarrow{-1} 5\,\vert\,\tau\,\nu\,11$		$(1,5)$, $z_{-1}=1$	mirror image of $5\,\vert\,\bar\tau\,\bar\nu\,11 \xrightarrow{1} 3\,\vert\,\bar\tau\,\bar\nu\,11$
(17)	$0\,\vert\,\tau\,\nu\,11 \xrightarrow{0} 1\,\vert\,\tau\,\nu\,11$		$(0,0)$, $z_0=1$	$D_r=c_r$
	$1\,\vert\,\tau\,\nu\,11 \xrightarrow{-1} 2\,\vert\,\tau\,\nu\,11$		$(1,1)$, $z_0=1$	mirror image of $1\,\vert\,\bar\tau\,\bar\nu\,11 \xrightarrow{1} 0\,\vert\,\bar\tau\,\bar\nu\,11$
	$2\,\vert\,\tau\,\nu\,11 \xrightarrow{0} 1\,\vert\,\tau\,\nu\,11$		$(0,2)$, $z_{-1}=1$	$D_r=c_r$
	$1\,\vert\,\tau\,\nu\,11 \xrightarrow{1} 0\,\vert\,\tau\,\nu\,11$		$(1,1)$, $z_0=1$	$D_r=c_r$
(18)	$0\,\vert\,\tau\,\nu\,11 \xrightarrow{2} 2\,\vert\,\tau\,\nu^x\,10$	$y_0=0$	$(2,0)$, $z_1=0$	$z_1=1\oplus(1\wedge\neg y_0)=y_0$; $D_r=10$, and z_0 changes
	$0\,\vert\,\tau\,\nu\,11 \xrightarrow{2} 5\,\vert\,\tau\,\nu\,01$	$y_0=1$	$(2,0)$, $z_1=1$	$z_1=y_0$; $D_r=(0,z_0)=01$
(19)	$5\,\vert\,\tau\,\nu\,01 \xrightarrow{1} 0\,\vert\,\tau\,\nu\,\varepsilon'$		$(1,5)$, $z_{-1}=0$	$\varepsilon'=D_r=(1\oplus(1\wedge\neg y_0),1)=(y_0,1)$
(20)	$3\,\vert\,\tau\,\nu\,10 \xrightarrow{3} 4\,\vert\,\tau\,\nu\,10$	$y_0=1$	$(3,3)$, $z_{-1}=1$	$\pi=1\oplus(\neg z_0\wedge y_0)=0$; $D_r=(\neg\pi,z_0)=10$, $z_1'=\pi=0=z_1$, and z_0 does not change
(21)	$4\,\vert\,\tau\,\nu\,00 \xrightarrow{2} 5\,\vert\,\tau\,\nu\,00$	$y_{-1}=1$	$(2,4)$, $z_0=0$	$D_r=(0,1\oplus(\neg0\wedge y_{-1}))=00$, and z_0 does not change
	$4\,\vert\,\tau\,\nu\,00 \xrightarrow{2} 5\,\vert\,\tau\,\nu^x\,01$	$y_{-1}=0$	$(2,4)$, $z_0=0$	$D_r=(0,1\oplus(\neg0\wedge y_{-1}))=01$, and z_0 changes
(22)	$5\,\vert\,\tau\,\nu\,00 \xrightarrow{3} 4\,\vert\,\tau\,\nu\,00$	$y_{-1}=1$	$(3,5)$, $z_2=1$	$z_1=\neg z_{-1}=1$, $y_1=y_{-1}\oplus f(w_0,y_0,0)=\neg y_{-1}=0$ and $z_2=z_0\oplus\neg y_1=1$; $D_r=(z_{-1},0)=00$, $z_1'=z_1$, and z_0 does not change
(23)	$5\,\vert\,\tau\,10\,01 \xrightarrow{3} 4\,\vert\,\tau\,00\,00$		$(3,5)$, $z_2=1$	$z_1=1$, $y_1=1\oplus f(w_0,0,1)=1$ and $z_2=z_0\oplus\neg y_1=1$; $D_r=00$, $z_1'=z_1$, z_0 changes, and $\nu^x=(1\oplus[w_0\,0\neq01],0)=00$
(24)	$0\,\vert\,\tau\,\nu\,01 \xrightarrow{2} 5\,\vert\,\tau\,\nu\,01$	$y_0=0$	$(2,0)$, $z_1=1$	$z_1=0\oplus(1\wedge\neg y_0)=1$; $D_r=(0,z_0)=01$
(25)	$5\,\vert\,\tau\,00\,00 \xrightarrow{3} 1\,\vert\,\tau\,10\,11$	$w_{-1}=1$	$(3,5)$, $z_2=0$	$z_1=1$, $y_1=f(w_0,0,0)=1$ and $z_2=z_0\oplus\neg y_1=0$; so $z_1'=0\neq z_1$. $w_1=w_{-1}\oplus f(a_0,w_0,0)=0$ and $\delta=[01\neq01]=0$; hence $y_0'=0$, $w_{-1}'=w_{-1}$, $D_r=(\neg y_0',1)=11$ and $y_{-1}'=y_1\oplus f(w_0,0,1)=1$
(26)	$5\,\vert\,\tau\,\nu\,01 \xrightarrow{3} 1\,\vert\,\tau'\,00\,11$	$\nu=(\neg w_0,1)$	$(3,5)$, $z_2=0$	$z_1=1$, $y_1=\neg w_0\oplus f(w_0,1,1)=0$ and $z_2=z_0\oplus\neg y_1=0$; so $z_1'=0\neq z_1$. $\delta=[w_1\,0\neq01]=1$; hence $y_0'=0$, $\tau'=(w_{-1}\oplus[a_0w_0\neq01],\,w_0)$, $D_r=(\neg y_0',1)=11$ and $y_{-1}'=y_1\oplus f(w_0,0,1)=0$

TABLE 1. The arrows of Steps 1–3 in the proof of Proposition 7.19. The eight arrows of (15) form a closed chain, and so do the four arrows of (17).

- The mirror images of the last two items are: $\nu = 11$ and $w_{-1} = 1$; $\nu = 10$ and $w_{-1} = 0$; $\nu = 01$ and $w_0 = 1$. So $\mathsf{C}(\tau, \nu)$ contains a good configuration whenever $\nu \in \{01, 10\}$, and whenever $\nu = 11$ and $\tau \neq 00$; and then, by the third item, $5\,|\,\tau\,\nu\,01$ is good if $y_0 = 1$.
- $\nu = 00$. Let $\tau^\circ = (u, w_0)$ with $(u, w_0) \neq (0, 0)$. Then $5\,|\,\tau^\circ\,\mu\,01$ with $\mu = (\neg w_0, 1)$ is good, by the last item, since $(\tau^\circ, \mu) \neq (00, 11)$; and by (26) it reaches $1\,|\,\tau'\,00\,11 \in \mathsf{C}(\tau', 00)$ with $\tau' = (u \oplus [a_0 w_0 \neq 01], w_0)$. For $w_0 = 1$ and $u \in \{0, 1\}$ these are the cells $\tau' = 01$ and $\tau' = 11$; for $w_0 = 0$ and $u = 1$, $\tau' = 00$. The mirror image gives $\tau' = 10$.

Step 4: from sector to sector. For sectors write $\Sigma \to \Sigma'$ if a configuration of Σ reaches a configuration of Σ'; then, by Step 1, every configuration of Σ reaches every configuration of Σ', and $\to$ is transitive. We write $\tau.\mu$ for $\Sigma(\tau, \mu)$. We claim

$$0\,|\,\tau\,\nu\,01 \xrightarrow{2} 2\,|\,\tau\,\nu^x\,10 \quad \text{if } y_0 = 1; \tag{27}$$

$$3\,|\,\tau\,\mu\,11 \xrightarrow{3} 4\,|\,\tau'\,\mu'\,01, \tag{28}$$

where, for $\tau = (w_{-1}, w_0)$ and $\mu = (y_{-1}, y_0)$: if $w_0 = 0$, then $(\tau', \mu') = (\tau, \mu)$ if $w_{-1} = y_{-1} \neq y_0$, and $\tau' = (\neg w_{-1}, 0)$, $\mu' = (\neg y_{-1}, \neg y_0)$ otherwise; if $w_0 = 1$, then $(\tau', \mu') = (\tau, \mu)$ if $y_{-1} = 1$ and $w_{-1} = \neg(a_0 \wedge y_0)$, and $\tau' = (w_{-1} \oplus a_0, 1)$, $\mu' = (y_{-1}, \neg y_0)$ otherwise. Indeed, (27) is the row $(2, 0)$ with $z_1 = 0 \oplus (1 \wedge \neg y_0) = 0$. In (28), the row $(3, 3)$ with $z_{-1} = 1$ has $\pi = 1$, so $z'_1 = 1 \neq z_1$ and $D_r = 01$, and by Lemma 7.18(b) and (14), with $y_1 = y_{-1} \oplus (\neg w_0 \wedge y_0)$, $w_1 = w_{-1} \oplus f(a_0, w_0, y_0)$ and $\delta = [w_1 y_1 \neq 01]$,

$$y'_0 = y_0 \oplus \delta, \qquad y'_{-1} = y_{-1} \oplus (\delta \wedge \neg w_0), \qquad w'_{-1} = w_{-1} \oplus \big(\delta \wedge [a_0 w_0 \neq 01]\big).$$

If $w_0 = 0$, then $y_1 = y_{-1} \oplus y_0$ and $w_1 = w_{-1} \oplus \neg y_0$, so $\delta = 0$ exactly when $w_{-1} = y_{-1} = \neg y_0$, and $[a_0 w_0 \neq 01] = 1$. If $w_0 = 1$, then $y_1 = y_{-1}$ and $w_1 = w_{-1} \oplus \neg(a_0 \wedge y_0)$, so $\delta = 0$ exactly when $y_{-1} = 1$ and $w_{-1} = \neg(a_0 \wedge y_0)$, and $[a_0 w_0 \neq 01] = a_0$.

These give four relations between sectors. In (27) let $\nu = \mu^x$, where $\mu = (y_{-1}, y_0)$ has $y_0 = 1$; then $0\,|\,\tau\,\nu\,01 \in \mathsf{A}(\tau, \mu^x) \subseteq \Sigma(\tau, \mu)$ and $2\,|\,\tau\,\mu\,10 \in \mathsf{D}(\tau, \mu) \subseteq \Sigma(\tau, \mu^y)$. In (28), $4\,|\,\tau'\,\mu'\,01 \in \mathsf{D}(\tau', \mu') \subseteq \Sigma(\tau', (\mu')^y)$, where $(\mu')^y$ is formed with τ'; put $J_{a_0}(\tau, \mu) = \tau'.(\mu')^y$. So, with the mirror images,

(S) $\tau.\mu \to \tau.\mu^y$ if $y_0 = 1$; (S̄) $\tau.\mu \to \tau.\mu^x$ if $y_{-1} = 1$;

(J) $\tau.\mu \to J_{a_0}(\tau, \mu)$; (J̄) $\tau.\mu \to \overline{J_{a_{-1}}(\bar\tau, \bar\mu)}$,

where the bar exchanges the two bits of both cells. By the definition of ν^y, $(\mu')^y = (y'_{-1},\ y'_0 \oplus [w'_{-1} y'_{-1} \neq 01])$ for $\tau' = (w'_{-1}, w'_0)$ and $\mu' = (y'_{-1}, y'_0)$: it is μ' with its second bit changed, unless $(w'_{-1}, y'_{-1}) = (0, 1)$, and then it is μ'. The following table gives, for each $\tau.\mu$, the cells $\tau'\,\mu'$ and, after the colon, the sector $J_{a_0}(\tau, \mu) = \tau'.(\mu')^y$; for $w_0 = 0$ they do not depend on a_0.

$\tau \setminus \mu$	00	01	10	11
00	10 11: 10.10	00 01: 00.00	10 01: 10.00	10 00: 10.01
10	00 11: 00.11	00 10: 00.10	10 10: 10.11	00 00: 00.01
01 ($a_0 = 0$)	01 01: 01.00	01 00: 01.01	01 11: 01.11	01 10: 01.10
11 ($a_0 = 0$)	11 01: 11.00	11 00: 11.01	11 10: 11.11	11 11: 11.10
01 ($a_0 = 1$)	11 01: 11.00	11 00: 11.01	11 11: 11.10	01 11: 01.11
11 ($a_0 = 1$)	01 01: 01.00	01 00: 01.01	11 10: 11.11	01 10: 01.10

For example, for $\tau.\mu = 00.00$ the exception $w_{-1} = y_{-1} \neq y_0$ does not hold, so $(\tau', \mu') = (10, 11)$, and $(\mu')^y = (1, 1 \oplus [11 \neq 01]) = 10$; for $\tau.\mu = 00.01$ it holds, so $(\tau', \mu') = (00, 01)$, and $(\mu')^y = (0, 1 \oplus [00 \neq 01]) = 00$.

For every c_{r-3}, the relations (J) and ($\bar{\text{J}}$) give the closed walk

$$00.00 \xrightarrow{\text{J}} 10.10 \xrightarrow{\text{J}} 10.11 \xrightarrow{\text{J}} 00.01 \xrightarrow{\bar{\text{J}}} 01.00 \xrightarrow{\bar{\text{J}}} 00.11 \xrightarrow{\text{J}} 10.01 \xrightarrow{\text{J}} 00.10 \xrightarrow{\bar{\text{J}}} 00.00,$$

which uses only the first two rows of the table: its steps ($\bar{\text{J}}$) are the mirror images of $00.10 \to 10.00$, $10.00 \to 00.11$ and $00.01 \to 00.00$. Its mirror image is the closed walk

$$00.00 \xrightarrow{\bar{\text{J}}} 01.01 \xrightarrow{\bar{\text{J}}} 01.11 \xrightarrow{\bar{\text{J}}} 00.10 \xrightarrow{\text{J}} 10.00 \xrightarrow{\text{J}} 00.11 \xrightarrow{\bar{\text{J}}} 01.10 \xrightarrow{\bar{\text{J}}} 00.01 \xrightarrow{\text{J}} 00.00.$$

Together they pass through all twelve sectors $\tau.\mu$ with $\tau \neq 11$; so any two of these satisfy $\Sigma \to \Sigma'$. If $a_0 = 1$, the last two rows of the table give the closed walks

$$01.00 \xrightarrow{\text{J}} 11.00 \xrightarrow{\text{J}} 01.00, \qquad 01.01 \xrightarrow{\text{J}} 11.01 \xrightarrow{\text{J}} 01.01, \qquad 01.10 \xrightarrow{\text{J}} 11.10 \xrightarrow{\text{J}} 11.11 \xrightarrow{\text{J}} 01.10,$$

so each of the four sectors with $\tau = 11$ lies on a closed walk through a sector with $\tau = 01$; if $a_{-1} = 1$, the mirror images of these walks do the same with $\tau = 10$. Finally let $c_{r-3} = 00$ and $\tau = 11$. Then

$$3\,|\,\tau\,01\,00 \xrightarrow{-3} 4\,|\,\tau\,01\,00 \xrightarrow{2} 5\,|\,\tau\,11\,01 \xrightarrow{-3} 4\,|\,\tau\,11\,01$$

by the mirror image of (22), by (21) with $(01)^x = 11$, and by the mirror image of (20); the first configuration lies in $\mathsf{D}(\tau, 01) \subseteq \Sigma(\tau, 00)$ and the last in $\mathsf{D}(\tau, 11) \subseteq \Sigma(\tau, 10)$, as $(01)^y = 00$ and $(11)^y = 10$ for $\tau = 11$. With the table, ($\bar{\text{S}}$) and (S) this gives the closed walk

$$11.00 \to 11.10 \xrightarrow{\text{J}} 11.11 \xrightarrow{\bar{\text{S}}} 11.01 \xrightarrow{\text{S}} 11.00.$$

To sum up: if $c_{r-3} \neq 00$, any two of the sixteen sectors satisfy $\Sigma \to \Sigma'$; if $c_{r-3} = 00$, this holds for any two of the twelve sectors with $\tau \neq 11$, and for any two of the four sectors with $\tau = 11$.

Conclusion. Let c' be as in the proposition. By Step 2, c lies in a sector or reaches a configuration in one; let this sector be $\tau_1.\mu_1$. By Step 3, c' lies in a sector or is reached from a configuration in one; let this sector be $\tau_2.\mu_2$. If $c_{r-3} \neq 00$, then $\tau_1.\mu_1 \to \tau_2.\mu_2$ by Step 4, and c reaches c'. Let $c_{r-3} = 00$. By Lemma 7.18(b), a move at a time $0 \leq m \leq 3$ does not change w_0, and it changes w_{-1} only if $[a_0 w_0 \neq 01] = 1$, that is, here, only if $w_0 = 0$; by the mirror image, a move at a time $-3 \leq m \leq 0$ does not change w_{-1}, and changes w_0 only if $w_{-1} = 0$. So no move at a time $|m| \leq 3$ changes whether the cell $r-2$ is 11. Hence $\tau_1 = 11$ if and only if $c_{r-2} = 11$, and $\tau_2 = 11$ if and only if $c'_{r-2} = 11$; by the assumption on c', $\tau_1 = 11$ if and only if $\tau_2 = 11$. So again $\tau_1.\mu_1 \to \tau_2.\mu_2$ by Step 4, and c reaches c'. □

The exceptions are genuine: *if $c \in \Omega_r$, $|m| \leq 3$ and $T_m c \in \mathcal{E}$, then $c \in \mathcal{E}$*; so $\mathcal{O}_3(c) \cap \mathcal{E} = \emptyset$ for $c \notin \mathcal{E}$. Since $\mathcal{E}$ is its own mirror image, it suffices to take $0 \leq m \leq 3$ and $D = T_m c \neq c$ in $\mathcal{E}$, and to go through the table of Lemma 7.18; recall that $D_{r-2} = 00$, $D_{r-1} = 11$, and that (ϑ', D_r) is $(0, 11)$, $(1, 11)$, $(2, 11)$, $(3, 10)$ or $(5, 01)$. The rows with $\vartheta' = 4$ do not occur; nor do those with $(\vartheta', D_r) = (3, 00)$, $(2, 10)$, $(1, 10)$, nor the row $(3, 5)$ with $z_2 = 0$, where $D_r = (\neg y'_0, 1) = 01$. In the rows $(1, 5)$ with $z_{-1} = 1$ and $(2, 4)$ with $z_0 = 1$, $D_r = c_r$ would be 10 with $\vartheta = 5$, respectively 11 with $\vartheta = 4$, the cell of the vacuum. In the rows with $m = 0$ and $\vartheta' = 1$, and in the row $(1, 1)$ with $z_0 = 1$, no cell changes, and c is $2\,|\,00\,11\,11$, $0\,|\,00\,11\,11$ or $1\,|\,00\,11\,11$. In the row $(1, 5)$ with $z_{-1} = 0$, $D_r = 11$ forces $z_0 = 1$, and $c = 5\,|\,00\,11\,01$. In the row $(2, 0)$ with $z_1 = 1$, $D_r = (0, z_0) = 01$ and $z'_0 = z_0$, so $c_{r-1} = 11$ and $z_{-1} = z_1 = 1$, and $c = 0\,|\,00\,11\,11$. In the row $(2, 4)$ with $z_0 = 0$, $D_r = 01$ forces $z_{-1} = 0$ and $y_{-1} = 0$, and then $y'_{-1} = y_{-1} \oplus [w_0 y_0 \neq 01] = [01 \neq 01] = 0$, which is not the case. So the five configurations of $\mathcal{E}$ with given cells on $(-\infty, r-3]$ leave their class but cannot be reached from it, and (F_3^{93}) fails exactly by $\mathcal{E}$; the condition on $(1, 1)$ is invariant, as was shown at the end of the proof.

Theorem 7.20. *Let $k \geq 4$ and $c \in \Omega_r$. Then $\mathcal{O}_k(c)$ is the set of $c' \in \Omega_r$, of any phase, with*

(i) *$c'_i = c_i$ for every $i \leq r - k$; and*

(ii) *if the exterior cell* c_{r-k} *is* $(0,0)$*, then* $c'_{r-k+1} = (1,1)$ *if and only if* $c_{r-k+1} = (1,1)$.

In particular $c \in \mathcal{O}_k(c')$ *for every* $c' \in \mathcal{O}_k(c)$.

We call the statement for a given k (F_k^{93}). It has the form of Theorem 4.10, with every phase of the vacuum allowed.

Proof. The base case $k = 4$. Let $\mathcal{C}$ be the set of $b \in \Omega_r$, of any phase, with $b = c$ on $(-\infty, r-4]$, let $\mathcal{S} \subseteq \mathcal{C}$ be the set described by (i) and (ii), and write $b_{r-4+j} = (u_j, v_j)$ for $j = 0, \dots, 4$, so that $(u_0, v_0) = c_{r-4}$, (u_1, v_1) is the tip and $b_r = (u_4, v_4)$ the edge cell. Write $b \to b'$ if $b' \in \mathcal{O}_4(b)$; this relation is transitive.

Invariance. Every element of $\mathcal{O}_4(c)$ lies in $\mathcal{S}$. Property (i) is Lemma 7.10(b). For (ii), let $u_0 = v_0 = 0$ and consider one move at a time $|m| \le 4$ applied to some $b \in \mathcal{C}$. By Lemma 7.10(b) the tip changes only if $|m| = 4$. By Lemma 7.15(a) (with $j = 3$ and $j = 2$), a move at time 4 changes neither $v_1 = Y_b(0, r-3)$ nor $Y_b(1, r-3) = u_1 \oplus f(0, v_1, v_2)$, and $f(0,1,c) = 1$; so if $v_1 = 1$, then u_1 does not change. By Lemma 7.10(c), a move at time -4 is the ρ-conjugate of one at time 4; it changes neither u_1 nor, if $u_1 = 1$, v_1. So $u_1 v_1$ is invariant.

Transfer. By Lemma 7.10(c), $F^{-1} T_m F = T_{m+1}$, so $\mathcal{O}_3(b) \subseteq \mathcal{O}_4(b)$, $F^{-1}\mathcal{O}_3(Fb) \subseteq \mathcal{O}_4(b)$, and $\rho\,\mathcal{O}_4(b) = \mathcal{O}_4(\rho b)$. The cells $r-4$, $r-3$, $r-2$ of $Fb \in \Omega_r$ are $(v_0, \cdot)$, (v_1, α) and (v_2, β), where $\alpha = u_1 \oplus f(v_0, v_1, v_2)$ and $\beta = u_2 \oplus f(v_1, v_2, v_3)$. Let $\tilde{b} \in \Omega_r$, of any phase, agree with Fb on $(-\infty, r-3]$, with $\tilde{b}_{r-2} = (x, x')$ and $\tilde{b}_{r-1} = (x'', \cdot)$. Then $b' = F^{-1}\tilde{b} \in \mathcal{C}$ by Lemma 7.2(b), the rows 0 and 1 of $Y_{b'}$ are the x- and y-rows of $\tilde{b}$, and (2) at $(0, r-3)$ and $(0, r-2)$ gives

$$b'_{r-3} = \big(\alpha \oplus f(v_0, v_1, x),\, v_1\big), \qquad b'_{r-2} = \big(x' \oplus f(v_1, x, x''),\, x\big).$$

By Proposition 7.19 for Fb, whose exterior cell and tip at level 3 are (v_1, α) and (v_2, β), $b \to b'$ whenever $\tilde{b} \notin \mathcal{E}$ and, if $(v_1, \alpha) = (0,0)$, $[\tilde{b}_{r-2} = (1,1)] = [(v_2, \beta) = (1,1)]$.

Step 1 (same tip). Let $b, b' \in \mathcal{C}$ have the same tip t, and if $t = (0,0)$ let $[b_{r-2} = (1,1)] = [b'_{r-2} = (1,1)]$. Then $b \to b'$. If $b' \notin \mathcal{E}$, this is Proposition 7.19 for b, whose exterior cell at level 3 is t. Let $b' \in \mathcal{E}$, so $(u'_2, v'_2) = (0,0)$ and $(u'_3, v'_3) = (1,1)$, and let b'' be b with its cell $r-2$ replaced by $(1,0)$; then $b \to b''$, as $b'' \notin \mathcal{E}$. Since $v'_2 = v''_2 = 0$ and the tips agree, Fb' and Fb'' agree on $(-\infty, r-3]$; their cells $r-2$ are $(0, \cdot) \ne (1,1)$, and $(Fb')_{r-1} = (1,\, 1 \oplus f(0,1,v'_4)) = (1,0)$, so $Fb' \notin \mathcal{E}$. By the transfer with $\tilde{b} = Fb'$, $b'' \to b'$.

Step 2 (tip $(u_1, 0) \to (\neg u_1, 0)$*).* Every $b \in \mathcal{S}$ with $v_1 = 0$ has $b \to b'$ for some b' with tip $(\neg u_1, 0)$; if $u_1 = 1$, b' can be chosen with $b'_{r-2} = (1,1)$, and also with $b'_{r-2} \ne (1,1)$. By Step 1 we may first change b_{r-2}: if $u_1 = 0$ and $(u_2, v_2) = (0,1)$ we replace it by $(1,0)$, and if $u_1 = 1$ we choose v_2 freely. Now $\alpha = u_1 \oplus f(v_0, 0, v_2) = u_1 \oplus \neg v_2$. If $(v_1, \alpha) = (0,0)$, the tip (v_2, β) of Fb is not $(1,1)$: $v_2 = 1$ and $\beta = u_2 \oplus f(0,1,v_3) = \neg u_2 = 1$ would mean $(u_2, v_2) = (0,1)$ and $\alpha = u_1 = 0$, which we excluded. Take $\tilde{b}$ with $\tilde{b}_{r-2} = (\neg v_2, v_2)$, which is neither $(0,0)$ nor $(1,1)$, and the other cells to the right of $r-3$ and the phase arbitrary. The transfer gives $b \to b'$ with $b'_{r-3} = (\alpha \oplus f(v_0, 0, \neg v_2), 0) = (\neg u_1, 0)$ and $b'_{r-2} = (v_2 \oplus f(0, \neg v_2, x''), \neg v_2)$, which is $(1,1)$ if $v_2 = 0$ and $(x'', 0)$ if $v_2 = 1$.

Step 3 (tip $(0, v_1) \to (0, \neg v_1)$*).* The ρ-image of Step 2: every $b \in \mathcal{S}$ with $u_1 = 0$ has $b \to b'$ with tip $(0, \neg v_1)$, and if $v_1 = 1$ with either value of $[b'_{r-2} = (1,1)]$.

Step 4. If $v_0 = 1$, every $b \in \mathcal{S}$ with $v_1 = 1$ has $b \to b'$ with tip $(\neg u_1, 1)$: here $\alpha = u_1 \oplus f(1,1,v_2) = u_1 \oplus \neg v_2$, the exterior cell $(1, \alpha)$ of Fb is not $(0,0)$, and $\tilde{b}_{r-2} = (\neg v_2, 1)$ gives $b'_{r-3} = (\alpha \oplus f(1,1,\neg v_2), 1) = (\neg u_1, 1)$. If $u_0 = 1$, the ρ-image gives $b \to b'$ with tip $(1, \neg v_1)$ for every $b \in \mathcal{S}$ with $u_1 = 1$.

Conclusion. For a tip t let $A_t = \{b \in \mathcal{S} :\ b_{r-3} = t\}$, and split $A_{00} = A_{00}^1 \cup A_{00}^0$ according to whether $b_{r-2} = (1,1)$. By Step 1, $b \to b'$ for any two elements of one of A_{01}, A_{10}, A_{11}, A_{00}^0, A_{00}^1. By Step 2 every element of A_{00} reaches A_{10} and every element of A_{10} reaches both A_{00}^0 and A_{00}^1;

by Step 3 every element of A_{00} reaches A_{01} and every element of A_{01} reaches both. So $b \to b'$ for all $b, b' \in A_{00} \cup A_{01} \cup A_{10}$. If $c_{r-4} = (0,0)$, $\mathcal{S}$ is this union or A_{11}. Otherwise $v_0 = 1$ or $u_0 = 1$, $\mathcal{S}$ also contains A_{11}, and Step 4 gives $b \to b'$ in both directions between A_{11} and A_{01} (if $v_0 = 1$) or A_{10} (if $u_0 = 1$). So $b \to b'$ for all $b, b' \in \mathcal{S}$; with the invariance, $\mathcal{O}_4(c) = \mathcal{S}$, which is (F_4^{93}).

The step. Let $k \geq 5$ and assume (F_{k-1}^{93}). Let $\mathcal{C}$ be the set of $b \in \Omega_r$, of any phase, with $b = c$ on $(-\infty, r-k]$. For $b \in \mathcal{C}$ write $b_{r-k+j} = (u_j, v_j)$ for $j = 0, \ldots, 3$, where $(u_0, v_0) = c_{r-k}$, and $\pi(b) = (u_1, v_1, u_2, v_2, u_3, v_3)$; as $k \geq 5$, the cells $r-k+1, \ldots, r-k+4$ are not the edge cell. For $\ell \in \{-1, 0, 1\}$ let $\sim_\ell$ be the relation on $\mathcal{C}$ of reachability by moves at the times $|m - \ell| \leq k-1$. By Lemma 7.10(c), $b \sim_\ell b'$ if and only if $F^\ell b' \in \mathcal{O}_{k-1}(F^\ell b)$, so by (F_{k-1}^{93}) each $\sim_\ell$ is an equivalence relation. The three windows together are $|m| \leq k$, so $\mathcal{O}_k(c)$ is the class of c under the equivalence relation generated by $\sim_{-1}$, $\sim_0$, $\sim_1$.

We claim that $b \sim_\ell b'$ if and only if $(\pi(b), \pi(b')) \in \mathcal{R}_\ell$, where, writing ab for $a \wedge b$ and $f = f_{93}$, the relations $\mathcal{R}_\ell \subseteq \{0,1\}^6 \times \{0,1\}^6$, which depend on (u_0, v_0), are defined as follows, with $\pi(b) = (u_1, \ldots, v_3)$ and $\pi(b') = (u_1', \ldots, v_3')$:

- $\mathcal{R}_0$: $(u_1', v_1') = (u_1, v_1)$, and (u_2', v_2', u_3', v_3') is arbitrary, except that $u_2'v_2' = u_2v_2$ if $(u_1, v_1) = (0,0)$.
- $\mathcal{R}_1$: with $\alpha = u_1 \oplus f(v_0, v_1, v_2)$ and $\beta = u_2 \oplus f(v_1, v_2, v_3)$: $v_1' = v_1$, $u_1' = \alpha \oplus f(v_0, v_1, v_2')$ and $u_2' = \beta' \oplus f(v_1, v_2', v_3')$ for some $\beta' \in \{0,1\}$, with v_2', v_3', β', u_3' arbitrary, except that $v_2'\beta' = v_2\beta$ if $(v_1, \alpha) = (0,0)$.
- $\mathcal{R}_{-1}$: with $\gamma = v_1 \oplus f(u_0, u_1, u_2)$ and $\delta = v_2 \oplus f(u_1, u_2, u_3)$: $u_1' = u_1$, $v_1' = \gamma \oplus f(u_0, u_1, u_2')$ and $v_2' = \delta' \oplus f(u_1, u_2', u_3')$ for some $\delta' \in \{0,1\}$, with $u_2', u_3', \delta', v_3'$ arbitrary, except that $\delta' u_2' = \delta u_2$ if $(\gamma, u_1) = (0,0)$.

For $\ell = 0$ this is (F_{k-1}^{93}) applied to b, whose exterior cell and tip at level $k-1$ are (u_1, v_1) and (u_2, v_2); for $\ell = 1$ it is (F_{k-1}^{93}) applied to Fb, whose exterior cell and tip at level $k-1$ are (v_1, α) and (v_2, β), translated back by F^{-1} as follows; $\ell = -1$ is the same with the rows exchanged.

The configuration $F^\ell b$ lies in Ω_r, its cells are $(Y_b(\ell - 1, i), Y_b(\ell, i))$, and its exterior cell and tip at level $k-1$ are those used in the definition of $\mathcal{R}_\ell$. For $\ell = 1$, with $\alpha = u_1 \oplus f(v_0, v_1, v_2)$ as in $\mathcal{R}_1$, F^{-1} maps the set of configurations $\tilde{b}$, of any phase, in Ω_r, that agree with Fb on $(-\infty, r-k+1]$ bijectively onto the set of $b' \in \mathcal{C}$ with $v_1' = v_1$ and $u_1' = \alpha \oplus f(v_0, v_1, v_2')$. Indeed, if $b' \in \mathcal{C}$, then $Fb' \in \Omega_r$ by Lemma 7.2(b), Fb' agrees with Fb on $(-\infty, r-k-1]$, and its cells $r-k$ and $r-k+1$ are $\big(v_0,\ u_0 \oplus f(Y_b(0, r-k-1), v_0, v_1')\big)$ and $\big(v_1',\ u_1' \oplus f(v_0, v_1', v_2')\big)$, while those of Fb are the same with v_1, u_1, v_2 in place of v_1', u_1', v_2'; so Fb' agrees with Fb on $(-\infty, r-k+1]$ if and only if $v_1' = v_1$ and $u_1' = \alpha \oplus f(v_0, v_1, v_2')$. Conversely, if $\tilde{b}$ is such, then $b' = F^{-1}\tilde{b} \in \Omega_r$ by Lemma 7.2(b), and b' agrees with b on $(-\infty, r-k]$ because F^{-1} has radius 1, so $b' \in \mathcal{C}$. Row 0 of b' is the row of x-bits of $\tilde{b}$, and $Y_{b'}(-1, i) = Y_{b'}(1, i) \oplus f(Y_{b'}(0, i-1), Y_{b'}(0, i), Y_{b'}(0, i+1))$; at $i = r-k+2$ this gives the formula of $\mathcal{R}_1$ for u_2', and for $i \geq r-k+3$, for given row 0, it is a bijection between the y-bits of $\tilde{b}_i$ and the x-bits of b_i'. The phase of $\tilde{b}$ is free, and so is that of b', and $\tilde{b}_r$ differs from its vacuum cell if and only if b_r' does, by Lemma 7.2(b). So (F_{k-1}^{93}) for Fb says that $b \sim_1 b'$ exactly when $(\pi(b), \pi(b')) \in \mathcal{R}_1$, the cells $r-k+4, \ldots, r$ of b' and its phase being free subject to $b' \in \Omega_r$. The cases $\ell = 0$ and $\ell = -1$ are the same. Hence abstract paths lift to chains of moves and chains of moves project to abstract paths, and $\mathcal{O}_k(c) = \{c' \in \mathcal{C} :\ \pi(c') \approx \pi(c)\}$, where $\approx$ is the equivalence relation generated by the $\mathcal{R}_\ell$.

We determine the classes of $\approx$; they depend on (u_0, v_0). We use $f(a, 0, c) = \neg c$, $f(a, 1, 1) = \neg a$, $f(0, 1, c) = 1$ and $f(1, 1, c) = \neg c$, all from (12). Write states as $(u_1, v_1, u_2, v_2, u_3, v_3)$, and divide $\{0,1\}^6$ into the five blocks

$$C_{ab} = \{(u_1, v_1) = (a, b)\} \quad \big((a,b) \neq (0,0)\big), \qquad C_{00}^e = \{(u_1, v_1) = (0,0),\ u_2v_2 = e\} \quad (e \in \{0,1\}).$$

By $\mathcal{R}_0$, any two states of the same block are related, so each block lies in one class. The following pairs are related, for every (u_0, v_0) unless stated otherwise; the free bits $v_2', \beta', \dots$ are those of the definitions of $\mathcal{R}_{\pm 1}$.

(E1) C_{10} and both blocks C_{00}^0, C_{00}^1, by $\mathcal{R}_1$. Let $(u_1, v_1) = (1, 0)$; then $\alpha = 1 \oplus f(v_0, 0, v_2) = v_2$ and $u_1' = \alpha \oplus \neg v_2'$. If $v_2 = 1$, then $(v_1, \alpha) \neq (0, 0)$, there is no exception, and $v_2' = 0$ gives $(u_1', v_1') = (0, 0)$ with $u_2' v_2' = 0$. If $v_2 = 0$, then $(v_1, \alpha) = (0, 0)$ and the exception requires $v_2' \beta' = v_2 \beta = 0$; take $v_2' = 1$ and $\beta' = 0$, so $u_1' = 0$ and $u_2' = \beta' \oplus f(0, 1, v_3') = 1$, a state of C_{00}^1.

(E2) C_{01} and C_{00}^0, by $\mathcal{R}_{-1}$. Let $(u_1, v_1) = (0, 1)$ and $u_2 = 1$; then $\gamma = 1 \oplus f(u_0, 0, 1) = 1$, there is no exception, and $v_1' = 1 \oplus f(u_0, 0, u_2') = u_2'$; take $u_2' = 0$.

(E3) If $u_0 = 1$: C_{10} and C_{11}, by $\mathcal{R}_{-1}$. Let $(u_1, v_1) = (1, 0)$; as $u_1 = 1$ there is no exception, and $v_1' = \gamma \oplus f(1, 1, u_2') = \neg u_2 \oplus \neg u_2'$; take $u_2' = \neg u_2$.

(E4) If $v_0 = 1$: C_{01} and C_{11}, by $\mathcal{R}_1$. Let $(u_1, v_1) = (0, 1)$; as $v_1 = 1$ there is no exception, and $u_1' = \alpha \oplus f(1, 1, v_2') = \neg v_2 \oplus \neg v_2'$; take $v_2' = \neg v_2$.

By (E1) and (E2), the four blocks other than C_{11} lie in one class. If $(u_0, v_0) \neq (0, 0)$, then (E3) or (E4) joins C_{11} to them, and there is one class. If $(u_0, v_0) = (0, 0)$, then $u_1 v_1$ is invariant: with $v_0 = 0$ and $v_1 = 1$, $\mathcal{R}_1$ gives $u_1' = u_1 \oplus f(0, 1, v_2) \oplus f(0, 1, v_2') = u_1$, and with $u_0 = 0$ and $u_1 = 1$, $\mathcal{R}_{-1}$ gives $v_1' = v_1$; and $\mathcal{R}_0$ keeps (u_1, v_1). So the classes are $C_{11} = \{u_1 v_1 = 1\}$ and the union $\{u_1 v_1 = 0\}$ of the other four blocks. With the lifting above, this gives (F_k^{93}). □

The base case cannot be lowered to $k = 3$, but it fails there only by the set $\mathcal{E}$ of Proposition 7.19.

Corollary 7.21. (H_k) *holds for every* $k \geq 1$.

Proof. By Propositions 7.12, 7.16 and 7.17 it suffices to show that (H_k) implies (H_{k+1}) for $k \geq 3$. Let $h \in \mathcal{H}$, $c \in \Omega_r$ and $t \in \mathbb{Z}$; replacing c by $F^t c$ we may take $t = 0$. There is a composite O of moves at the times $|m| \leq k$ with $Y_{Oc}(0, r-k+1) = Y_{Oc}(0, r-k+2) = 1$: take $c' \in \mathcal{C}$ with the phase of c, with c'_{r-k+1} equal to $(1, 1)$ if $c_{r-k} = (0, 0)$ and $c_{r-k+1} = (1, 1)$ and to $(0, 1)$ otherwise, with $c'_{r-k+2} = (0, 1)$, and with $c'_i = c_i$ for $i \geq r - k + 3$; then $c' \in \mathcal{O}_k(c)$ by Theorem 7.20 if $k \geq 4$, and by Proposition 7.19 if $k = 3$, since $c'_{r-1} = (0, 1)$ and so $c' \notin \mathcal{E}$. By Lemma 7.10(b), Oc and Ohc agree with c and with hc on $(-\infty, r-k]$, and by Lemma 7.13, $Ohc = hOc$. Put $Y = Y_{Oc}$ and $G' = Y_{hOc} \oplus Y$. Then $G'(0, r-k) = Y_{hc}(0, r-k) \oplus Y_c(0, r-k)$, and by (H_k) for Oc, G' vanishes on the columns $\geq r - k + 1$ at all times. The relation (2) for both diagrams at $(0, r-k+1)$ gives, as in the proof of Lemma 4.6,

$$0 = f\big(Y(0, r-k) \oplus G'(0, r-k), 1, 1\big) \oplus f\big(Y(0, r-k), 1, 1\big),$$

and $f(a, 1, 1) = \neg a$ gives $G'(0, r-k) = 0$, that is $Y_{hc}(0, r-k) = Y_c(0, r-k)$. □

7.6. Conclusion for the class of 79.

Proof of Lemma 7.7. Let $h \in \mathcal{H}$ have local rule γ of radius R, let $w = (w_{-R}, \dots, w_R) \in \mathcal{A}^{2R+1}$, and let c equal w on $[-R, R]$ and V_0 elsewhere. If $c = V_0$, then $\gamma(w) = (hV_0)_0 = (V_0)_0 = w_0$. Otherwise $c \in \Omega_r$ for some $r \in [-R, R]$, and (H_{R+1}) (Corollary 7.21) gives $(hc)_i = c_i$ for $i \geq r - R$, in particular for $i = 0$. So $\gamma(w) = (hc)_0 = w_0$ for every w, and $h = \mathrm{id}$. □

Theorem 7.22. $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2$ *for* $f \in \{79, 93, 186, 242\}$, *and* $(a, b) \mapsto \sigma^a F_f^b$ *is an isomorphism.*

Proof. Let $g \in \mathrm{Aut}(X_{F_{93}})$. By Proposition 7.6, $h = F^{-3b} \sigma^{-a} F^{-j} g$ fixes K^- and K^+, so $h = \mathrm{id}$ by Lemma 7.7, and $g = F^j \sigma^a F^{3b} \in \langle \sigma, F \rangle$. If $\sigma^a F^b = \mathrm{id}$, then F^b fixes the constants, so $b = 3n$, and $\sigma^a F^{3n}$ maps K^- to $\sigma^{a+n} K^-$ and K^+ to $\sigma^{a-n} K^+$; as neither kink is periodic, $a + n = a - n = 0$. For 79, 186 and 242 use the conjugacies of Section 2.3. □

No computer is used in the proof for the class of 79. Proposition 7.19 is used twice, for (H_4) in Corollary 7.21 and for the base case (F_4^{93}).

8. Discussion

8.1. **Scope of the method.** The proof uses three properties of a rule, beyond the normalisation by offsets. (1) A *stuck edge*: a spatially periodic background B (the constant $(0,0)^{\mathbb{Z}}$ for 110, 42 and 46, either constant $(\beta,\beta)^{\mathbb{Z}}$ for 66, the vacuum for 93) such that a configuration that agrees with B to the right of a cell r and differs from it at r keeps this property under F and F^{-1}, with the same r. (2) *Clean probes*: kinks between B and a second fixed configuration of some $\sigma^a F^b$ which, glued next to the edge, leave the configuration again of the form (1) after a bounded time, change it only near the edge, and are themselves fixed by a normalised block map. (3) *Controllability*: the moves produced by the probes reach, for every depth k, a configuration from which column $r-k$ can be read off through a permutive entry of the spacetime relation. Properties (1) and (2) give (H_1) and the commutation of h with the moves; property (3) gives the induction.

For the classes of 110, 14, 46, 24, 189 and 79 all three hold; for 189 they hold for the rule 66, whose automaton has the same fourth power. For 66, property (3) is transported from 46 by a factor map (Section 6). For a rule without any background with property (1) the construction cannot start: there is no edge at which to glue a kink, and the debris of a probe is not a configuration of the same kind. For the classes of 25 and 191 we observed, in exploratory computations that are not part of this paper, that backgrounds and probes exist, but that the orbits of the moves they produce at small depth do not reach the configurations required by (CTRL_k). This is an observation, not a result proved here, and it concerns only the probes that were tried. Whether $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle$ for these rules is open.

8.2. **Endomorphisms.** Lemma K does not assume invertibility, and it gives the following description of the endomorphisms of X_F, that is, of the block maps commuting with F.

Corollary 8.1. *Let $f \in \{110, 42, 46\}$, $F = F_f$, and let g be a block map commuting with F. Then $g\eta_0$ is $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ or a kink η_a, and $g\rho\eta_0$ is $(0,0)^{\mathbb{Z}}$, $(1,1)^{\mathbb{Z}}$ or a kink $\rho\eta_b$. If $g\eta_0 = \eta_a$ and $g\rho\eta_0 = \rho\eta_b$, then $a \equiv b \pmod 2$ and $g = \sigma^\alpha F^\beta$ with $\alpha = -\frac{1}{2}(a+b)$, $\beta = \frac{1}{2}(a-b)$.*

Proof. Since g commutes with σ and F, it maps the fixed points of σF to fixed points of σF, and those of $\sigma^{-1}F$ to fixed points of $\sigma^{-1}F$; by Lemma 3.1, which holds for 42 and 46 by Lemma 5.1(1), this gives the first statement. Suppose $g\eta_0 = \eta_a$ and $g\rho\eta_0 = \rho\eta_b$. As g commutes with σ, $g\eta_p = \eta_{p+a}$ and $g\rho\eta_p = \rho\eta_{p+b}$ for all p. By Proposition 3.2, $\sigma^{\alpha'}F^{\beta'}$ shifts the two families by $\beta' - \alpha'$ and $-\alpha' - \beta'$.

If $a+b$ is even, take (α,β) as in the statement; then $h = F^{-\beta}\sigma^{-\alpha}g$ is a block map commuting with F that fixes η_0 and $\rho\eta_0$, so $h = \mathrm{id}$ by Lemma K, which holds for 110, 42 and 46 by Theorems 4.14, 5.10 and 5.15. Hence $g = \sigma^\alpha F^\beta$.

Suppose $a+b$ is odd, and put $\alpha = -\frac{1}{2}(a+b+1)$, $\beta = \frac{1}{2}(a-b-1)$. Then $g_1 = F^{-\beta}\sigma^{-\alpha}g$ fixes every η_p and maps $\rho\eta_p$ to $\rho\eta_{p-1}$. So g_1^2 maps $\rho\eta_p$ to $\rho\eta_{p-2}$, as does σF, and both fix every η_p; by Lemma K applied to the block map $F^{-1}\sigma^{-1}g_1^2$, $g_1^2 = \sigma F$. Now g_1 maps fixed points of σ to fixed points of σ, so it induces a map ϕ on the set of the four constant configurations, which commutes with the transposition π induced by F (proof of Proposition 3.4). Since σ acts trivially on constants, $\phi^2 = \pi$. In particular ϕ is a bijection, so ϕ lies in the centralizer of π in $\mathrm{Sym}(4)$, every element of which has order at most 2; hence $\phi^2 = \mathrm{id} \neq \pi$, a contradiction. So $a+b$ is even. □

By conjugation (Section 2.3) the corollary transfers to the other rules of the classes of 110, 14 and 46, with the kinks replaced by their images. It leaves the endomorphisms that map

a kink to a constant configuration. The two constant maps onto $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$ are such endomorphisms.

Lemma K_{93} does not assume invertibility either, and the class of 79 has the following analogue.

Corollary 8.2. *Let $F = F_{93}$, and let g be a block map commuting with F. If neither gK^- nor gK^+ is a constant configuration, then $g \in \langle \sigma, F \rangle$.*

Proof. As in the proof of Proposition 7.6, g maps the four constant configurations, the fixed points of σ, to constant configurations, and the induced map ϕ commutes with the permutation π induced by F, which is the 3-cycle $(0,0)^{\mathbb{Z}} \mapsto (0,1)^{\mathbb{Z}} \mapsto (1,0)^{\mathbb{Z}}$ and fixes $(1,1)^{\mathbb{Z}}$. Hence $\phi((1,1)^{\mathbb{Z}})$ is fixed by π, so it is $(1,1)^{\mathbb{Z}}$; and ϕ maps the 3-cycle either onto itself, as a power π^j, or entirely to the fixed point $(1,1)^{\mathbb{Z}}$.

In the second case gK^-, which agrees with $g((0,0)^{\mathbb{Z}}) = (1,1)^{\mathbb{Z}}$ on a right half-line, lies in Fix_1, and agrees on a left half-line with gV_0, a configuration of period 2. The configurations of period 2 are the four constants, the six vacua and the six points of $\mathcal{W}$ (proof of Proposition 7.6). By Lemma 7.5(c), gK^- is then constant, or one of the $K^-_{j'}$; but the right constants of the $K^-_{j'}$ are $(0,0)^{\mathbb{Z}}$, $(0,1)^{\mathbb{Z}}$ and $(1,0)^{\mathbb{Z}}$ (Lemma 7.5(b)). So gK^- is constant, which is excluded.

In the first case $g' = F^{-j}g$ fixes every constant configuration, and $g'K^- = F^{-j}gK^-$ is not constant. As above, $g'K^- \in \mathrm{Fix}_1$ agrees with $g'V_0$ on a left half-line and with $(0,0)^{\mathbb{Z}}$ on a right half-line; by Lemma 7.5(c), $g'V_0$ is neither a constant nor in $\mathcal{W}$, so it is a vacuum V_{ω_0}, and $g'K^- = K^-_{j'}$ for some j'. Likewise $g'K^+ \in \mathrm{Fix}_{-1}$ agrees with V_{ω_0} on a left half-line and with $(0,1)^{\mathbb{Z}}$ on a right half-line, so $g'K^+ = K^+_{j''}$ for some j''. From here the proof of Proposition 7.6 applies without change and gives $a, b \in \mathbb{Z}$ such that $h = F^{-3b}\sigma^{-a}g'$ fixes K^- and K^+. By Lemma 7.7, which does not assume that h is invertible, $h = \mathrm{id}$, so $g = F^j\sigma^a F^{3b}$. □

The constant map onto $(1,1)^{\mathbb{Z}}$, which F fixes, commutes with F and maps both kinks to a constant configuration.

Question 8.3. Let $f \in \{110, 42, 46\}$. Is every block map commuting with F_f either an element of $\langle \sigma, F_f \rangle$ or one of the two constant maps onto $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$?

For $f = 110$ and radius at most 4 the SAT enumeration announced in [12] finds no other block map commuting with F_{110}. By Corollary 8.1, a positive answer is equivalent to the statement that an endomorphism mapping η_0 or $\rho\eta_0$ to a constant configuration is constant.

8.3. **Reversing and extended symmetries.** Following [1], an *extended symmetry* of the $\mathbb{Z}^2$-action (σ, τ) on X_F is a homeomorphism ψ of X_F such that $\psi\sigma^a\tau^b\psi^{-1} = \sigma^{a'}\tau^{b'}$ with $(a', b') = M(a, b)$ for a matrix $M \in \mathrm{GL}(2, \mathbb{Z})$ independent of (a, b); the extended symmetries form a group that contains $\mathrm{Aut}(X_F)$ as a normal subgroup, the kernel of $\psi \mapsto M$. Since $\rho F \rho = F^{-1}$ and $\rho\sigma = \sigma\rho$ for every f, the time reversal ρ induces an extended symmetry with $M = \mathrm{diag}(1, -1)$. For the rules of Theorem 1.1 the extended symmetry group therefore contains $\langle \sigma, F_f \rangle \rtimes \langle \rho \rangle \cong \mathbb{Z}^2 \rtimes \mathbb{Z}/2$, with ρ acting by $(a, b) \mapsto (a, -b)$. For 46 the class has two elements, and $\kappa \circ \mathrm{rev}$ conjugates F_{46} to itself and σ to σ^{-1} (Section 2.3); so for 46 the group contains $\mathbb{Z}^2 \rtimes (\mathbb{Z}/2)^2$, with matrices $\mathrm{diag}(\pm 1, \pm 1)$. For the classes of 110, 14 and 79, which have four elements, no composite of κ and rev maps f to itself. For $f \in \{66, 189\}$ the composites that map f to itself are id and κ, which lies in $\mathrm{Aut}(X_{F_f})$; as ρ commutes with κ, the extended symmetry group contains $(\langle \sigma, F_f \rangle \times \langle \kappa \rangle) \rtimes \langle \rho \rangle$.

Question 8.4. Is the extended symmetry group of $X_{F_{110}}$ equal to $\langle \sigma, F_{110} \rangle \rtimes \langle \rho \rangle$? Which matrices M occur for the rules of Theorem 1.1?

8.4. **Open problems.**

(1) Questions 8.3 and 8.4.
(2) Is F_{110} computationally universal?
(3) Determine $\mathrm{Aut}(X_{F_f})$ for the classes of 25 and 191 (see Section 8.1), and for the open classes in which no configuration has a stuck edge, of which there are at least 10 [12]. The absence of a stuck edge does not decide the group: the classes of 94 and 95 have none, and $\mathrm{Aut}(X_{F_{94}}) = \langle \sigma, F_{94} \rangle$, while $\mathrm{Aut}(X_{F_{95}})$ contains every finite group [13].
(4) Give a uniform criterion on f under which the induction of Theorem 4.10 closes.

Appendix A. Computer checks

No computer is used in any proof in this paper. This appendix specifies computer checks that confirm statements proved by hand; where a check is not exhaustive, this is said. Code and output accompany the paper; the directory `code/` contains, for each confirmation, the program, the command that runs it, and its output. Every program checks its claim and reports a mismatch if there is one. Throughout, a cell (x, y) is coded as $2x + y$, a configuration near the edge r is stored as two bit-words (the rows -1 and 0 of its diagram), bit d of a word standing for the column $r - d$, and the cells to the right of r are $(0, 0)$.

A.1. **Confirmations for the classes of 110, 14 and 46.** The following checks are not used in the proofs. The orbits of the moves on the light-cone window $[r - 3k + 1, r]$, computed by union–find with every move evaluated under four different paddings, are those of Theorems 4.10, 5.4 (with equality in (B)) and 5.13 for $k \le 4$, and on the window $[r - k, r]$ with several fillings to the left for $k \le 10$ (110, 42) and $k \le 8$ (46). The proofs of Theorems 4.10 and 5.4 were run as algorithms: for every configuration on $[r - k - 2, r]$ with $k \le 5$ (and three random fillings further left), and for random configurations with $k \le 8$, they produce explicit words of moves, which, replayed with the tables of Lemmas 4.4 and 5.2, end at the predicted configuration. Independently, with moves computed from probes and debris, every step of these proofs (the configuration $\tilde{b}$ lies in $\mathcal{O}_{k-1}(Fb)$, and $F^{-1}\tilde{b}$ has the stated tip) was checked against orbits computed by search, for $k \le 5$. Moves computed directly from probes and debris, by gluing $\rho\eta_{r+2}$ and separating the outgoing kink at the time $n + 2 + \varepsilon$ of Lemma 4.3(b), agree with the tables of Lemmas 4.4 and 5.2 on 1500 random configurations per rule, and the bounce formula of Lemma 4.1(c) holds in all cases tested. The conjugacy $\Psi F_{14} \Psi^{-1} = F_{42}$ with (7), the collision formula of Lemma 5.8 for $N \le 30$, and the invariance of the bond bits of 46 on all values of four consecutive cells were also checked. The figures are drawn from exact simulations, and the formulas they illustrate are checked in the same run.

A.2. **Confirmations for the classes of 24 and 189.** The following checks are not used in the proofs either. The formulas (8), the allowed triples, and the identity of Lemma 6.5(a) were checked on all arguments; the operator identities of Lemmas 6.1(a),(b) and 6.5(a), on 200 random configurations; the kinks and Lemma 6.3(b),(c), Lemma 6.5(c) and the gluing of Lemma 6.7(a) on random data; and the bounce lemma (Lemma 6.9), in all its parts, in 500 random cases with both colours. On 400 random tailed configurations, $\delta M_m(b)$ and $\delta \bar{M}_m(b)$ coincide with the moves of 46 computed from probes and debris of 46, and $M_m(b)$, $\bar{M}_m(b)$ agree with b on $(-\infty, r - |m|]$ (Lemmas 6.14 and 6.13(d)). A search of the fixed points of $\sigma^n F^n$ that are $(0, 0)$ to the left of a window finds, on the first 14 cells of the window, exactly the restrictions of $(0, 0)^{\mathbb{Z}}$ and of the $K_{p,0}$, for $F = F_{66}$ with $1 \le n \le 5$ and for $F = F_{189}$ with $n = 4$ (Lemma 6.4). Directly for 66, for $k \le 3$, a composite of moves at the times $|m| \le k - 1$ reaches a configuration whose y-row differs at $r - k + 1$ and $r - k + 2$, from 60 random tailed configurations.

A.3. **Confirmations for the class of 79.** No computer is used in the proof for the class of 79. The folder `code/c79` contains programs that confirm statements proved by hand in Section 7, together with their output; the script `run_all.sh` runs them in about seven minutes in Python on one core. Throughout, a cell is the integer $2x+y \in \{0,1,2,3\}$, the vacuum is V of Section 7.1, the cell of V_ω at i is $2V(\omega-1,i)+V(\omega,i)$, the edge is $r=0$ (by σ^r), and F, F^{-1} are applied to finite words, a word of length n giving a word of length $n-2$, so that no boundary values are invented. Every assertion listed below is checked by the program, which stops if it fails.

A.3.1. *The move graphs* Γ_L^{93}. The programs work with finite graphs of the moves, which are defined as follows.

Level-L states. For $L \ge 1$ and $c \in \Omega_r$ put $\pi_L(c) = (\vartheta(c), c_{r-L}, \dots, c_r)$. The possible values are the pairs (ϑ, w) with $w \in \mathcal{A}^{L+1}$ whose last cell differs from the cell of V_ϑ at 0; there are $6 \cdot 3 \cdot 4^L = 18 \cdot 4^L$ of them.

Lemma A.1 (window)**.** *Let $c \in \Omega_r$ have phase ω, let $m \in \mathbb{Z}$, $\omega' \in \mathbb{Z}/6$ and $D = S_m^{\omega'}(c)$.*

(a) *D agrees with c on $(-\infty, r-|m|]$, and its cells on $[r-|m|+1, r]$ depend only on ω, ω' and the cells $c_{r-|m|}, \dots, c_r$.*

(b) *Let $L \ge 1$ and $|m| \le L$. The values $Y_c(t,r)$ for $-L-1 \le t \le L$, and $\pi_L(T_m c)$, depend only on $\pi_L(c)$.*

Proof. (a) If $m = 0$, then $D = c \sqcup_r V_{\omega'}$ and there is nothing to prove. Let $m \ge 1$. By Lemma 7.2(a), applied with Λ a vacuum, $Y_D(t,i) = V(t+\omega', i)$ and $Y_c(t,i) = V(t+\omega, i)$ for $i > r$. The states at time m of D and c agree on $(-\infty, r]$; as F^{-1} has radius 1, the states at time $t \le m$ agree on $(-\infty, r-(m-t)]$, so the rows $t \le m-2$ of Y_D and Y_c agree on the columns $i < r-(m-2-t)$, and the rows $m-1, m$ agree on the columns $\le r$. Let

$$\Delta = \{(t,i) :\ -1 \le t \le m-2,\ r-(m-2-t) \le i \le r\}.$$

We show by downward induction on t that $Y_D(t,i)$, $(t,i) \in \Delta$, depends only on ω, ω' and c on $[r-m, r]$. By (2), $Y_D(t,i) = Y_D(t+2,i) \oplus f(Y_D(t+1,i-1), Y_D(t+1,i), Y_D(t+1,i+1))$. Each of the four arguments is at a point (s,i') with $s \in \{t+1, t+2\} \subseteq [0,m]$ and $i' \ge i-1 \ge r-(m-s)$. Either $i' > r$, and the value is $V(s+\omega', i')$; or $(s,i') \in \Delta$, and the induction applies; or $i' \le r$ and $(s,i') \notin \Delta$, and then $Y_D(s,i') = Y_c(s,i')$, which, as $s \ge 0$, depends only on c on $[i'-s, i'+s] \subseteq [r-m, \infty)$, where c is known from ω and $c_{r-m}, \dots, c_r$. Rows -1 and 0 of Δ are the columns $\ge r-m+1$ and $\ge r-m+2$; so D agrees with c on $(-\infty, r-m]$, and the cells $D_i = (Y_D(-1,i), Y_D(0,i))$, $r-m+1 \le i \le r$, depend only on ω, ω' and c on $[r-m, r]$. For $m \le -1$ use ρ: by the proof of Lemma 7.10(c), $S_m^{\omega'}(c) = \rho S_{-m}^{2-\omega'}(\rho c)$, where $\rho c \in \Omega_r$ has phase $2-\omega$ and ρ acts cellwise.

(b) For $t \ge 0$, $Y_c(t,r)$ depends on c on $[r-t, r+t]$, and for $t \le -1$ it is the x-bit of the cell r of $F^{t+1}c$, which depends on c on $[r-|t|+1, r+|t|-1]$; to the right of r the cells are those of the vacuum of phase $\vartheta(c)-3r$. This gives the first statement. Whether T_m acts at c, and the phase ω' of $T_m c$, depend on $\vartheta(c)$ and on z_{m-1} or z_m, which are among these values. Then $T_m c = S_m^{\omega'}(c)$, and by (a) its cells on $[r-L, r]$ depend only on ω, ω' and c on $[r-|m|, r] \subseteq [r-L, r]$. □

The interval $[r-|m|, r]$ in (a) cannot be replaced by $[r-|m|+1, r]$: for $m=-1$, $\vartheta=3$ and c ending in $c_{r-1}c_r = (0,1)(1,0)$ or $(1,0)(1,0)$, the new edge cells are $(1,0)$ and $(1,1)$.

So for $L \ge 1$ the moves at the times $|m| \le L$ define a directed graph Γ_L^{93} on the level-L states, with an edge from $\pi_L(c)$ to $\pi_L(T_m c)$ for every move; by Lemma 7.10(b) a move at a time $|m| \le L$ does not change c on $(-\infty, r-L]$.

By Lemma A.1(b), for $|m| \le L$ the level-L state $\pi_L(T_m c)$ and the edge values $Y_c(t,0)$, $-L-1 \le t \le L$, are functions of $\pi_L(c)$; the program `engine.py` computes these functions, and `graphs.py` the graphs, for $L \in \{2,3,4\}$. They are used only in the confirmations below. *Model.*

Values in $\{0,1,*\}$, with $*$ unknown; f and $\oplus$ are extended so that a value is known exactly when it is the same for all completions of the unknown arguments. The diagram of c is computed on the rows $|t| \le L+3$ and the columns ≤ 0 from the cells w on $[-L,0]$, with $*$ on every column $< -L$ and the vacuum of phase ϑ on the columns > 0. For each time m with $|m| \le L$ and $m+\vartheta \equiv 0,2$, the new phase is read from z_{m-1} or z_m, and the diagram of $T_m c$ inside the cone where it differs from that of c (rows $t \le m-2$, columns $\ge -(m-2-t)$ if $m \ge 1$; rows $t \ge m+1$, columns $\ge -(t-m-1)$ if $m \le 0$) is recomputed from the state at time m with the new vacuum, outside the cone the diagram of c being used. *Certified.* All edge values z_t, $-L-1 \le t \le L$, all new phases and all cells of $T_m c$ on $[-L,0]$ are known; the cell $-L$ is unchanged and the new edge cell is not a vacuum cell. Since a value in three-valued logic is known only if it is the same for all completions of the unknown cells, the known values are those of every c with the given level-L state; so the graphs computed are the graphs Γ_L^{93}.

A.3.2. *Confirmations.* The following programs in `code/c79` confirm statements that are proved by hand in Section 7; they are not used in the proofs.

- `check_F3_hand.py`: Lemma 7.18 and the proof of Proposition 7.19, on Γ_3^{93} (1152 states, 2688 move edges at the times $|m| \le 3$). The table and part (b) of Lemma 7.18, and the rule $T_{-m} = \rho T_m \rho$, are evaluated literally for every state. Then the proof is followed step by step: every arrow of Table 1 and of Step 4, and its mirror image; the sixteen sectors of twelve states for each cell c_{r-3}, sixty-four in all, each strongly connected; the case distinctions of Steps 2 and 3 in the order of the text, which lead from every irregular state to a sector, and from sectors to every irregular state outside $\mathcal{E}$; the formula (28), the table of J_{a_0} and the closed walks of Step 4. The pairs of states joined by the proof are exactly the 295,272 pairs of Proposition 7.19. With the argument `core`, the same checks on Γ_3^{93} built by `core.py`, an exact simulation of whole configurations with three random left parts per state and move, which are required to give the same result (`out_check_F3_hand.txt`, `out_check_F3_hand_core.txt`).
- `c_F3.py`: Proposition 7.19 and the statement after its proof, by breadth-first search on Γ_3^{93}. For every state ξ, the set $R(\xi)$ of states reachable from ξ is compared with the set $S(\xi)$ of states with the exterior cell of ξ that satisfy the condition on $(1,1)$ of Proposition 7.19: $S(\xi) \setminus \mathcal{E} \subseteq R(\xi) \subseteq S(\xi)$ for all 1152 states, and $R(\xi) \cap \mathcal{E} = \emptyset$ for every $\xi \notin \mathcal{E}$, where $\mathcal{E}$ stands for the set of the 20 level-3 states of its elements, five for each exterior cell (`out_c_F3.txt`).
- `check_F3_second.py`: the search of `c_F3.py` repeated on Γ_3^{93} built twice more, by a second three-valued engine written independently of `engine.py` (`engine2.py`, `graphs2.py`) and by exact simulation of whole configurations with three random left parts per state and move (`exact2.py`), which are required to give the same result; the two graphs are identical, and on both the statement and its sharpness hold (`out_check_F3_second.txt`).
- `check_kinks.py`: the vacuum, the F-orbits on the sixteen configurations of period 2, the identities of Lemma 7.5(a), the values $p^-(j)$, $q^+(j)$ and the half-line on which K_j^- is constant for $-12 \le j \le 12$, and, for Lemma 7.5(c), the graph of $\mathrm{Fix}_{\pm 1}$ whose edges are the words $w \in \mathcal{A}^7$ with $(F^3 w)_0 = w_{3\pm 1}$: after pruning it has 107 vertices, its recurrent components are the four constant loops, the three 2-cycles of the vacua and one component of 37 vertices containing the 2-cycles of the points of $\mathcal{W}$, which has no exit, and each vacuum cycle has exactly one path to a constant loop (`out_check_kinks.txt`).
- `check_gluing.py`: the identity of Lemma 7.11 for $r \in \{0,1\}$, every phase, every edge cell and $-3r-35 \le j \le -3r$, for every input sequence in column $r-1$, by a simulation that branches exactly where column r reads column $r-1$ (216 cases, 24,896 branches); and `check_farglue.py`: the same identity, and its time reversal, on random

whole configurations (40 forward and 40 time-reversed probes) (`out_check_gluing.txt`, `out_check_farglue.txt`).

- `check_engine.py`: 630 moves of `engine.py` on random configurations against an exact simulation of $S_m^{\omega'}$ on the whole configuration; and `check_graphs_exact.py`: the graphs Γ_3^{93} and Γ_4^{93} built by exact simulation of $F^{\pm 1}$ with two random fillings to the left of the window, which are required to give the same result, are identical to those of `engine.py` (2688 and 13,824 edges) (`out_check_engine.txt`, `out_check_graphs_exact.txt`).
- `check_closure64.py`: the classes of $\approx$ in the step of Theorem 7.20, by a closure on the 64 states of $\{0,1\}^6$ for each exterior cell (`out_check_closure64.txt`).
- `check_H2H3.py`: the proofs of Propositions 7.16 and 7.17, run as recipes (`recipe_H2.py`, `recipe_H3.py`): at time 0, for every level-L state, every rule (G1)–(G3), (K1)–(K3), every identity (Y), (Z), (W), every case of Lemma 7.15(b) and every block type that the proofs use is checked in three-valued logic, which certifies it for all configurations with that state; $L = 4$ suffices for (H_2), and $L = 6$, with 2 states completed at $L = 7$, for (H_3). The same recipes are run, at the times $|t| \le 3$, on 1500 random whole configurations with an exact simulation (`recipe_states.py`; `out_check_H2H3.txt`).
- `check_F4steps.py`: the identities of the transfer and of Steps 1, 2 and 4 in the proof of (F_4^{93}) from Proposition 7.19, for all level-4 states and random left parts; and the closure: the relation generated by Proposition 7.19 applied to b, Fb and $F^{-1}b$ joins each of the five sets of Theorem 7.20 at $k = 4$ into one strongly connected class (`out_check_F4steps.txt`).
- `check_bisim.py`: the greatest bisimulation $\mathcal{G}_L$ on Γ_L^{93} with respect to the edge values, that is the largest relation on level-L states whose pairs (ξ, ξ') have the same ϑ and the same edge values $Y(t, 0)$, $-L - 1 \le t \le L$, and are mapped to pairs in the relation by the move at every time $|m| \le L$. Starting from all 30,784 pairs with the same ϑ and edge values at $L = 3$, pairs with an image outside the current relation are removed until nothing changes: $\mathcal{G}_3$ is the diagonal, while $\mathcal{G}_2$ has 1660 pairs on 288 states. Together with Lemma 7.13 this gives a second proof of (H_4) (`out_check_bisim_L3.txt`, `out_check_bisim_L2.txt`).
- `check_F4_graph.py`: (F_4^{93}) directly: each of the five sets of Theorem 7.20 at $k = 4$ (1152, 1152, 1152, 864, 288 states) is strongly connected and closed in Γ_4^{93} (4608 states, 13,824 edges); for $k = 3$ the same test fails for four of the five sets (`out_check_F4_graph.txt`).

Use of AI tools

Large language models (Anthropic's Claude models) were used extensively in this work: to propose and develop proofs, to design, write and run the computer searches and their independent re-implementations, to check proofs and computations, and to draft the manuscript. The models are not authors. The author has reviewed the content and takes full responsibility for it. All code and computational output accompany the paper (Appendix A).

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